Connection Charges and the Private Provision of Utility Infrastructure*

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November 9, 2026

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Utilities expanding their networks to connect new users typically charge them either a flat fee or a distance-based connection charge. Under the latter, the first user to connect in a neighborhood funds a network extension that lowers their neighbors' connection charges. This externality creates a dynamic private provision game in which neighbors have strategic incentives both to coordinate and share infrastructure costs, and to delay adoption to free-ride on a neighbor's anticipated extension. I reconstruct the expansion of 177 electricity distribution networks in rural Rwanda. Using their evolving geography, I measure and instrument anticipated reductions in distance to the grid to estimate how strategic incentives affect adoption and cost incidence. Anticipating a 10% reduction in distance to the grid over the next eight weeks lowers the weekly probability of connecting by 0.91 percentage points, providing evidence of strategic delay. Among neighbors who connect within eight weeks of each other, the share connecting jointly rises from 32% to as much as 81% as they share more infrastructure. Customers with higher willingness to pay choose to connect earlier and pay higher charges, generating an elasticity of realized connection charges to willingness to pay of up to 0.24. Distance-based connection charges therefore create competing incentives for strategic delay and cooperation, while inducing endogenous price discrimination through the order in which customers choose to connect.

*I thank Rwanda's Ministry of Infrastructure for providing information and data. This paper benefited from early discussions with Gabriel Gonzalez Sutil and Joel Mugyenyi, whom I thank for insight and support. I thank my advisors, formal and informal, for their guidance: Scott Barrett, Eric Verhoogen, Daniel Björkegren, Gautam Gowrisankaran, Geoffrey Heal, Vijay Modi, and Jeffrey Shrader. I also thank participants in the Development and Sustainable Development Colloquia at Columbia University, participants at the Northeast and Heartland Workshops for helpful feedback, and Jin Hyun Park for excellent research assistance.

1 — Introduction

Suppose two immediate neighbors live near an existing utility network whose upfront connection charge increases with distance. Whichever neighbor connects first does so by paying a high price for the utility to extend the network toward them. The extension brings the second neighbor closer to the network, so they face a lower connection charge. This positive externality creates a *second-mover advantage*, a strategic incentive to wait for thy neighbor to connect first. Despite this incentive to delay adoption, such *distance-based connection charges* are common across physically networked utilities, ranging from water, natural gas, and fiber optics to electricity transmission and distribution. As each network extension can bring many users closer to the network, strategic delays can compound, slowing aggregate infrastructure expansion and progress toward any economic policy goal that depends on it.

Utilities choose distance-based connection charges because they pass through (some of) the marginal supply costs of lines, pipes, and poles to new users. For utilities, network extensions are a marginal cost of connecting first movers and a sunk cost of connecting second movers. But for users, the utility's network extension is a local public good. Distance-based connection charges and the externality they create generate a private provision of a public good game over each network extension. Users who are willing to pay nevertheless choose between adopting unilaterally, coordinating to share costs with neighbors, or waiting to free-ride on them. Strategic incentives inefficiently delay adoption, but the game's equilibria may also offer an equity benefit. Of the two neighbors, if one has a higher willingness to pay or *connection value* than the other, who is more likely to connect first?[1] As in a war of attrition over the private provision of a public good (Bliss and Nalebuff 1984), the higher-value neighbor has more to lose by waiting, so in equilibrium they choose to connect first. By connecting first, they *cross-subsidize* their lower-value neighbors. Inasmuch as connection values are related to income, these cross-subsidies would be directed toward the relatively poor. Distance-based charges, therefore, may allocate the brunt of infrastructure costs to users who value connections the most, and may be a mechanism for utilities to price-discriminate.

Utilities implicitly choose a mechanism when they set the structure of connection charges. Prospective users observe the mechanism and choose whether to connect, and their adoptions determine how the network grows and subsequent connection charges. How should connection charges be designed? The answer to this normative question depends on a positive one: How strategic are connection decisions among prospective users of utility infrastructure? If they do not connect strategically, distance-based charges simply pass through marginal costs. If they do, distance-based connection charges trade off inefficient strategic delay against equitable cross-subsidies. Policy goals, be they utility financial outcomes, network growth, or welfare, may all be primitively determined by connection charge structure.

The present paper focuses on the positive question. First, I develop a model showing how connection charges, values, and the geography of existing and prospective infrastructure combine to determine whether users find it optimal to either free-ride or cooperate with neighbors. I use the model to show that high-value neighbors connect first in equilibrium and to predict how anticipated network extensions create a wave-like pattern in adoptions. Second, with a dearth of literature on connection policy design, I briefly review and describe the landscape of utilities' revealed connection charge fee structures. I find that the modal

1. I use willingness to pay and connection value interchangeably throughout the paper.

alternative to a marginal-cost distance-based charge is an average-cost flat fee, and that some utilities pair distance-based charges with ex post first-mover refunds, cooperatives, or subscription models. These cooperation and compensation policies are consistent with how the model predicts utilities can increase connections. Third, I turn to empirics. The empirical setting is electricity distribution networks in rural Rwanda, where the utility uses a distance-based connection charge. There, I test for strategic behavior by seeing whether users delay adoption in anticipation of externalities, and I quantify the degree of price discrimination alongside other theoretical predictions of the model. In ongoing work, I nest a cooperation game within a dynamic structural model, estimate it, and use it to evaluate counterfactual real-world and theoretical connection-charge policies.

The first part of this paper maps how distance-based connection charges turn connection decisions into a dynamic public-good provision game on a network. The utility starts the game by selecting a connection charge policy and the root of the network. Prospective users decide whether to connect given the charge design and the network's current extent. Each connection extends the network, lowering the charges that users further along the extension face; then the game repeats. Users cannot exclude each other from connecting to the network and can only make binary connection choices. This makes the game related to but distinct from the literature on the private provision of public goods, which generally studies private contributions of arbitrary size to (networked) public goods in a static setting (Warr 1983; Bergstrom, Blume, and Varian 1986; Allouch 2015). It is also distinct from network formation games (Jackson and Wolinsky 1996; Bala and Goyal 2000; Haller 2016), which let players choose which network linkages to form. Finally, it is distinct from infrastructure games, where participants contribute piecewise Shapley cost shares toward the construction of infrastructure (Littlechild and Owen 1973; Littlechild and Thompson 1977; Márkus, Pintér, and Radványi 2012).

The main theoretical innovation is that dynamics reconcile the strategic complements of potential and cooperative infrastructure games, where users who do not contribute are excluded, with the strategic substitutes of voluntary public-good provision, where they are not (Littlechild and Owen 1973; Littlechild and Thompson 1977; Bergstrom, Blume, and Varian 1986; Bliss and Nalebuff 1984). I restrict cooperation to be within periods and free-riding to be across periods. This timing is natural in my setting because neighbors forming a coalition to induce a network extension must coordinate their adoption. By contrast, a user who free-rides must wait until the utility builds the extension before they can connect at a lower charge. As in the potential- and infrastructure-game literatures, Shapley cost shares result from shared extensions (Monderer and Shapley 1996, Sec. 6; Littlechild and Owen 1973; Littlechild and Thompson 1977; Márkus, Pintér, and Radványi 2012), but then the network expands, and other users can free-ride on the group's provision. In the model, some pairs of neighbors may face strategic complements while other pairs have strategic substitutes. Complements occur when coordination costs are low, when users share more infrastructure, and when infrastructure is expensive relative to connection values. While strategic complements predict that large groups of buildings will choose to connect jointly, strategic substitutes predict a snowball effect, where one first mover causes a chain reaction of subsequent adopters downstream through the distribution network graph.

The effectiveness of the mechanism depends on whether potential utility users actually behave strategically. The next part of the paper tests for strategic behavior in a case study of Rwandan electricity

distribution networks. Sub-Saharan African electricity distribution has particularly high stakes, where electricity access targets set by Sustainable Development Goal 7 and Mission 300 confront harsh fundamentals in low willingness to pay and high capital costs (Lee, Miguel, and Wolfram 2020b). Rwanda has a relatively clean institutional setting to observe the game: a distance-based policy is used by default, last-mile distribution network expansion is mostly triggered by customer connection requests, and *timestamped* geospatial electricity distribution infrastructure and customer data were previously publicly available. I use these data to reconstruct the expansion of 177 rural distribution networks over time, which allows me to observe several measures of distance and the corresponding connection charges for each building at each grid state. As a result, I can observe both sides of the externality: who provisions infrastructure and who subsequently benefits. I encode these and other measurements into a prospective customer-by-week panel.

Testing and identifying whether strategic incentives causally affect adoption behavior is substantively difficult because of five empirical challenges. First, each prospective adopter optimizes the best moment to connect by anticipating future neighbor adoption and how it endogenously reduces their current connection charge. To measure these price declines, I adapt the theoretical adjacency matrix from Elliott and Golub (2019) for an empirical setting. At each state of the grid, I simulate a hypothetical network extension for each unconnected building, and measure how much this extension would lower each neighbor's connection charge. This creates a time-varying *externality matrix* that lets me observe all potential externalities that could have been induced at a given moment, regardless of which neighbor actually adopts. Interacting the externality matrix with future neighbor adoptions allows me to reconstruct the realized decline in connection charges for each building.

Second, current (the coming week) and future (following weeks) neighbor adoption create different strategic incentives. Specifically, current neighbor adoptions create mixed incentives: to adopt through cost-sharing, and to delay by free-riding on the induced extension. By contrast, future neighbor adoptions induce delay regardless of whether users wait to coordinate with or free-ride on them. I use the maximum distance (or connection charge) change induced by current neighbor adopters and future neighbor adopters over the next eight weeks as my two primary endogenous variables, which I call *current externality exposure* and *future externality exposure*, respectively. I estimate the impact of these externality exposures on adoption hazards while controlling for, among other variables, the static connection charge each prospective customer is facing at each point in time.

Third, own behavior affects neighbors' connection charges and therefore own future price declines, a simultaneity and reflection problem (Manski 1993). I instrument future externality exposure by adapting the peers-of-peers instrument of Bramoullé, Djebbari, and Fortin (2009) to a nonlinear geographic network context. The identifying variation comes from the neighbor with the largest mean peers-of-peers building footprint, specifically the size of the externality they would induce on the focal adopter. I censor the peers-of-peers whose adoption would provide a network provision externality to the focal customer and break the exclusion restriction. For current externality exposure, I use a network-based shift-share instrument, interacting the share of common infrastructure with exchange rate shifts. Both instruments only use variation from the beginning of the period, so strategic incentives are identified only from externalities that are anticipable as prospective customers make connection decisions.

Fourth, utilities may purposely build the grid so that other customers are closer, creating an endogeneity

problem in the network itself. I construct the externality matrix in several ways, by using realized or least-cost-planned network extensions, a "unilateral" least-cost network extension, and a straight-line network extension. The latter two network extensions remove the demand pull of other anticipated users on the infrastructure network's topology, building on the geographic instruments of Lipscomb, Mobarak, and Barham (2013), Faber (2014), and Banerjee, Duflo, and Qian (2020). Finally, social peer effects may confound identification of connection charge externalities. I use a second network control to separate these effects, using a bilateral inverse-distance matrix that controls for the similarity between social and infrastructure network structure (Blume et al. 2015). My preferred specification uses the straight-line network extension and includes social peer effects as a control.

I use these empirical strategies to test for strategic behavior. I find that strategic incentives are both significant and quantitatively important; an anticipated 10% reduction in distance to the grid over the next eight weeks lowers the weekly adoption hazard by 0.91 percentage points. This effect is large. The mean weekly hazard is just 0.379 percentage points, because there are long stretches where no one adopts. Therefore, the point estimate is 2.4 times the mean weekly hazard, with delays concentrated when the grid is expanding. To ensure that strategic incentives are not driven by supply-side activity or economies of scale, I conduct a placebo test and find that nearby network extensions that do not provide externalities do not result in the same strategic delay. I conclude that strategic behavior slows utility adoption.

I then move to complementary empirical exercises motivated by theory. Using a dyadic regression among neighbor pairs that connect within eight weeks of each other, I find that when they share nearly all of their infrastructure, where the theory predicts strategic complements, 66% to 81% of them jointly adopt by connecting in the same week, depending on whether I use shared cost or distance as a metric. When they share only a minimal proportion of infrastructure and, according to theory, are in more of a strategic substitutes regime, this drops to 32% of customer pairs who jointly adopt. Finally, I show that among customers with grid electricity, those with larger buildings or higher electricity consumption pay higher connection charges in the cross-section. Connection charges have about a 0.2 elasticity with indicators that proxy for willingness to pay such as electricity consumption and about a 0.05 elasticity with indicators that proxy for wealth such as housing value and footprint area. As a uniform flat fee does not vary with income, the results indicate selection of high-value customers into private provision. If income elasticities for grid willingness to pay are about 0.23, as in Sievert and Steinbuks (2020), then the willingness-to-pay (WTP) and income elasticities I estimate are consistent with each other. Given that the mechanism uses no information about prospective customers aside from their revealed preferences, it is an effective price discrimination mechanism.

While the model can apply to any physically networked utility, the paper's empirical setting lends itself to the literature on electricity access in developing countries. Few papers study connection pricing institutions; instead, the literature has focused primarily on the impacts of providing electricity infrastructure on development outcomes, with substantially mixed evidence.[2] The limited literature that studies pricing has focused on the level of connection charges rather than the mechanisms within them. Notably, Lee,

2. For example, see Lee, Miguel, and Wolfram (2020a), Khandker, Barnes, and Samad (2009, 2013), Lipscomb, Mobarak, and Barham (2013), Dinkelman (2011), Chakravorty, Emerick, and Ravago (2016), Grogan and Sadanand (2013), Rud (2012), Masselus et al. (2024), van de Walle et al. (2013, 2017), Burlig and Preonas (2024), and Jeuland et al. (2021).

Miguel, and Wolfram (2020b) estimate the demand for grid connections in a randomized controlled trial, and Blimpo, McRae, and Steinbuks (2018) argue that low electricity consumption means that Sub-Saharan utilities struggle to generate revenue from consumption tariffs, forcing them to charge high upfront connection charges. If policymakers focus on the price level as the barrier to electrification, the natural prescription for expanding electricity access is to increase public or concessional subsidies. By contrast, the market-design focus I take in this paper suggests that policymakers can use economic mechanisms that harness strategic incentives to more cheaply achieve their objectives. The present paper may also explain empirical regularities observed in previous studies. For example, Figueiredo Walter and Moneke (2024) show that electricity access tends to be bimodal across rural Zambian villages, with industrial users present in high-access regions, and their mechanisms section highlights that cost externalities may explain this pattern. This paper suggests that the Zambian utility's distance-based connection charges may precisely be the mechanism which produces the bimodal pattern they observe.

Overall, I find that in Rwandan electricity distribution networks, prospective utility customers behave in ways consistent with strategic behavior, that outcomes predicted by my theoretical analysis are reflected in the data, and that the magnitudes of both strategic behavior and theoretical consequences are considerable. While upfront charges may seem innocuous, these findings suggest that they are not. How utilities price entry into their networks is a critical policy and market design choice.

2 — A Network Expansion Game

This section presents a toy model of the strategic game created by distance-based utility connection policies. Despite its simplicity, the toy model incorporates network cost, geometry, timing, a bargaining subgame, and policy pricing. This section presents the game, its equilibria, and propositions relevant to the empirical exercises.

2.1. Positive Externalities and Cross-Subsidies

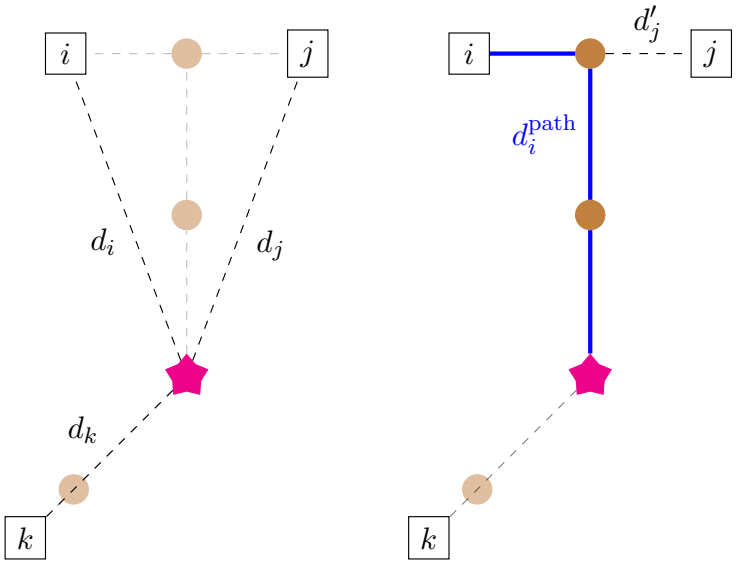
Figure 1 visualizes the externality in an infrastructure network graph.[3] Three potential players, i, j, and $k \in \mathcal{I}$, are depicted as squares. I refer to them as *players* or *adopters* throughout and as *buildings* or *prospective customers* in the empirical analysis. The left panel is the initial state of infrastructure. The magenta star is the *access point* on the existing network closest to the three players, a point from which the utility can extend the network further. The utility has a least-cost plan minimizing the total cost of connecting all players, whose *junction nodes* are light brown. These represent poles or pumps for wire- and pipe-based networks, respectively. Players do not observe the plan but may have beliefs over it. Customers observe their closest points on the existing infrastructure, their ex ante straight-line distances to those points, denoted d_i, d_j, and d_k, and the spatial configuration of neighbors around them.

Customers understand the *connection charge policy* $c(\cdot)$ chosen by utilities, regulators, or governments.[4]

3. Network nodes may be players, junction points, or sources, and edges represent the relevant linear infrastructure, such as pipes or wires. For my electricity-distribution application, nodes are buildings, poles, and transformers; edges are low-voltage lines.

4. I do not assume differentiability because the policies are typically step functions.

Figure 1 — Distance-based connection charges create externalities.



(a) Before adoption (b) After player i adopts

Notes. Panel (a) shows the initial state, in which players i, j, and k observe their Euclidean distances d_i, d_j, and d_k to their closest points on the existing infrastructure (magenta star). Ex ante connection charges are based on each player's hypothetical unilateral extension path. The utility has an expansion plan based on spatial configuration (muted colors), with the planned node and edge extensions shown in light brown. Panel (b) shows the state of the network after player i connects. The utility installs nodes (dark brown circles) and connects them with edges (blue lines) to reach i. This is the network-path distance d_i^{path}. This extension reduces player j's distance to the network d'_j, creating a positive pecuniary externality and cross-subsidy equal to $c(d_j) - c(d'_j)$. Player k remains unaffected and receives no externality because they are not en route or downstream from the expansion that player i induces.

Under a *distance-based policy*, player i pays $c(d_i)$; under a *materials-based policy*, they pay $c(d_i^{\text{path}})$. The former charges customers for the *minimum technical solution*, even if the utility chooses to extend the grid in a way amenable to connecting neighbors. The latter charges customers for whatever the utility actually builds; therefore, the realized network-path distance d_i^{path} may be longer than the straight-line distance d_i because of obstacles or the utility's least-cost plan. The charge may also depend on other variables such as player characteristics.

Without loss of generality, the right panel of Figure 1 assumes that i connects first. In response to i making a connection request and paying for it, the utility installs the two nodes and three edges to connect player i to the existing infrastructure. Player i's connection request creates a positive *externality* by provisioning infrastructure that makes j closer to the network. The extension shortens j's distance to the closest access point from d_j to d_j' and lowers their connection charge from $c(d_j)$ to $c(d_j')$. The size of the externality or *cross-subsidy* is $c(d_j) - c(d_j')$. Who receives the cross-subsidy depends on the order of connections. Under a per-node materials-based charge, costs are allocated to i, j, and k in a $(2, 0, 1)$ ratio if i connects first and $(0, 2, 1)$ if j connects first. Under a flat-fee policy, costs are allocated $(1, 1, 1)$. To achieve an equal allocation $(1, 1, 1)$, a distance-based policy would require i and j to cooperatively adopt and split the cost.

Realized connection-charge-induced externalities and cross-subsidies occur only between players who ultimately share infrastructure. Consider player k. As they have no neighbors en route or downstream from the planned network extension, they are isolated and would not induce any positive externalities or cross-subsidies upon adoption. Ex ante to adoption, however, the anticipated prospect of receiving externalities is uncertain. Players do not observe the utility's least-cost plan; if the utility had counterfactually expanded the grid toward i westwards instead of northwards, k might have received an externality from i. With incomplete information on the least-cost plan, strategic incentives may arise between neighbors who anticipate they are either en route to, or downstream of, a neighbor's network expansion.

The externality is *pecuniary, positive, local, directed, dynamic, state-dependent, and policy-induced.* I justify these properties as follows:

- **Pecuniary:** The externality affects prices.
- **Positive:** Extensions only decrease neighbors' connection charges.
- **Local:** It affects only en-route or downstream players relative to the network extension.
- **Directed:** Suppose that i were connected to the first node instead of the second in Figure 1. In this case, their adoption would reduce half of j's cost, but j's adoption would reduce all of i's cost in relative terms.
- **Dynamic and state-dependent:** Each infrastructure extension changes the state of the utility's infrastructure network. The externality from i to j would be smaller if the first junction node already existed. As infrastructure expands, possible externalities decrease in both magnitude and number.
- **Policy-induced:** The externality sizes depend on the shape of the policy $c(\cdot)$. Positive externalities arise only if $c(\cdot)$ is weakly increasing with distance.

The anticipated externality matrix I develop for the reduced-form analysis inherits these properties.

I refer to the utility's supply-side capital costs as *connection costs* and the upfront fees faced by prospective players as *connection charges*.[5] Distance-based connection charges pass through marginal utility network-expansion costs to players, but flat fees do not. Let C^m denote the utility's *total cost of shared infrastructure* for a bilateral pair of neighbors i and j who have strategic interactions.[6] Suppose the utility expects to recoup *total player charges for shared infrastructure* $C = \alpha C^m$ in connection charge revenue, where the remaining $(1-\alpha)C^m$ is subsidized with government subsidies or usage charges. Then the total amount C recovered through connection charges satisfies the bounds

$$\max\{c(d_i), c(d_j)\} \leq C \leq c(d_i) + c(d_j). \tag{1}$$

In the lower bound, C must cover at least the subsidized cost of connecting the more distant player. In the upper bound, there are weakly positive savings from sharing infrastructure instead of building two unilateral extensions. I let $M_{ij} = M_{ji} = c_i + c_j - C$ represent the symmetric *shared infrastructure savings*.

2.2. Strategic Network Game with Distance-Based Charges

I now focus on the strategic incentives between i and j in a two-player game, assuming they share some infrastructure. The players choose whether to adopt or not in each period t. A *pure strategy* selects $s_{it} \in \{\text{Adopt}, \text{Wait}\}$, while a *mixed strategy* adopts with probability $\sigma_{it} \in [0,1]$. I abstract from distances and let players observe a *unilateral* or *first-mover connection charge* $c_i = c(d_i)$ and $c_j = c(d_j)$ and *second-mover connection charges* $c_i' = c(d_i')$ and $c_j' = c(d_j')$ at the beginning of the game. Players' private *connection values* V_i, V_j, *outside options* W_i, W_j, and all connection charges are assumed to be nonnegative: $V_i, V_j, W_i, W_j, c_i, c_j, c_i', c_j' \geq 0$. For exposition, I initially assume that, conditional on the first mover adopting, the other player subsequently connects: $s_{it+1} \mid (s_{-it} = \text{Adopt}) = \text{Adopt}$. This assumption holds if and only if each player is willing to pay their residual connection charge in the following period: $V_i - c_i' > W_i$ and $V_j - c_j' > W_j$, as there are no remaining strategic incentives. Table 1 shows the normal-form payoffs of the network game under this setup.

If a *leader* unilaterally adopts, they incur charges c_i or c_j to receive their connection values V_i or V_j. The *follower* connects in the second period at the *cross-subsidized connection charge* c_i' or c_j' and receives payoffs $\beta(V_i - c_i')$ or $\beta(V_j - c_j')$. The value of second-period outcomes is discounted at rate $\beta \in [0,1]$. Greater pure time preferences, a longer wait between periods in calendar time or slower utility responses, or risk-averse preferences all make β smaller.

A participation subgame is nested within the *(Adopt, Adopt)* cell. If the two players cooperatively adopt, they incur a *bargaining cost* ρC, where $\rho \geq 0$, and divide the total shared-infrastructure cost through *cost shares* that satisfy $\kappa_i + \kappa_j = (1+\rho)C$. I model κ using Shapley cost shares, which are a standard solution concept for cooperative games and also arise as an equilibrium outcome in non-cooperative bargaining games with transferable utility (Shapley 1953; Hart and Mas-Colell 1989, 1996). The Shapley cost share of

5. These terms follow electricity industry parlance (Golumbeanu and Barnes 2013), but I use them generically.

6. In the figure, it would represent the total cost of the two junction nodes and the edges that connect them.

Table 1 — Payoff matrix for the two-player dynamic networked public-good provision game

		Player j	
		Adopt	**Wait**
Player i	**Adopt**	$V_i - \kappa_i,\ V_j - \kappa_j$	$V_i - c_i,\ \beta(V_j - c'_j)$
	Wait	$\beta(V_i - c'_i),\ V_j - c_j$	$\beta W_i,\ \beta W_j$

Notes. Player values are V_i and V_j. Unilateral connection charges are $c_i = c(d_i)$ and $c_j = c(d_j)$. Cooperative connection charges κ_i and κ_j are the result of bargaining over the shared extension. If player i waits while player j adopts first, player i's subsequent connection charge is $c'_i = c(d'_i)$, and similarly for j. The parameter $\beta \in [0, 1]$ weights second-period payoffs relative to first-period payoffs. The outside options are W_i and W_j. These payoffs assume that if one player adopts in the first period, the other player is willing to pay the connection charge: $V_i - c'_i > 0$ and $V_j - c'_j > 0$.

player i conditional on reaching *(Adopt, Adopt)* is

$$\kappa_i = \frac{1}{2}\left[(1+\rho)C + c_i - c_j\right] = c_i - \frac{1}{2}\left[M_{ij} - \rho C\right]. \tag{2}$$

I derive the first equality of Equation 2 in Appendix C.1.[7] The second equality of Equation 2 substitutes in M_{ij}. Cooperation saves each player half the shared infrastructure savings, $\frac{1}{2}M_{ij}$, at a bargaining cost of $\frac{1}{2}\rho C$. Conditional on reaching *(Adopt, Adopt)*, Shapley cost shares, unlike Shapley values, do not depend on willingness to pay. Cooperation may not be optimal if the bargaining cost exceeds the infrastructure savings, $\rho C > M_{ij}$. Therefore, utilities can facilitate coordinated adoption by reducing ρ. Joint adoption requires both players to adopt simultaneously, so it is contemporaneous. By contrast, the benefits from free-riding accrue in the future, are anticipated at t, and are discounted to the present in $\beta(V_j - c'_j)$.

It will be helpful to define special cases for distance-based charges and outside options.

Definition 1 (Mechanical Distance-Based Policy)**.** Infrastructure costs are passed through at a flat pass-through rate α. Let c_i^m and c_j^m denote the utility's initial infrastructure costs of connecting players i and j so that the corresponding connection charges are $c_i = \alpha c_i^m$ and $c_j = \alpha c_j^m$. Second movers pay the residual charge, $c'_i = \alpha c_i^{m\prime} = C - c_j = \alpha(C^m - c_j^m)$. Then for all $k \in \{i, j\}$, the externality, total connection charges, and Shapley cost shares satisfy

$$M_{ij} = c_k - c'_k \qquad C = c_k + c'_{-k}, \qquad \kappa_k = \frac{1}{2}[(c_k + c'_k) + \rho(c_k + c'_{-k})].$$

Savings are symmetric in absolute terms but asymmetric in relative terms because neighbors may be upstream or downstream of each other.[8] In the mechanical baseline, infrastructure cost spillovers are fully

7. The reader may ask what would happen if I reversed the design, centering the participation game and assigning the free-riding value as the opportunity cost of the participation game. This would imply that the participation game concerns how to split the consumption value of utility service rather than the connection cost. It would also make the repeated game less tractable.

8. Three example graphs may help illustrate. If the two players' paths do not overlap, then $M_{ij} = 0$ and there is no externality. If one path is fully nested within the other, so player i uses no infrastructure that would not also be used by player j, then $M_{ij} = c_i$ and $C = c_j$: if j adopts first, player i's connection is free, $c'_i = 0$, and the size of j's cross-subsidy to i is i's initial cost $c_i - c'_i = c_i$; if i adopts first instead, player j's second-mover charge is the residual cost $c'_j = c_j - c_i = C - M_{ij}$, the total less the shared

passed through at rate α to connecting players. Therefore, α never changes whether an externality exists, the direction of cross-subsidies, or whether infrastructure savings cover the bargaining cost.

Consider two options for the game's temporal structure:

Definition 2 (One-Shot Game)**.** The game ends if both players wait, so the outside option value is $W_i = W_j = 0$. Second movers still receive $\beta(V_k - c'_k)$.

Definition 3 (Repeated Game)**.** In a stationary *repeated game*, if both players wait, the same game is played again next period. The outside option W_k is the undiscounted one-period-ahead equilibrium payoff with present value βW_k.

In the following subsection, I analyze when each cell of Table 1 is a pure-strategy Nash equilibrium (PSNE) in the one-shot game. Next, I present mixed-strategy Nash equilibria (MSNE) with a generic outside option, resembling a chicken game. Finally, I analyze MSNE in a repeated game with private values, which resembles a war of attrition.

2.3. Pure-Strategy Nash Equilibria in a One-Shot Game

In this subsection, I assume the game is one-shot. Adopting is a dominant strategy for k if cooperation beats the second-mover payoff, $V_k - \kappa_k \geq \beta(V_k - c'_k)$, and unilateral adoption beats the outside option, $V_k - c_k \geq 0$. Waiting is a dominant strategy if players are unwilling to pay the unilateral upfront connection charge, $0 \geq V_k - c_k$, and the second-mover payoff is preferred to cooperative adoption, $\beta(V_k - c'_k) \geq V_k - \kappa_k$. The game at its core consists of two comparisons: the static incentive of whether connection values exceed the unilateral connection charge, and the strategic incentive of whether cooperation exceeds second-mover surplus.

From the point of view of utilities that target connection quantities as a key performance indicator, such as electricity access measures in electricity distribution, *(Adopt, Adopt)* is ideal because both players connect immediately, achieving player surplus $V_i + V_j - (1 + \rho)C$. It is a PSNE if one period of connection value exceeds the difference between cooperative and second-mover charges $(1 - \beta)V_i \geq \kappa_i - \beta c'_i$, a demanding constraint when the second period may be as little as a week away. The right-hand side is $\kappa_i - \beta c'_i = c_i - \beta c'_i - \frac{1}{2}(M_{ij} - \rho C)$, so for joint adoption to occur, the discounted externality $c_i - \beta c'_i$ must be small, and cooperation costs must not be too high.[9]

A *(Wait, Wait)* PSNE may often be relevant when both players are unwilling to pay the upfront costs, $0 \geq V_k - c_k$. This PSNE is inefficient when there are gains from cooperation, $V_k - \kappa_k \geq 0 \geq V_k - c_k \forall k \in \{i, j\}$, but no player deviates to adopt. There can be a prisoner's dilemma if waiting is weakly dominant and free-riding beats cooperation in $\kappa_k - \beta c'_k \geq (1 - \beta)V_k \quad \forall k \in \{i, j\}$. There can also be a stag hunt if *(Wait, Wait)* and *(Adopt, Adopt)* are both equilibria; low connection rates can be the result of a coordination failure. Under distance-based charges, low connection rates could reflect strategic incentives rather than static WTP.

infrastructure. In a Y-shaped graph, both players have some private infrastructure and a shared trunk, so $0 < M_{ij} < \min\{c_i, c_j\}$: the first mover pays $c_i > M_{ij}$ and the second pays the residual cost of the total less the trunk $c'_j = c_j - M_{ij}$.

9. Under the mechanical distance-based policy, this expression is $\frac{1}{2}c_i + (\frac{1}{2} - \beta)c'_i + \frac{1}{2}\rho C$, because the size of the externality is the shared infrastructure.

The most interesting behavior is in the *(Adopt, Wait)* or *(Wait, Adopt)* cells. It is a PSNE if at least one player is willing to pay the full initial cost of connection, $V_i - c_i \geq 0$, but the other player prefers to wait:

$$c_j - \beta c'_j - \frac{1}{2}(M_{ij} - \rho C) = \kappa_j - \beta c'_j \geq (1-\beta)V_j. \tag{3}$$

Because the condition compares upfront connection charges or cooperation costs to just one period of flow benefits, it is not demanding. This PSNE nests three special cases. First, when Equation 3 holds and the follower is willing to pay the upfront charge, there is a *chicken game* in which the leader provisions the public infrastructure and the follower free-rides. I will return to analyzing the chicken game's MSNE in the following subsection. Second, when Equation 3 holds and the follower is unwilling to pay the upfront charge unilaterally, it produces *price discrimination* that expands connections in equilibrium. Finally, free-riding and price discrimination can occur simultaneously, in which case the charge can redistribute without expanding equilibrium quantities. I call $\beta(V_i - c'_i)$ the *second-mover payoff* when the second mover cannot afford the initial cost and the *free-riding payoff* when they can.

Free-riding occurs if both players are willing to pay the initial connection charge, $V_i - c_i \geq 0$ and $V_j - c_j \geq 0$, but one chooses to *Wait* because $\kappa_j - \beta c'_j > (1-\beta)V_j$. The leader provisions the shared infrastructure unilaterally, while the follower saves the externality cost, $c_k - c'_k$, even though it can pay the initial charge. Free-riding delays adoption but does not prevent it. Players wait for their neighbor to connect, which lowers their connection charge, and then take advantage of the cross-subsidy immediately afterward.

Price discrimination can expand the extensive margin of who can connect to utility infrastructure. Suppose high type j is willing to pay $V_j - c_j \geq 0$, a unilateral connection charge higher than a flat fee, but low type i is willing to pay only a second-mover connection charge

$$V_i - c'_i \geq 0 \geq V_i - c_i. \tag{4}$$

High type j *selects* into early adoption and connects at a high price, which creates a cross-subsidy of size $c_i - c'_i$ to i, allowing them to connect at a low price as a second mover. Low type i would not connect unilaterally. This price discrimination happens without coercion, as it is not from j's benevolence that i receives a cheap connection, but instead from j's own self-interest. Price discrimination can increase the quantity of connections.

Price discrimination can also redistribute on the intensive margin. Without loss of generality, suppose players i's and j's net surpluses from unilateral adoption are ordered $V_i - c_i \geq V_j - c_j$. Lemma 1 shows that the parameter region in which *(Wait, Adopt)* is a Nash equilibrium is contained within the parameter region in which *(Adopt, Wait)* is a Nash equilibrium. If one of these is a unique asymmetric equilibrium, the higher-surplus neighbor adopts first in it.

Lemma 1 (Higher Net-Surplus Players Lead If There Is a Unique Asymmetric Equilibrium)**.** Without loss of generality, suppose that i has a higher unilateral adoption surplus $V_i - c_i \geq V_j - c_j$. If the first-mover relative surplus is larger than the second-mover relative surplus, $(V_i - c_i) - (V_j - c_j) \geq \beta\left[(V_i - c'_i) - (V_j - c'_j)\right]$, the support of *(Wait, Adopt)* being a PSNE is contained within the support of *(Adopt, Wait)* being a PSNE.

Therefore, if there is a unique equilibrium, it is *(Adopt, Wait)* and player i is the first mover. If there are multiple equilibria, joint player surplus is weakly greater in the equilibrium where the higher-value player adopts first.

Proof. See Appendix C.2.

The conditions for Lemma 1 are weak. Some ordering of net surplus will generically hold: $V_i - c_i \geq V_j - c_j$. The second condition $(V_i - c_i) - (V_j - c_j) \geq \beta \left[(V_i - c_i') - (V_j - c_j')\right]$ holds as long as i is not much further from the existing infrastructure in connection-charge terms than their difference in values. While support containment does not necessarily mean that the higher-surplus player will cross-subsidize the lower-surplus player in general, there are certain parameter supports where they will. If one further assumes positive income elasticities and symmetric costs, Lemma 1 shows that the cross-subsidy is directed, redistributing from wealthier to poorer neighbors. Later, Proposition 2 strengthens this lemma to show that when the game is repeated, there is a unique symmetric cutoff equilibrium in which price discrimination occurs.

2.4. Mixed-Strategy Equilibria

In this subsection, I no longer assume that the game is one-shot and assume arbitrary continuation values. A chicken game exists when both players' unilateral surplus exceeds their continuation values, $V_k - c_k > \beta W_k$, but both players have incentives to free-ride via Equation 3. An MSNE can then occur where players randomize between adopting and waiting. I limit the following discussion to MSNE predictions that differ from those of the one-shot PSNE. These include strategic complements and substitutes, a refinement of Lemma 1 that makes price discrimination the unique equilibrium, and how players should react when they anticipate a future neighbor externality.

A mixed-strategy Nash equilibrium exists if both players are unilaterally willing to pay and free-riding benefits exceed savings from cooperative adoption. As mixed-strategy equilibria imply indifference between strategies conditional on the neighbor's mixed strategy, i's equilibrium strategy is a function of j's primitives

$$\sigma_i = \frac{\overbrace{V_j - c_j - \beta W_j}^{\text{WTP given outside option}}}{\underbrace{\beta(V_j - c_j')}_{\text{free-riding benefit}} - \underbrace{\tfrac{1}{2}(M_{ij} - \rho C)}_{\text{cooperative savings}} - \underbrace{\beta W_j}_{\text{outside option}}}. \tag{5}$$

Comparative statics of Equation 5 show that adoption increases in neighbor value V_j, decreases in the neighbor's outside option W_j and standalone charge c_j, if c_j', M_{ij}, and C are held fixed. If cooperation is useful, $M_{ij} \geq \rho C$, then adoption also decreases in the discount factor.

My first proposition is that the MSNE can result in both *strategic substitutes and strategic complements*.

Proposition 1 (Strategic Complement and Substitute Regimes)**.** Assume that continuation values W_i are fixed. There are strategic complements if and only if

$$\frac{1}{2} \underbrace{(M_{ij} - \rho C)}_{\text{cooperation benefits}} + \underbrace{\beta W_i}_{\text{outside option}} > \underbrace{\beta(V_i - c_i')}_{\text{free-riding payoff}}, \tag{6}$$

and strategic substitutes if the opposite holds.

Proof. Strategic complements occur if and only if the cross-partial of payoffs to own and neighbor strategies is positive; strategic substitutes occur if it is negative. Player i's payoff is $\pi_i(\sigma_i, \sigma_j) = \sigma_i\sigma_j(V_i - \kappa_i) + \sigma_i(1-\sigma_j)(V_i - c_i) + (1-\sigma_i)\sigma_j\beta(V_i - c_i') + (1-\sigma_i)(1-\sigma_j)\beta W_i$. Its derivative is the marginal value of adoption,

$$\frac{\partial \pi_i}{\partial \sigma_i} = \sigma_j(V_i - \kappa_i) + (1-\sigma_j)(V_i - c_i) - \sigma_j\beta(V_i - c_i') - (1-\sigma_j)\beta W_i.$$

Taking the second derivative in neighbor strategies gives the cross-partial

$$\frac{\partial^2 \pi_i}{\partial \sigma_i \partial \sigma_j} = \underbrace{\left[(V_i - \kappa_i) - \beta(V_i - c_i')\right]}_{\text{cooperate or free-ride?}} - \underbrace{\left[(V_i - c_i) - \beta W_i\right]}_{\text{unilaterally adopt or continue?}} = \tfrac{1}{2}(M_{ij} - \rho C) + \beta W_i - \beta(V_i - c_i').$$

The last equality substitutes $\kappa_i = c_i - \frac{1}{2}(M_{ij} - \rho C)$. □

Proposition 1 shows how primitives interact to define a *strategic regime*. When the gains from sharing infrastructure M_{ij} are large relative to bargaining costs ρC, players have an incentive to coordinate and adopt jointly. When the outside option W_i is large, players would delay to coordinate adoption rather than adopt unilaterally. By contrast, second-mover payoffs $\beta(V_i - c_i')$ create an incentive to free-ride and favor strategic substitutes. Recall that β represents how the second-period payoffs are discounted relative to the first, and so increases with a responsive utility, short time between periods, low discount rates, and low risk aversion. All these would create strategic substitutes, as would lower pass-through.[10]

All primitives enter Proposition 1, so strategic regimes will vary a lot across neighborhoods and even within bilateral dyadic pairs of neighbors, and across utility types, jurisdictions, contextual, spatial, and social network settings. Given the Adopt and Wait strategies, the strategic regime can be observed in relative adoption timing. Strategic complements predict *synchronous* adoption. Coalitions bargain internally over how to share infrastructure-extension costs and collectively negotiate the extension with the utility. The utility then connects all coalition members to utility infrastructure at once. Strategic substitutes predict *sequential* adoption. The first mover provisions the network extension, and then their neighbor observes a lower cost and adopts only then. These patterns will allow me to distinguish free-riding from cooperation and strategic substitutes from complements when I turn to empirics.

Neither strategic substitutes nor strategic complements are necessarily preferable to the other. Although strategic substitutes create a dynamic efficiency cost through free-riding, the equity benefits resulting from Lemma 1 may mean that more players can connect in equilibrium. Furthermore, strategic substitutes do not necessarily mean that connections are slow; the sequence may result in a "snowball" effect once the first mover induces the initial infrastructure (Dahl, Løken, and Mogstad 2014). Conversely, strategic complements are not necessarily good. They can result in multiple equilibria with high and low adoption,

10. A higher pass-through rate (smaller subsidies) expands the strategic-complements regime. Under a mechanical baseline policy, for example, each infrastructure-related charge in Equation 6 is proportional to α, so the cross-partial equals $\alpha\left[\frac{1}{2}(M_{ij}^m - \rho C^m) + \beta c_i^{m\prime}\right] - \beta(V_i - W_i)$. With the spatial structure of shared infrastructure M_{ij}^m and the second-mover cost $c_i^{m\prime}$ held fixed, the cross-partial is zero when $\alpha_i^* = \beta(V_i - W_i)/\left[\frac{1}{2}(M_{ij}^m - \rho C^m) + \beta c_i^{m\prime}\right]$. The network switches from strategic substitutes to strategic complements when α moves above this threshold.

and players may not connect because they are stuck in a coordination failure.[11] However, both strategic incentives delay network adoption relative to what static incentives alone would predict. Under strategic complements, a player delays to coordinate with neighbors who may not yet be ready. Under strategic substitutes, they delay in the hope that the neighbor will move first.

2.5. Price Discrimination and Anticipation in the War of Attrition

Assume now a repeated game in Definition 3, and that connection values are private. This results in a war of attrition. I show two results. First, Proposition 2 refines Lemma 1 to show that with private values, there exists a symmetric cutoff equilibrium in which the high-type neighbor selects into adopting first. Under strategic substitutes, this cutoff equilibrium is unique. Second, I describe the shape of mixed-strategy equilibria around an anticipated neighbor adoption in Proposition 3, a result that provides testable hypotheses for my empirical exercises.

For simplicity, I assume symmetric charges: $c_i = c_j = c$, $c'_i = c'_j = c'$, and $\kappa_i = \kappa_j = \kappa$. This reflects Figure 1 and does not change the qualitative results.[12] Connection values V_k are known only to player k and are now random values drawn independently from a common distribution F that has positive density f on $[\underline{V}, \overline{V}]$. The charges c, c', and κ, the discount factor $\beta < 1$, and the distribution F are all common knowledge, but individual connection values V_i, V_j are private. Let $W(V_k)$ replace the continuation value W_k for player k, so that I can assume it is differentiable in the connection value and further that it satisfies the condition $\beta W'(V_k) < 1$: higher connection values raise adoption payoffs more than they raise the discounted continuation value. Then let $s_j(V_j) \in \{\text{Adopt}, \text{Wait}\}$ denote a neighbor's decision rule, now a function of their connection value draw, and let j's mixed strategy be $\sigma_j = \Pr\big[s_j(V_j) = \text{Adopt}\big]$. Player i's net benefit from adopting is

$$\Delta_i(V_i; \sigma_j) \equiv \frac{\partial \pi_i}{\partial \sigma_i} = \sigma_j \underbrace{\big[(V_i - \kappa) - \beta(V_i - c')\big]}_{\text{cooperate or free-ride?}} + (1 - \sigma_j) \underbrace{\big[(V_i - c) - \beta W(V_i)\big]}_{\text{unilaterally adopt or continue?}}. \tag{7}$$

Equation 7 shows that i performs a cost-benefit analysis when deciding whether to adopt. They compare cooperation to free-riding and unilateral adoption to continuing the game, and weight these comparisons by neighbor adoption probabilities. The neighbor's adoption probability σ_j determines the relative importance of static and strategic incentives. The following proofs in Lemma 2 and Proposition 2 adapt war-of-attrition and single-crossing intuition from the dynamic public good provision game of Bliss and Nalebuff (1984).

Lemma 2 (Best Responses Are Cutoff Strategies)**.** Player i's best response to a neighbor adoption probability σ_j is a cutoff, a threshold $V^*(\sigma_j)$ such that player i adopts if and only if $V_i \geq V^*(\sigma_j)$. When the cutoff is interior,

$$\frac{dV^*}{d\sigma_j} = -\frac{\partial \Delta_i / \partial \sigma_j}{\partial \Delta_i / \partial V_i}, \tag{8}$$

is positive under strategic substitutes and negative under strategic complements. An increase in the

11. More precisely, anticipating that a neighbor will soon be ready to adopt raises W_k and decreases the payoff from adopting now: $\frac{\partial^2 \pi_i}{\partial \sigma_i \partial W} = -(1 - \sigma_j)\beta < 0$. If bargaining were instantaneous, there would be no delay.

12. Asymmetric charges would produce player-specific cutoff strategies. The results continue to hold conditional on charges.

neighbor's adoption probability σ_j under strategic substitutes decreases own adoption by raising the cutoff. Under strategic complements, it increases adoption by decreasing the cutoff.

Proof. See Appendix C.3.

Proposition 1 says the strategic regime determines whether prospective customers adopt jointly or wait in response to higher neighbor adoption probabilities. While it follows directly from the definitions of strategic substitutes and complements, the lemma's form helps to prove Proposition 2.

Proposition 2 (Higher-Value Players Connect First)**.** Given a fixed continuation-value function $W(V)$, a symmetric cutoff equilibrium exists. If strategic substitutes hold, it is unique. If strategic complements hold, there may be more than one symmetric cutoff equilibrium. In any such equilibrium, players with $V_k \geq V^*$ connect first, while players with $V_k < V^*$ wait. Therefore, adoption decisions are weakly increasing in connection value, so high-connection-value players select into adopting earlier at the first-mover charge, while low-connection-value players wait to face the second-mover charge.

Proof. See Appendix C.4.

The proposition sharpens Lemma 1.[13] In Lemma 1, multiple equilibria need not involve price discrimination; here, there may still be multiple equilibria, but every cutoff equilibrium price-discriminates. Specifically, whenever a lower-value player adopts, their higher-value neighbor will too, so the higher-value neighbor is weakly more likely to adopt first. Proposition 2 shows that price discrimination is an equilibrium outcome.[14]

Finally, Proposition 3 describes what happens in an event study around a foreseeable neighbor's connection. It forms the basis of testable hypotheses for my primary reduced-form exercises. I compare two versions of the war of attrition. I refer to the game described in Proposition 2 as the *baseline war* and compare it against a *foreseeable event war*. The foreseeable event is that a third player k will connect at time T^k, and both players i and j foresee it. Player k provides both i and j with an externality, decreasing their symmetric connection charge from c to $c^k \in [c', c]$. By comparing the option value of waiting for the foreseeable event against that of the baseline war, the proposition describes an adoption pattern shaped like a wave.

Proposition 3 (Wave-Shaped Adoption Hazards)**.** I assume strategic substitutes, that k is more upstream relative to the existing infrastructure than the bilateral pair i, j, so that the externality is of intermediate size $c^k \in [c', c]$ and $\kappa^k > \kappa$, and that the continuation values of the baseline war $W(V)$ are fixed so that beliefs about neighbor values do not update.

13. There, in the one-shot game with complete information, high-value players will weakly tend to connect first because a unique equilibrium may exist where they do so. However, when multiple chicken equilibria exist, the lemma says nothing about who adopts first.

14. In Bliss and Nalebuff (1984), low-cost, high-net-benefit players voluntarily provided a shared good first because they had more to lose when the game continues into the next period. Here, although charges are not voluntary, the intuition is the same. High-value players select into early adoption because they have more to lose from delaying connection. They pay for earlier, more expensive connections that provision the shared infrastructure. This cross-subsidizes low-value players, who wait for the lower second-mover charge. If low-value players would not otherwise connect, the cross-subsidy expands connections.

Both wars have a unique symmetric cutoff equilibrium because of Proposition 2. Let V^*_{t,T^k} and V^* denote the cutoff strategies of the foreseeable event war and the baseline war respectively; the latter is stationary. And let $\sigma^*_{t,T^k} = 1 - F(V^*_{t,T^k})$ and σ^* denote the corresponding adoption probabilities.

I collectively refer to the inequalities $\beta^{T^k-t}(\overline{V}-c^k) \geq \overline{V}-c$ and $\beta(\overline{V}-c') > \overline{V}-\kappa$, as a freeze condition, because these make waiting a dominant strategy in both wars for even the upper-bound private value. The first inequality holds if at time t, waiting for the anticipated event beats unilateral adoption. The second inequality holds if free-riding beats cooperation in the stationary baseline. By comparison to the baseline, the foreseeable event war affects adoption as follows:

1. *Baseline unaffected.* If the freeze condition does not hold, and $\beta^{T^k-t}(V^* - c^k) \leq \beta W(V^*)$, the anticipated event does not change current payoffs and k's anticipated adoption has no effect. Cutoffs are equal in both wars $V^*_{t,T^k} = V^*$, and adoption is unaffected.

2. *Increasing delay ahead of the externality.* If k's anticipated adoption makes waiting more attractive $\beta^{T^k-t}(V^* - c^k) > \beta W(V^*)$ but is not strong enough to make the freeze condition hold, then cutoffs increase $V^*_{t,T^k} \geq V^*$. Players reduce adoption to wait for the event $\sigma^*_{t,T^k} = 1-F(V^*_{t,T^k}) \leq 1-F(V^*) = \sigma^*$. Because $\beta^{T^k-t}(V^* - c^k)$ increases into the event, adoption decreases into it.

3. *Frozen adoption, an optional phase.* When the freeze condition holds, waiting is optimal for every type $V \in [\underline{V}, \overline{V}]$ because holding out for player k weakly dominates unilateral or joint adoption. Both players wait, and current adoption is zero.

4. *Joint adoption with k at time T^k.* When player k adopts, player i can join k's connection request and pay the Shapley cost share κ^k. Joining k's coalition to adopt at time T^k is preferred to waiting one more period and connecting at the post-event charge c^k if $V_i - \kappa^k \geq \beta(V_i - c^k)$. This means that i has a higher value than a threshold $V_i \geq V^{\text{joint}}_{T^k} \equiv (\kappa^k - \beta c^k)/(1-\beta)$, which will allow me to segment the private value distribution. Let a *survivor cutoff*, the lowest adoption cutoff players faced before T^k, be denoted $V^{\text{surv}}_{T^k} \equiv \min_{s<T^k} V^*_{s,T^k}$. The players remaining just before T^k are those with $V_i < V^{\text{surv}}_{T^k}$. Joint adoption with k, conditional on reaching the event, occurs with positive probability $\big(F(V^{\text{surv}}_{T^k}) - F(V^{\text{joint}}_{T^k})\big)/F(V^{\text{surv}}_{T^k})$ when $V^{\text{joint}}_{T^k} < V^{\text{surv}}_{T^k}$.

5. *Realization of free-riding after time T^k.* A player who delayed or froze adoption for k's externality $c - c^k \geq 0$ can free-ride, connecting at charge c^k after k adopts. Let $W^k(V_i)$ denote the continuation value of the war that resumes after the event. As in the baseline war, adopting beats waiting if $V_i - c^k \geq \beta W^k(V_i)$. These sequential adopters have connection values $V_i < V^{\text{joint}}_{T^k}$, so that waiting one additional period is preferred to joining k.

6. *A new war at a higher baseline.* If neither i nor j specifically delayed adoption to receive the free-riding benefit, then a lower c^k replaces the original connection charge c, and a bilateral war resumes at lower cost. Holding $W(V)$ and c' fixed, let $V^{*,k}$ denote a new symmetric cutoff. As $c^k \leq c$, the new cutoff is weakly lower, $V^{*,k} \leq V^*$, and the new baseline adoption probability is weakly higher, $\sigma^{*,k} = 1 - F(V^{*,k}) \geq 1 - F(V^*) = \sigma^*$. In the resumed war, players adopt exactly when $V_i \geq V^{*,k}$.

Overall, when the cutoffs are ordered $V^{*,k} \leq V_{T^k}^{\text{joint}} \leq V_{T^k}^{\text{surv}} \leq V^*$, the phases sort connection values into regions by cutoff. Highest-value players $V_i \geq V_{T^k}^{\text{surv}}$ adopt unilaterally. High-value players in $[V_{T^k}^{\text{joint}}, V_{T^k}^{\text{surv}})$ adopt jointly with k at time T^k. Lower-value players in $[V^{*,k}, V_{T^k}^{\text{joint}})$ free-ride and connect after. Lowest-value players $V_i < V^{*,k}$ remain unconnected and resume the war.

Proof. See Appendix C.5.

Proposition 3 shows a wave-like pattern around a neighbor externality. Graphically, adoptions decline steeply into the event, spike as joint adoption occurs, and regress to a higher baseline. I show an event study with this pattern in section 5. Empirically, it provides the basis for reduced-form predictions regarding neighbor externalities. I will also present a dyadic regression that shows the empirical magnitude of Proposition 1, and a cross-section that observes Proposition 2's price discrimination and selection. I now turn to typical connection costs and charges in electricity distribution, before turning to the case study.

3 — Electricity Connection Costs and Charges

Electric utilities participate in at least four markets. Just as the wholesale electricity market is split into transmission interconnections and electricity generation, retail electricity markets are split into distribution connections and electricity consumption. For brevity, I refer to the market for retail electricity connections as the infrastructure *connections market* and to retail electricity consumption as the *electrons market.* Because the two policy bookends, flat fees and distance-based charges, reflect the choice of whether to pass on supply-side costs, I describe the supply-side costs of the connections market here.

3.1. Physical Spatial Marginal Connection Costs

Table 2 — Connection materials and marginal costs increase with the distance to existing infrastructure

Distance	Description	Additional materials	Marginal cost
Within 30–50 m	Service drop	Drop, meter, wiring	<USD 100
50–200 m	Under-grid / within coverage	Several poles, lines	USD 100–2,500
200–600 m	Last mile / transformer communities	More poles, lines	USD 2,500–10,000
Further	Off-grid (new transformer required)	Transformer	≫USD 10,000

Notes. Indicative costs for new marginal consumers at each distance bracket in East Africa. Exact distances depend on utility choices such as pole heights, voltage choice, pole costs, exchange rates, and other factors. Terms and costs are sourced from the existing energy and development economics literature: Blimpo and Cosgrove-Davies (2019) for coverage, Lee et al. (2016) for transformer communities and under-grid, and Chakravorty, Emerick, and Ravago (2016) for "last mile."

The cost of marginally expanding the distribution grid is typically a non-decreasing, stepwise function of distance from the closest existing infrastructure.[15] The existing academic literature has used jargon such as "service drop," "under-grid," "last mile," and "off-grid" to denote different distance-cost ranges. I

15. Capacity constraints can generate additional jumps in marginal cost.

summarize these in Table 2.[16] There are two edge cases in which supply-side costs may not increase with distance. Existing infrastructure may reach capacity and require a non-marginal reinforcement to serve an additional customer. Alternatively, line losses may make a new separate grid the least-cost technical solution. Otherwise, marginal connection costs increase with the materials required that correspond to the prospective customer's distance from an access point.

Expanding the grid in a particular direction lowers the utility's supply costs of connecting consumers in that direction. Supply-side costs are poorly summarized by a stable one-dimensional marginal-cost or supply curve because costs depend on the grid state and direction of expansion. Zvoleff et al. (2009) attempt to model the supply curve by assuming the planner's least-cost sequence of connections. The resulting curve is nonlinear and poorly behaved, visually appearing as a random walk, because the next consumer in a least-cost sequence can be near or far from the existing network depending on spatial clustering. The intersection of demand and such a supply curve in one dimension is meaningless because ordering consumers by declining WTP would not match the supply-side geographic order. I use evolving isocost regions to provide a tractable description of the relationship between supply costs, distances, spatial configuration, and the state of the grid.

Figure 2 shows that isocost regions evolve as the distribution network expands. From the high-voltage backbone network, a substation steps the high voltage down to medium-voltage feeders, from which transformers step the voltage down further to low voltage.[17] When a utility installs a *transformer*, consumers within range fall into the yellow "last mile" where connections are feasible.[18] Electricity distribution networks expand with *network extensions* that emanate from the transformer. In an overground distribution network, the extensions are discrete because wires hang from poles spaced 30–60 meters apart. The distance between poles depends on terrain, obstacles, pole height, and how much wires droop. As network extensions emanate outward, consumers are marginally cheaper to connect. Ultimately, the *minimum supply cost* is reached when customers sit within a 30–50 meter *service drop* of a pole. At this point, the utility needs to drop a wire from the pole to a meter, from which the household wires its appliances. Service-drop coverage is represented by the date-colored regions surrounding poles in Figure 2.

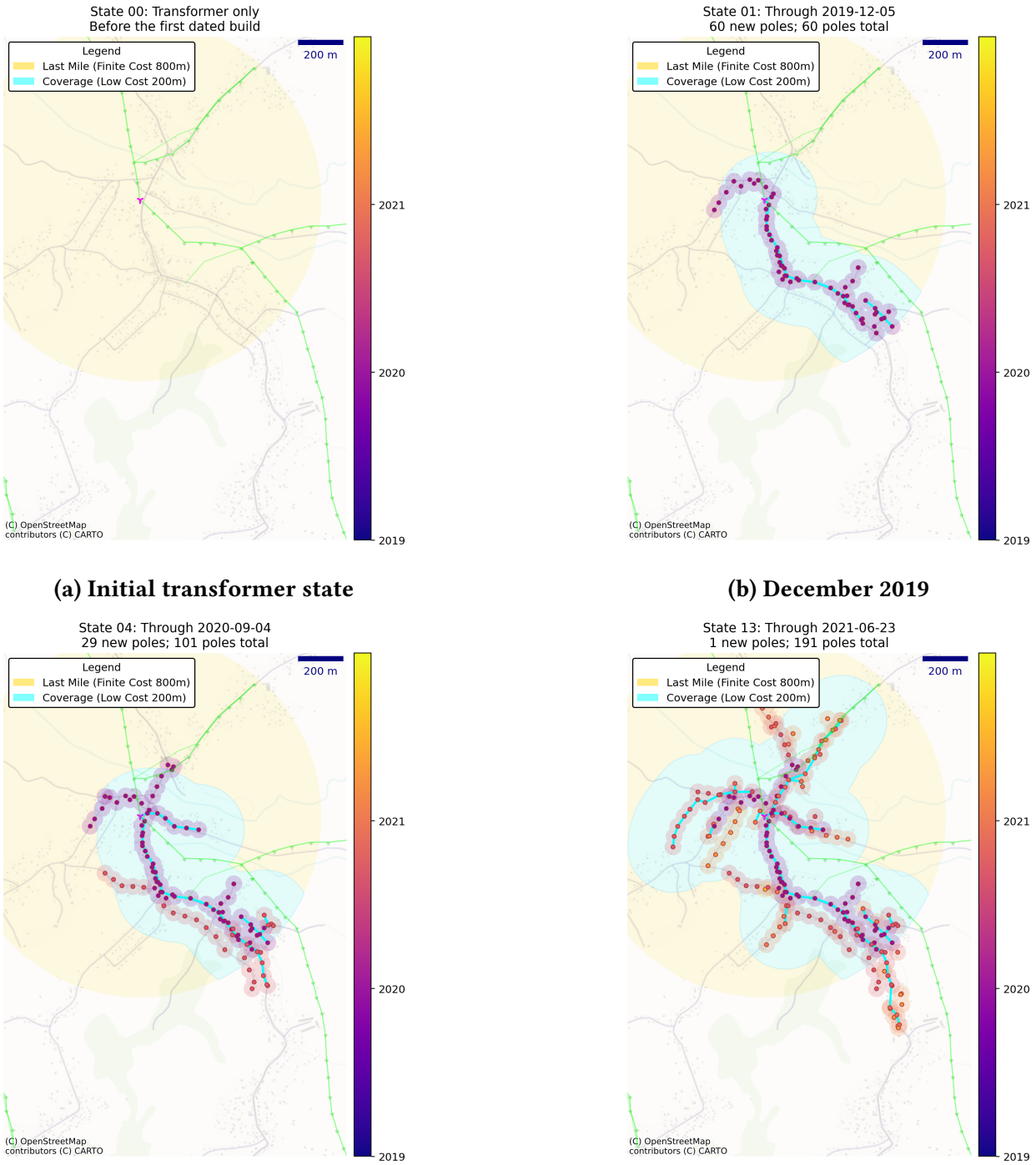
3.2. Recovering Distribution Capital Costs

Utilities fund distribution network expansion through three channels: cross-subsidies from the electrons market, direct subsidies, and connection charges. If they are vertically integrated, they can "recover" revenue from the downstream electrons market to cross-subsidize "fixed costs" in the connections market. Direct subsidies from government finance, multilateral development grants, or concessional loans are typically allocated as a flat subsidy per consumer or piece of infrastructure. Connection charges cover the remainder; how distribution network capital costs are funded determines whether consumers pay for distribution capital upfront, through subsequent electricity consumption, or through future taxes.

16. Throughout, "off-grid" as a distance category means a location far enough from existing infrastructure that connection requires a new distribution transformer, distinct from "off-grid providers" such as solar home systems and microgrids, a competing technology considered in the counterfactual analysis.

17. Development banks and external donors typically fund "backbone" transmission grids across Sub-Saharan Africa.

18. "Last mile" is a misnomer: a typical distribution network has a half-mile radius. By feasible, I mean without unilaterally paying for transmission infrastructure costs, which are several times higher than distribution ones per unit distance.



(a) Initial transformer state

(b) December 2019

(c) September 2020

(d) June 2021

Figure 2 — Spatial rollout and supply isocost regions, 2019–2021

Notes. The initial transformer-only state and three dated states of the sampled Gakoro Health Center distribution network in Musanze. Light yellow is the last-mile region; light blue is the under-grid region; the smaller date-colored regions show 37-meter service-drop coverage around poles. The dated panels show December 5, 2019; September 4, 2020; and June 23, 2021. Their expanding blue and date-colored regions show how successive extensions make new locations cheaper to connect. Data sources: Google Open Buildings, OpenStreetMap, CartoDB/Contextily, Rwanda Energy Group, and Rwanda's Ministry of Infrastructure GIS portals.

Most countries outside Sub-Saharan Africa primarily subsidize connections with cross-subsidies from the electrons market. In most developed countries with built-out grids, connections are effectively free within a coverage area. Distribution network expansions are typically treated as systemwide fixed costs, recouped via "fixed cost recovery" in recurring retail tariffs (Joskow 2007; Borenstein 2016).[19] Such cost recovery is not necessarily out of reach for all developing countries. Golumbeanu and Barnes (2013) describe how rural electrification cooperatives in Bangladesh provided loans to finance infrastructure and recovered the loans from consumers via electricity tariffs. Upfront, consumers only needed to pay a small service connection fee (USD 23). Most utilities worldwide forgo connection market revenue and collect it elsewhere.

However, Sub-Saharan African utilities have substantially less scope to recover distribution capital costs through electricity consumption. Trimble et al. (2016) report that all but two Sub-Saharan African utilities charge consumption tariffs below the marginal cost of electricity generation. Blimpo, McRae, and Steinbuks (2018) show that low and elastic electron demand forces utilities to charge high connection charges to generate revenue. This explains why connection charges tend to be higher in Sub-Saharan Africa than anywhere else in the world (Golumbeanu and Barnes 2013).

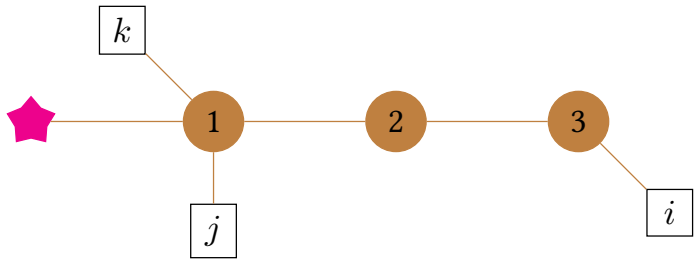
3.3. Connections Bundle Private and Public Infrastructure

Electricity distribution networks, encompassing the infrastructure from the transformer to the consumer, are neither purely private nor purely public for three reasons. First, distribution infrastructure becomes congestible and rivalrous as it approaches capacity. Second, the utility can exclude users from service, so the network is technically a congestible club good. But excludability is limited in practice. Utilities typically have legal obligations and responsibilities to serve on a non-discriminatory basis under a universal service mandate. Further, once service has commenced, utilities are typically obligated to continue providing service and can exclude only consumers who do not pay their bills (Rossi 2000). Fundamentally, customers cannot exclude each other. While hypothetically one could exclude a neighbor by cutting their electricity line, laws prohibit this. So shared distribution infrastructure is effectively a public good: non-rival among its users and excludable by the utility, though not by customers against each other. Third, and most importantly for this paper, *upstreamness* determines how widely each specific component is shared. The most downstream meters and service drops are customer-specific private goods. The most upstream components—transformers, feeders, and transmission infrastructure—serve many users. The network components between these endpoints lie on a public good spectrum.

Public-ness depends on the distribution tree's network structure. Consider the following infrastructure network, where service emanates from the source through the brown junction nodes to users i, j, and k. Figure 3 shows that downstream poles are more private and less public because of the neighborhood's spatial configuration relative to a network access point. Users j and k require only the first junction node, but i requires two additional poles. Although all three nodes are non-rival once constructed, nodes 2 and 3 are *de facto* private to i because no other user's path includes them. Meanwhile, node 1 is much more public. Heuristically, downstream components tend to be more exclusive, and upstream components will

19. While out of scope for this paper, one could show that increasing fixed costs increases economies of scale, entrenching the natural monopoly. Regulation becomes more essential; it raises regulated consumption tariffs, especially under rate-of-return regulation.

Figure 3 — Distribution infrastructure as a hierarchical public good



Notes. A single branch emanates from the closest point on the existing network (magenta star). Nearby users j and k attach at junction node 1, while reaching the distant user i requires nodes 1–3. Nodes 2 and 3 are *de facto* private to i because no other user's path includes them. In an electricity distribution network, the source may be a transformer or an existing pole, and the junction nodes represent poles.

tend to be more widely shared.

The figure shows that flat fees cross-subsidize across space, and distance-based charges endogenously cross-subsidize across time. An average-cost flat fee splits the cost of all three nodes equally among i, j, and k, with weights $(1, 1, 1)$. Because i is the only customer using nodes 2 and 3, a flat fee would implicitly transfer part of i's marginal network extension cost to j and k. A materials-based charge instead assigns the cost of infrastructure components according to who connects first. If i connects first, i pays for all three nodes, and j and k then do not have to pay for any poles. If, by contrast, j or k connects first, the first mover pays for the first pole. The second mover then does not need to provision a pole to connect, and the downstream customer i would still need to pay for poles 2 and 3. The order of connections endogenously creates externalities and determines who pays.

3.4. Average-Cost and Marginal-Cost Network Connection Policies

This subsection compares which costs utilities pass through in distance-based charges, the rationales utilities and policymakers give for choosing average- or marginal-cost policies, institutional variation within distance-based connection charges, and accompanying policies or clauses that acknowledge the market failures studied in this paper. I compiled the connection policies and strategy documents, where available, from utilities and regulators across Sub-Saharan Africa and other developing countries. I describe qualitative findings below.

Regardless of policy, all Sub-Saharan connection charge policies charge consumers for private electricity meters. Except via community-led adoption policies, they do not charge grid electricity adopters for transmission, feeder, or transformer infrastructure.[20] The main disagreement is whether to charge consumers for low-voltage distribution network extensions. I found that about a third of countries pool distribution network infrastructure costs and charge an average-cost policy. In contrast, about half of countries pass these costs on to customers in a distance-based policy. The remainder use hybrid policies or apply different policies in different areas.[21]

I call a policy an *average-cost policy* or *flat-fee policy* if supply-side distribution network connection

20. Funding from multilateral banks typically targets transmission infrastructure (Blimpo and Cosgrove-Davies 2019).

21. I will provide a full appendix in a future version of the paper.

costs are not passed on to consumers through connection charges. In Lee, Miguel, and Wolfram (2020b), for example, all consumers by default would pay a flat rate of USD 398 at baseline, as long as they were within 600 meters of a distribution transformer. Today, Kenya uses a flat USD 120 connection charge. Other countries that use a flat-fee connection charge include Botswana, Morocco, Vietnam, Ecuador, Guyana, Angola, and Mozambique. To set an average-cost policy, utilities choose an exogenous "penetration rate" constraint in their least-cost planning algorithm, subtract the share of cross- or direct capital subsidies, and then pool the remaining cost among the expected adopters (Parshall et al. 2009). A significant limiting factor for utility planning is demand estimates over space, which allow the planner to spatially allocate infrastructure (Lee et al. 2021). The clearest economic arguments for a flat fee are lower administrative costs and ex ante fairness through a single market price.

Distance- or materials-based charges pass through supply-side connection costs to the consumer. In policy documents, the main rationale given for distance-based policies is to make connections "cost-reflective." Other policy design concerns include simplicity, transparency, consistency, transaction costs, and early network expansion (Woods 2019; Zambia Electricity Regulation Board 2023). Cost reflectivity decentralizes to customers the decisions about whether and where the network expands. As remote customers see higher costs, they choose whether it is worth it. Hence, the spatial extent of distribution networks becomes a function of revealed preference and should improve infrastructure allocation.

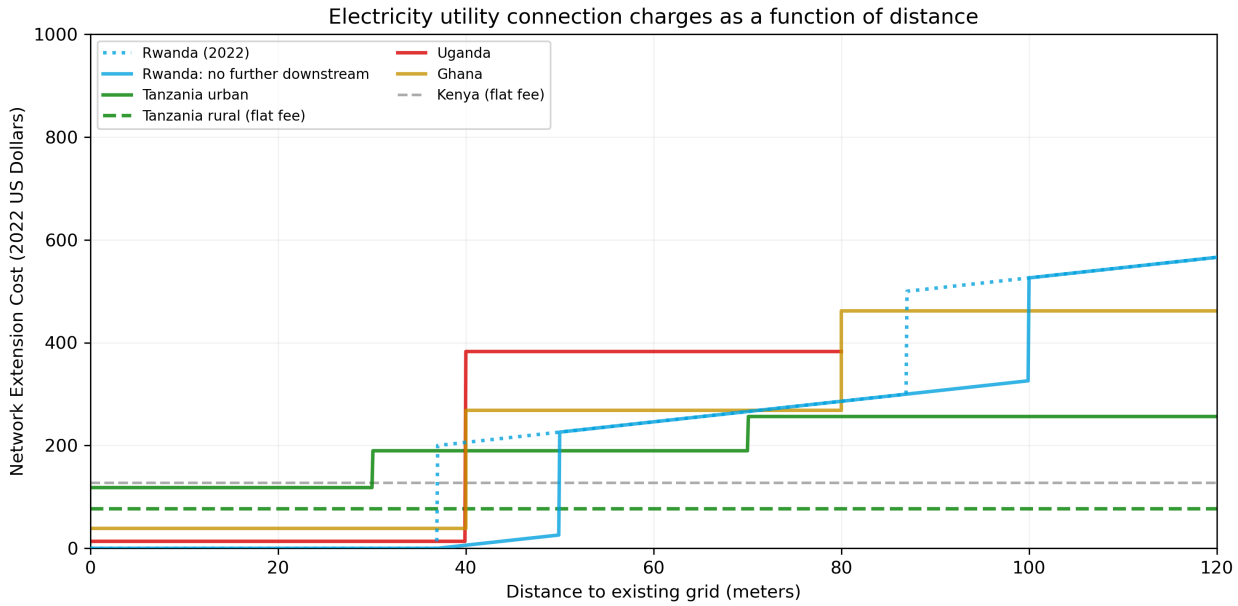
I plot connection charges as a function of distance for selected countries in Figure 4. The visual takeaway is that distance-based charges tend to be piecewise step functions. They start from a low value within the service-drop range, covering the cost of meters, wiring, and service-drop cables. Discrete jumps occur when additional poles are necessary to extend the grid, with jump size depending on material costs and per-pole subsidies. Unlike other countries' policies, Rwanda's policy slopes upward because it charges consumers USD 2 per meter of line starting from the service drop.

I observed three additional stylized facts about distance-based charge design when constructing Figure 4. First, published distance-based connection-charge schedules were not the norm. Instead, most countries depend on an ad hoc engineer's survey of the required network extension. Second, how charges are calculated may depend on distance or materials. Ghana and Uganda index the cost to the number of poles, whereas Rwanda and urban Tanzania index it to distance. Finally, all cost-distance functions found on utility websites were piecewise linear, even though utilities theoretically could smooth them while retaining some degree of cost reflectivity. I found no academic literature on these design tradeoffs, but policy and consulting documents divulged the following.

Why do some countries require quotations while others publish distance- or pole-based schedules? The apparent advantage of quotations is that they pass construction-cost risk to consumers. Quoting each network extension separately allows the utility to pass through any heterogeneous supply-side costs from transportation, terrain, procurement, eminent-domain, or wayleave to new consumers (Woods 2019). By contrast, a schedule offers transparency, allowing consumers to make connection decisions without requesting a utility quotation.

Why do some utilities that pass on costs base connection charges on materials, while others base them on distance? Policy documents highlight that utilities should charge customers for the *minimum technical solution* needed to connect the household, not the real infrastructure that the utility builds (Woods 2019;

Figure 4 — Connection charges in Sub-Saharan African countries with transparent distance-based connection schedules



Notes. Connection charge policies collected from utility websites in 2022, converted to USD using calendar-year average exchange rates from the World Bank API (PA.NUS.FCRF). For Ghana and Uganda, the geographic component is an explicit function of poles required rather than distance; I assume these are required at 40-meter intervals. Where credit loans were available directly from the utility, the figure shows the upfront minimum charge. Kenya and rural Tanzania use flat fees, shown for comparison. Rwanda's policy is partially transparent online, with distance categories and materials denoted but not exact prices. I supplemented this with information from Rwanda's Ministry of Infrastructure, which shared that the utility charges consumers for the extension above 37 meters at USD 2 per meter of line and USD 200 per pole. Whether a pole is needed within 50 meters depends on whether there are downstream customers. Additional Rwandan contextual and cost details are provided in subsection 3.5.

Zambia Electricity Regulation Board 2023). Utilities have incentives to place poles that follow a least-cost electrification plan that minimizes the total cost of providing electricity to the entire neighborhood, rather than to connect the adopting consumer at their least cost. Using a distance-based charge over a materials-based charge ensures that utility decisions that warp the grid toward neighbors are not passed through to consumers.

Why do utilities use stepwise rather than smoothed connection charge schedules? Utilities face a substantial political economy challenge when using distance-based policies because they must justify to consumers why different consumers should pay different connection charges for the same service (Woods 2019). Using a stepwise function makes material costs tangible, promotes transparency, and therefore helps justify connection costs to consumers. Perceived fairness would seem to be an important constraint on consumers' private provision of shared infrastructure.

I now describe how institutional choices are implemented in Rwanda, the setting for my empirical case study.

3.5. Rwanda's Connection Market

In 2009, only 6% of Rwandan households had access to on-grid electricity. Substantial electrification has occurred since then; REG recorded 55.9% on-grid access by June 2023, and the World Bank's World Development Indicators show 74% electricity access (including off-grid technologies) at the end of 2023. I briefly review Rwanda's distance-based policy design and other pertinent institutional features, focusing exclusively on policy and institutional facts used to calibrate connection charges from the data, motivate empirical design choices, or implicitly nod at strategic incentives.

The 2017 "New Connection Policy" codified Rwanda's distance-based connection policy, though REG used a distance-based charge informally before that (Rwanda Energy Group 2017). The online connection process portal shows that distance boundaries are transparent to new customers but describes the specific charge as the "cost of extension." It indicates a four-working-day response time for connections within 100 meters of the grid and a ten-working-day response time beyond 100 meters. I use these service standards to motivate an otherwise unobserved weekly delay between connection request and installation in section 4. The discussion below draws from the "New Connection Policy" unless otherwise noted.

The baseline minimum charge, which the utility calls the "Connection Fee," is 56,000 Rwandan francs (RWF) or USD 56.[22] It covers the cost of a meter, service cable, and "connection accessories." However, the effective minimum upfront connection charge is zero (Dabalen et al. 2025), because customers can take up to RWF 56,000 in credit, repaying it at a 100% markup on their electricity-consumption tariff with zero interest. But the credit only applies to the cost of meters, cables, and connection accessories, not poles or lines, so upfront connection charges are zero only if the prospective customer is within a 37-meter service drop from an existing pole. From 37 to 50 m, customers pay for the "cost of extension" but are nevertheless connected "with existing infrastructure." Rwanda's Ministry of Infrastructure reported that this is an RWF 2,000/m electricity-line charge. When it would help downstream customers, the utility gives a quotation

22. I study the period from October 2018 to November 2021. Dollar equivalents use the approximate reference conversion of RWF 1,000 to USD 1. The Rwandan franc depreciated during the study period, and exchange-rate exposure is the basis of my secondary shift-share instrument.

for an electricity pole within this range. This discretion is explicit in Figure 4.

From 50 to 100 m, the utility passes through part of the supply cost of one pole. Poles are spaced 55 m apart in engineering reticulation standards (Rwanda Energy Group 2020). In the cleaned geospatial data I describe in the next section, I observed a median pole-to-pole span of 49 m. REG places poles to follow roads, engineering standards, local geography, and anticipated demand. Roads are followed to ensure that poles are placed on public land. The connection charge is based on distance rather than materials, so customers pay for the minimum technical solution. The policy emphasizes that customers "will not stop new applicants from being connected on such extension lines," indicating that the utility is aware that it is asking customers to privately provision public goods.

Beyond 100 m, the policy states that "For the applicant or a *group* of applicants beyond 100 meters from the point of electricity supply, a detailed survey will be conducted to determine the most appropriate extension line and point of connection, and a related bill of quantities will be produced." The emphasis on *group* is mine. Consistent with Proposition 1, it suggests that the utility is accustomed to cooperation when the costs of shared infrastructure are extremely large relative to income. Two poles cost USD 400, and Rwandan annual income in 2022 was USD 935.9 per capita (constant 2015 USD); five poles are more than annual income, and so will generally require cooperative provision. The policy neither prescribes nor suggests how costs might be shared within the group. The last-mile region, indicating a finite cost for customers, is also described by the policy when it says that "the Low Voltage line cannot be extended beyond 800 meters from the Distribution transformer." This gives me geographic bounds for which potential adopters are relevant when transformer placement starts the network game.

Except for the service drop line and meters, Rwanda's distance-based charge is almost the mechanical distance-based policy described in Definition 1. About a third of the supply-side connection costs get passed through to connection charges. Rwanda's Ministry of Infrastructure reported in 2022 that customers are charged USD 200 (RWF 200,000) for a pole, in addition to the aforementioned line charge of RWF 2,000 per meter. I consolidate documentary evidence I collected reflecting supply-side costs, demand-side connection charges, and subsidies in Appendix A. I found that supply-side connection costs for a pole fluctuate between USD 600 and USD 1,000 over time, and that USD 6 per meter for lines appears to match aggregate financial reports.

Consumer requests primarily drive grid extension, except when the utility subsidizes "massive connections" funded by the Electricity Access Rollout Program (EARP). The EARP pools financial resources from the World Bank and other multilateral development agencies to fund specific extension projects confined to specific localities. Typically, the EARP targets schools and other public infrastructure and electrifies the entire route from transformer to public service. Customers in areas funded by massive connections pay the fill-in upfront rate of zero.

Information provision helps customers understand the connection policy (Rwanda Energy Group 2017). The utility disseminates the policy through markets and district offices, radio and television broadcasts, religious ceremonies, and Umuganda. Umuganda is a monthly required community service that brings together a representative from each household at the village level to do a day of community service each month. These monthly meetings may decrease coordination costs in Rwanda. If cooperation costs are more about coordination than baseline cohesion, this would predict more strategic complements and fewer

strategic substitutes.

Via the Rwandan Least-Cost Electrification Plan, Rwanda effectively spatially delineates off-grid and on-grid regions to private mini-grid providers and the utility. While not an explicit policy, licenses for off-grid solar providers are limited to designated off-grid zones. This spatial delineation coordinates monopolistic on-grid and off-grid regions, diffusing electrification efforts over space. Therefore, the empirical exercises reflect a context with little competition between the grid and alternative sources of electricity.

4 — Data

I transform timestamped geospatial infrastructure data from Rwanda Energy Group and Rwanda's Ministry of Infrastructure into a prospective customer-week panel of distances, costs, externalities, and peer effects. The pipeline has three stages: geospatial cleaning, a least-cost planning algorithm, and a dynamic grid rollout algorithm. The geospatial cleaning removes duplicated transformers, poles, lines, and customers, matches each utility customer meter to a remotely sensed building polygon, and links them into a network graph. The least-cost planning algorithm adds connection paths for the remaining buildings in the vicinity of a distribution network. Non-adoption is a revealed preference, so all remotely sensed buildings are included in the empirical exercises. The dynamic grid rollout algorithm then constructs a panel from customer installation dates by chronologically adding customer-induced network expansions.

4.1. Raw Data

The primary raw data are from Rwanda Energy Group (REG)'s and Rwanda's Ministry of Infrastructure's Geographic Information Systems (GIS) Portals (Rwanda Energy Group and ESRI 2022; Ministry of Infrastructure (MININFRA) Rwanda and ESRI 2024).[23] Secondary raw data come from Google Open Buildings, OpenStreetMap, the Rwandan Land Use Authority, Earth Observation Group nightlights, the World Bank Multi-Tier Framework Survey, and the Global Economic Monitor. I describe each of the datasets and their uses:

- **REG Customer GIS data.** Two shapefiles of customer *meter* geolocations and associated metadata were available on Rwanda Energy Group's GIS portal. Each point has geospatial coordinates, administrative plot numbers, customer characteristics, meter serial numbers, and transformer names. Connection dates were available only in the substantially smaller file, which covers 4.96% of unique metered customers after deduplication (34,087 of 686,891). I limit the analysis to the 177 distribution networks where the smaller file accounted for more than 80% of installed meters. The customer data appear to be entered when the utility fulfills a customer's connection request and a technician installs a meter.
- **REG LV network GIS data.** I use the GPS locations and characteristics of LV poles, lines, and transformers to construct each distribution network's initial graph and train the least-cost planning algorithm. Across the datasets, the infrastructure metadata include the primary and secondary voltages and currents, capacities, materials, years of manufacture, manufacturer, lines, upstream substation and

23. Each World Bank funding round has targeted several million dollars of grant funding at geospatial data collection, as this helps with utility planning and modeling.

feeder, phases, utility branches, and more. The infrastructure data are collected by survey, and the data-generating processes for LV lines and poles are related. A surveyor can either create a dataset of poles and sequence them in a collection to form a line or collect a line directly, with the phone or tablet recording location every few seconds as the surveyor walks along the infrastructure. I use the combined universe of line and pole data to construct the infrastructure network. The infrastructure data cover the near-universe of electricity infrastructure across Rwanda in a snapshot that lags behind the customer data.

- **REG branch boundaries.** Branch boundaries are not perfectly aligned with administrative district boundaries. Branch membership is used to construct branch-by-week fixed effects that control for correlated shocks in the reduced-form regressions.

- **Google Open Buildings** provides a temporal "2.5D" raster and polygon shapefile. The .5D refers to building heights (Sirko et al. 2023). I spatially join the raster to polygons via polygon centroids to observe building heights and areas and create a binary panel of construction dates at yearly frequency.[24] A bipartite minimum-cost matching algorithm matches each meter to a building polygon within 50 meters using point-to-polygon distance (Virtanen et al. 2020). A large 50-meter range accommodates the rare cases where the utility places meters on electricity poles rather than buildings to deter theft. Matched meters and buildings are 3.7 meters apart on average, with 1.25% of matches above 37 meters. I use each building's "Google plus code" as a customer identifier. Matched buildings are adopters, and unmatched buildings are non-adopters.

- **The Rwandan Land Use Authority** provides geographic terrain data. I use rivers and a 10 m digital elevation model (DEM) (UNDP 2023). I construct terrain ruggedness indices and slope measures from the DEM. I use terrain ruggedness as a customer characteristic. Slope and waterways enter the cost surface of the least-cost planning algorithm.

- **OpenStreetMap roads.** I use roads primarily in the least-cost planning algorithm because the utility builds poles along roads to keep them on public land (OpenStreetMap Contributors 2026).[25]

- **Earth Observation Group Nightlights.** I use 2019 Visible Infrared Imaging Radiometer Suite (VIIRS) nighttime lights as a contextual control in the reduced form (Elvidge et al. 2021).

- **Exchange-rate data from World Bank Global Economic Monitor.** I construct a monthly imported-pole cost shifter for each utility branch using official exchange rates from the World Bank Global Economic Monitor. I interact exchange rates lagged by 18 months with each branch's pre-2019 shares of LV poles by manufacturer country to construct a shift-share instrument. I motivate the lag in section 6.

- **The World Bank Multi-Tier Framework (MTF) Surveys** of Rwanda in 2016 and 2022 are used to check the distributions of connection charges paid (Appendix A) and to assess price discrimination with alternative data. Survey respondents reported their stated willingness to pay and revealed payments

24. I do this by observing when heights and precision monotonically increase, then plateau.

25. OpenStreetMap provides substantially better coverage of Rwandan roads than Rwandan government sources.

for a connection to the grid, alongside many other variables on electricity demand, appliances, and alternatives (World Bank 2018a, 2022).

4.2. Geospatial Graph Construction

The first part of the pipeline converts raw geospatial data into a graph representation and data structure (Hagberg, Swart, and Schult 2007). The main goal is to encode the lines as edges and poles, customers, and transformers as nodes, satisfying Definition 4.

Definition 4 (Physical Hierarchical Network)**.** Electricity distribution networks are hierarchical. If a low-voltage pole or consumer is electrified, it must have a physically installed and electrified network path connecting the transformer to it.

Consumers require a transmission path from a generator to the distribution transformer and an LV path from the transformer to their premises to be connected to the grid. By hierarchical, I mean that there is a tier for transformers, poles, and meters: on their step-down side, transformers connect only to poles. Poles can connect upstream to transformers, connect horizontally to other poles, or connect downstream via service drops to meters. But meters do not connect to each other, nor do transformers.

I clean the geospatial data to create a network graph for each distribution network that fits Definition 4. I remove duplicates, cluster points, impute missing line data from pole data and pole data from line data, and check network validity. I use methods similar to those electrical engineers use to prepare infrastructure networks for power-flow simulations. Specifically, I clean metadata by normalizing null strings, ensuring consistent engineering units, and standardizing categories. I clean geospatial coordinates by harmonizing coordinate reference systems (CRS), dropping invalid geometries for all points in the infrastructure data, and substantially deduplicating them.[26]

Deduplication is challenging because poles are often recorded multiple times by different surveyors and because floating-point errors propagate through CRS conversions; Rwanda's Ministry of Infrastructure and Rwanda Energy Group had versions of the data in different CRS projections. I cluster low-voltage points within 7 meters and medium-voltage points within 12 meters, using the resulting centroids as representative points. To speed up lookup, I spatially index each deduplicated point on a hexagonal grid using H3, Uber's geospatial indexing library.[27] I spatially index lines as a sequence of H3 indices. Then I add each pole, transformer, and meter as a node to the graph, and low-voltage lines as edges between nodes.

Some infrastructure data are missing but necessary to ensure Definition 4. Where line data exist but pole data do not, I impute poles from points along the line, accounting for kinks and the distance between poles.[28] Where pole data exist but line data do not, I greedily create synthetic edges between unconnected poles, then prune them while preserving each pole's shortest path to the transformer. Service drops are not observed as data, so as the last stage of graph construction, I link customer meters using a waterfall

26. I use EPSG:3857 (Web Mercator) throughout. This is a projected coordinate reference system where longitude-latitude coordinates are expressed in meters. As Rwanda is close to the equator, at approximately 2° south, Web Mercator distances are negligibly inflated by 0.06% to 0.14% across the country.

27. Hexagonal cells perform better than a square grid because they ensure equal distances to every neighboring cell.

28. The data structure of a line is just a sequence of points.

approach. I connect each building to the closest pole within 10 meters, then to the most upstream pole within a 37-meter buffer, and then use a maximum distance of 50 meters.

After geospatial cleaning, I observe that more than 93% of low-voltage distribution networks outside Kigali are radial trees. I convert the remainder into trees by pruning cycles while preferring transformer-pole shortest paths. I check that each meter has a network path to the distribution transformer. For each matched or unmatched building, I observe nightlights, elevation, terrain ruggedness, building height, building area, and construction year. For matched buildings, I also observe residential, commercial, industrial, or public-service categories, as well as installation dates. The resulting graph has transformers, poles, and meters as nodes, and LV lines and service drops as edges.

4.3. Least-Cost Planning Algorithm

I develop and calibrate a least-cost planning (LCP) algorithm that constructs a network path to each unconnected building within range of the transformer and existing network. It plays several roles in my analysis: First, it allows me to touch up the infrastructure data for customers without a viable observed LV network path because of lags in infrastructure data collection. Second, it allows me to construct the minimum technical solution for each customer at every point in time. The corresponding connection charges are what customers should actually pay under Rwanda's connection charge policy. Third, I use it to extend the observed network to all non-adopters in the sample, simulating the utility's least-cost plan from the existing network. Finally, I also use it in a demand-stripped way for robustness checks and to consider an alternative instrument in subsection 6.4.

The least-cost planning algorithm's objective is to connect *demand points* to the existing LV network feasibly and at minimum cost. Essentially, it is a wrapper around a Steiner tree algorithm. Relative to a minimum spanning tree, a Steiner tree additionally allows for the construction of Steiner points — poles — that can connect demand points in space to the existing grid. In addition to material costs, the wrapper adds cost surfaces that account for terrain and engineering parameters, including reticulation standards. It takes three primary inputs: the network graph of existing infrastructure, which at minimum consists of a transformer; the demand points, which are buildings and/or meters to be connected; and a set of geographic features and parameters. The algorithm outputs a geospatial network graph, extending from the existing network, that represents the least-cost infrastructure necessary to connect the input demand points.

I train the algorithm to simulate how Rwanda's realized grid behaves. I develop it iteratively. In a manual outer loop, I tinker with functional forms of the cost surfaces and network by using the algorithm to construct a synthetic network from each transformer to the meters it connects. I visually compare the synthetic network against the real network, satellite backdrops, and various inputs to the cost surfaces, assessing whether the algorithm's output geospatial infrastructure qualitatively matches how the real distribution infrastructure is organized around terrain, other infrastructure, and buildings.

In the current version of the LCP, I assume poles cost USD 600, and lines cost USD 6 per meter on the supply side, following subsection 3.5. Feasibility constraints from the utility's reticulation standards are softened to a spline around reticulation-standard thresholds. I also embed the catenary equation to measure clearances and impose a cost based on the clearance of hanging lines over terrain. For terrain, two H3 hexagonal cost surfaces are assembled from flexible spline and kernel functions over geographic features:

roads, slope, waterways, and (optionally) buildings. Splines measure cost; kernels measure ideal setbacks. Separate cost surfaces are necessary because line and pole costs vary across space; for example, lines can cross waterways, but poles may not be constructed in them.

In an inner loop, I use gradient descent to optimize how parameters interact with geographic features or engineering standards to construct the cost surface and reticulation parameters. The objective compares the synthetic network against the ground truth by minimizing the root-mean-square error between bipartitely matched poles, with a penalty for mismatches. Across distribution networks, the least-cost planning algorithm results in a 1.00 (median 0.93) pole-count ratio and a root-mean-square distance of 45 (median 39) meters between bipartitely matched poles within each transformer. The algorithm is imperfect, but it creates a least-cost plan that replicates the average number of poles needed for each network.

After training it on real data, I use the least-cost planning (LCP) algorithm to extend the grid to nearby buildings to create a least-cost plan for non-adopters. I include an unmetered building as a potential electricity adopter only if it lies within 800 meters of its assigned transformer, following the utility's service-radius policy described in subsection 3.5, and if the building is within 150 meters of that transformer's observed grid, which implicitly assumes that far-off customers have no strategic effects. I assign each unmetered building to the transformer associated with its nearest observed-infrastructure node, and assign buildings on multiple networks to the network game in which they have the shortest building-to-transformer distance. Non-adopters become part of the game and estimation panels via Assumption 1.

Assumption 1 (Potential Customers and Network Assignment)**.** Each remotely sensed building footprint is a potential electricity connection location, assumed to be a player in the network game that treats its assigned transformer network as the one through which it would receive a grid connection. Matched buildings adopt, but unmatched buildings are treated as non-adopters through the end of the sample period. Adoption and non-adoption are both assumed to be revealed preferences.

This assumption determines the size of networks and assumes every remotely sensed building could potentially adopt electricity and can provision externalities.

4.4. Dynamic Grid Rollout Algorithm

I convert the static network graph into a panel by iterating through customer adoptions that induce network extensions. Assumption 2 is based on utility operations and allows me to impute each pole's latest possible installation date from customer adoptions.

Assumption 2 (Consumer-Driven Monotone Grid Expansions)**.** When not driven by EARP massive connections, low-voltage distribution infrastructure within each transformer network is constructed in response to customer connection requests. Once installed and electrified, infrastructure remains active, so the grid monotonically expands.

To verify Assumption 2, I quantify the number of raw poles that are not on a network path from a transformer to a consumer. I find that 6.9% of the poles exist without serving a consumer, but of these, 62% have a public service en route, indicating that the EARP likely funded them. So only 2.6% of poles detectably break Assumption 2 at endline. Given that the geospatial data are a snapshot of the network at the end of

2021, I have no reason to believe this would be higher within the panel period. When Assumption 2 breaks because the physical infrastructure predates the observed installation of the customer assigned to induce it, it biases estimates of strategic behavior toward zero; I explain this at the end of this section. Nevertheless, Assumption 2 is critical because it tells me which utility customer or group of customers induced each en-route pole, allowing me to assess who provisions infrastructure, and when each pole existed, allowing me to recompute connection charges and observe who benefits from it.

The last assumption is a convention that links connection requests to observed installations. This is necessary because the GIS data indicate installation dates, and I do not observe when customers make requests. Weekly periods provide a natural break because utility policy provides connections within four working days for consumers within 100 meters (about 2 poles) of the grid.[29] Within weeks, adoptions are also clustered: among transformer-weeks with multiple adopters, 73.6% of adopters connect on the same day. Most installations are on weekdays, and Sundays account for only 3.8% of installations. Therefore, I make Assumption 3.

Assumption 3 (Week-Ahead Connection Requests)**.** For network extensions that occur at t, I assume they were induced by connection decisions at $t-1$, and enter the grid state at the beginning of week t. I treat customers whose installations share infrastructure and occur in the same week as a coalition.

Assumption 3 rounds the four-working-day lag to five working days, or one weekly period. Given a four-working-day response time, customers that connect within the same week must have submitted an extension request just before the utility constructed the infrastructure. If they did so simultaneously, either they coordinated adoption, or the utility must batch customers because of scale economies. A placebo will rule out scale economies.

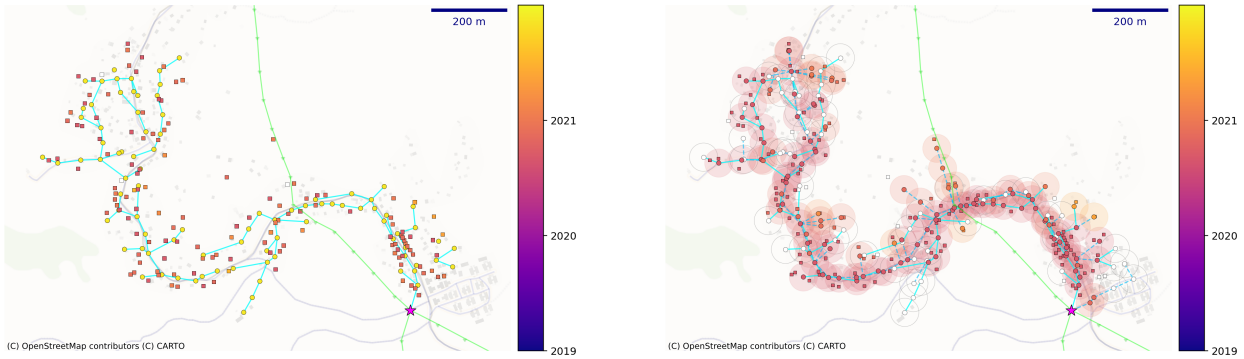
The dynamic grid rollout algorithm applies Definition 4, Assumption 2, and Assumption 3 to estimate when each piece of infrastructure was constructed. It proceeds as follows:

1. Initialize the grid from the transformer.
2. Iterate through each week in chronological order. If there is a network extension,
    i. Select the poles and edges necessary for each coalition of adopters.
    ii. Assign the coalition's minimum installation date to its induced network path infrastructure.
    iii. Save the set of infrastructure that existed at each discrete point in time as a grid state.
3. At each state, measure all consumer adoption incentives and instruments.

Figure 5 shows how the algorithm proceeds. In the left panel, poles lack installation dates and are colored by their raw endline record dates. The uncolored meters lack raw dates. Not all poles were connected to the network in the raw data. Geospatial cleaning connects dangling networks; for example, it reaches the poles in the bottom right that were initially unconnected by lines. Least cost planning extends the grid, for example, to reach the customer in the middle living under the LV line. The dynamic grid rollout algorithm imputes pole dates. In this grid, most of the infrastructure was built concurrently, but many more customers around the periphery connected and induced poles later.

29. For consumers more than 100 meters from the grid, the standard is ten working days.

Figure 5 — Raw inputs and date-imputation output for a sampled distribution network



(a) Raw observed data **(b) Completed graph and imputed dates**

Notes. Comparison of raw and algorithm-completed data for the Nkomero distribution network in Burera. The left panel shows observed low-voltage lines and poles, potential buildings, raw pole record dates, and observed meter-connection dates. The right panel adds the completed tree topology and colors poles by their imputed activation dates. Translucent circles show the 37-meter service-drop region around each dated pole. Squares are meters, circles are poles, colored by their installation date. The magenta star is the transformer; solid cyan lines are observed low-voltage paths, dashed cyan lines are completed paths, and green lines and triangles are medium-voltage infrastructure.

The dynamic grid rollout algorithm mechanically creates one-sided systematic measurement error. The imputed date for each pole is the latest possible installation date; it represents the first time I observe the pole to be used by customers. So connection charges c_{it} are overstated early in the network rollout, and mechanically shrink as imputed dates catch up to reality. It biases estimates in predictable directions. When estimating the effect of static incentives, this shrinks the effect of static costs toward zero. But it biases tests of strategic incentives against the model's predictions. Consider a cluster of consumers who share poles that physically exist but have not yet entered the imputed grid state. In truth, everyone is close to the grid, and so true externalities are nil. The estimated exposure to externalities is large for whoever connects first and zero after that. This generates a spurious positive bias between future externalities and adoption, against my hypothesis that future externalities would delay adoption. It also generates a spurious negative bias against cooperation, as I see phantom unbuilt shared infrastructure that provides no incentive to cooperate. When the first consumer in the cluster connects, I see an externality that should be followed by sequential adoption if free-riding exists. Without this externality, there would be no incentive to adopt sequentially after the first mover.[30] All these effects run opposite to my hypotheses: anticipation should delay adoption, cooperation should increase it, and free-riding sequential adoption should follow realized externalities.

I select a sample because of data quality concerns in the raw data, geospatial cleaning, LCP, and date-imputation pipeline. The raw data include 687,065 utility customer records over 7,079 total transformers (including HV-MV and step-up transformers); the current reconstruction algorithm retains 7,105 customers, of which 6,992 have dated connection events, and 42,287 potential building adopters across 177 transformer networks. I provide full details on how and why I select this sample in Appendix B.

30. I control for peer effects.

Qualitatively, the sample is primarily selected toward rural distribution networks with unusually complete installation date metadata. These high-coverage networks represent a little more than 8.5% of non-Kigali distribution networks and 1% of utility customers across Rwanda. Date coverage among metered customers in the selected networks is 98.4%. Relative to other reconstructed non-Kigali networks, the selected networks are smaller, less electrified, less illuminated, and more residential. They also require a smaller share of synthetic poles to connect never-adopter buildings, indicating better infrastructure coverage of nearby buildings. And by the end of the observed rollout, approximately 16.8% of potential building adopters in the sample were connected. The estimates therefore primarily describe relatively early-stage residential electrification between October 2018 and November 2021. The next section uses the reconstructed weekly grid states to measure the incentives facing each remaining unconnected building.

5 — Static Incentives and Motivating Evidence

I construct various measures of individual static incentives and dynamic strategic incentives at each weekly grid state. This section focuses on the former: the derivation of unilateral connection charges, social network peer controls, and grand-coalition Shapley cost shares from the grid states. At the end of the section, I present an event study showing the wave-like pattern predicted by Proposition 3. This pattern motivates the measures of strategic incentives that I construct and instrument in section 6.

5.1. Grid States and Adoption Status

Let $\mathcal{T}$ be the set of in-sample distribution transformers. Each $g \in \mathcal{T}$ has a static planned graph $\mathscr{G}_g = (V_g, E_g)$, where V_g contains transformer, pole, and building nodes and E_g contains distribution lines and service drops. Let $\mathcal{A}_g = V_g \cup E_g$ be the set of all node and edge elements in this planned network. Each element $e \in \mathcal{A}_g$ has a nonnegative supply-side cost w_e. The graph is the observed network for utility customers, augmented by the least-cost plan to connect non-adopting buildings, defined in Assumption 1. It is a directed tree network rooted at the transformer following Definition 4. The *grid state* $\mathcal{E}_{gt} \subseteq \mathcal{A}_g$ is the set of nodes and edges electrified at the beginning of week t. An edge $(u, v) \in E_g$ is active if and only if the two nodes it connects are active. The pair $(\mathscr{G}_g, \mathcal{E}_{gt})$ is the *current state of the grid* at time t. Because all paths, coalitions, and externalities are defined only within distribution tree networks, I suppress g and write the state of a distribution network as $(\mathscr{G}, \mathcal{E}_t)$.

Let $\mathcal{I} \subset V$ be the set of customers and $\mathrm{tx} \in V$ be the distribution transformer. The *network path* of customer i, $\mathrm{path}(i, \mathscr{G})$, is the unique alternating sequence of nodes and edges from tx to i inclusive,

$$\mathrm{path}(i, \mathscr{G}) = (v_0, e_1, v_1, \ldots, e_K, v_K), \qquad v_0 = \mathrm{tx}, \quad v_K = i.$$

Under Assumption 2, an electrified customer's network path must be electrified: $i \in \mathcal{E}_t \implies \mathrm{path}(i, \mathscr{G}) \subseteq \mathcal{E}_t$. The first installations observed within each network are fixed to occur at $t = 1$. The transformer is assumed to be installed the week prior, so $\mathcal{E}_0 = \{\mathrm{tx}\}$. Let $y^*_{it} = 1$ if customer i is already connected to the grid at the beginning of week t. The *connected set* is $Y^*_t = \{i \in \mathcal{I} : y^*_{it} = 1\}$, and the *unconnected set* is $R_t = \mathcal{I} \setminus Y^*_t$. The panel has one row for each weekly period a customer could make a connection decision.

Each network contributes observations from the week before its first adopter until its last adopter,[31] and each customer contributes observations for as long as they are unconnected at the beginning of the week.

5.2. Connection Decisions and Connecting Coalitions

At each observation, the customer $i \in R_t$ chooses whether to make a connection request $y_{it} \in \{0, 1\}$. Under Assumption 3, a connection request in week t means they are connected in week $t + 1$. Connections are permanent and move the adopting customer from R_t to Y_t^* at the end of the week: $y_{it} = 1 \implies y^*_{i,t+h} = 1$ for all $h \geq 1$. The beginning-of-week state $\mathcal{E}_t$ is predetermined by previous connection requests and is taken as given by all consumers when they make period-t choices. The *connecting set* of customers is $Y_t = \{i \in R_t : y_{it} = 1\}$. Customer i's *required network extension* at time t is the ordered subsequence of unbuilt elements on its network path, $\mathcal{B}_t(i) = \big(a \in \text{path}(i, \mathscr{G}) : a \notin \mathcal{E}_t\big)$.

Define customer i's *neighborhood* $\bar{Q}_{it} = \{j \in R_t : e^a(j, \mathcal{E}_t) = e^a(i, \mathcal{E}_t)\}$ as the maximal set of remaining customers who share unbuilt infrastructure with i. The collection of neighborhoods $\bar{\mathcal{Q}}_t$ partitions the set of unconnected customers R_t; each neighborhood is an unbuilt branch of the distribution-network tree. Each neighborhood shares an *anchor node*, which is the first unbuilt pole $e^a(i, \mathcal{E}_t)$ on their path. Via Assumption 3, customer installations within the same week whose required extensions share infrastructure are assumed to have requested a connection together, so let $Q = \bar{Q}_t \cap Y_t$ be called a *connection coalition*, and let $\mathcal{Q}_t$ be the set of all such connection coalitions at time t that partition Y_t. Any neighbor or coalition collectively providing an externality to i is therefore a subset of i's neighborhood $Q \subseteq \bar{Q}_{it}$, because customers in different neighborhoods do not share infrastructure. The required full path and network extension of coalition Q are $\text{path}(Q, \mathscr{G}) = \bigcup_{i \in Q} \text{path}(i, \mathscr{G})$ and $\mathcal{B}_t(Q) = \text{path}(Q, \mathscr{G}) \setminus \mathcal{E}_t$. The total supply-side cost of the coalition's network extension is $C(Q, \mathcal{E}_t) = \sum_{e \in \mathcal{B}_t(Q)} w_e$. The utility extends the grid just enough to connect each customer by aggregating the extensions required by all adopting coalitions in $\mathcal{Q}_t$, $\mathcal{E}_{t+1} = \mathcal{E}_t \cup \bigcup_{Q \in \mathcal{Q}_t} \mathcal{B}_t(Q)$. As grid expansion is monotone under Assumption 2, connection charges weakly decrease for all consumers as the grid expands.

5.3. Anticipated and Realized Network Extensions

A *network-extension method* $m \in \{\text{straight}, \text{lcp}, \text{path}\}$ assigns poles and lines to unconnected customers at a given state. I refer to these as unilateral *straight-line* network extensions, unilateral *least-cost-planning* (LCP) network extensions, and the *realized* or *network-path* extensions, respectively. I define the three extension methods as follows:

- The straight-line network extension draws a line from the nearest electrified pole in $\mathcal{E}_t$ to the adopting customer i. The line $\mathcal{L}_{it}^{\text{straight}}(\{i\}, \mathcal{E}_t)$ is the geometric object from the existing grid to i's service-drop pole, which has raw straight-line distance d_{it}, and represents the shortest possible distance over which the utility could connect customer i.

31. Across all transformers, the last observed adoption occurs in November 2021, but this is not uniform across utility branches. I want to avoid assuming that weeks at the end of the recorded installations are zero-adoption weeks, so I truncate each distribution-network panel at its last observed installation.

- The unilateral least-cost-planning (lcp) extension is the least-cost network path from the existing grid $\mathcal{E}_t$ to an "individual" customer singleton coalition $\{i\}$. In contrast to the straight-line network extension, it incorporates the effects of geography and terrain on the network topology. Its construction uses the least-cost-planning algorithm described in subsection 4.3. Excluding other buildings outside of the coalition from the algorithm removes their demand pull on the distribution network topology planned by the utility. I measure it as $\mathcal{L}_{it}^{\text{lcp}}(\{i\}, \mathcal{E}_t)$ and $\mathcal{P}_{it}^{\text{lcp}}(\{i\}, \mathcal{E}_t)$.

- The *network-path* extension follows the realized or planned distribution infrastructure that the utility would use to connect customer i. This method calculates the shortest path on the distribution tree and returns the lines and poles $\mathcal{L}_{it}^{\text{path}}(\{i\}, \mathcal{E}_t)$ and $\mathcal{P}_{it}^{\text{path}}(\{i\}, \mathcal{E}_t)$, together denoted $\mathcal{B}_{it}(\{i\}, \mathcal{E}_t)$, that the utility would actually build to connect i at each point in time. Unlike the previous extensions, the network path incorporates utility planning based on projected demand from other customers in the neighborhood.

Expectations over how the utility would expand the network to provide electricity can be very similar to or very different from the utility's realized network path. Both the straight-line and LCP paths remove the demand pull of other customers on the realized utility network topology. Relative to the straight-line method, the unilateral LCP rule allows customers' expectations to incorporate terrain and other geographic construction constraints. Figure 6 illustrates a selected case in which a target adopter's hypothetical network extension substantially differs because of hilly terrain, which the straight-line extension ignores. The least-cost path goes around the obstacle but takes a shortcut relative to the real network path used by the utility, which goes around a switchback and uses substantially more infrastructure.

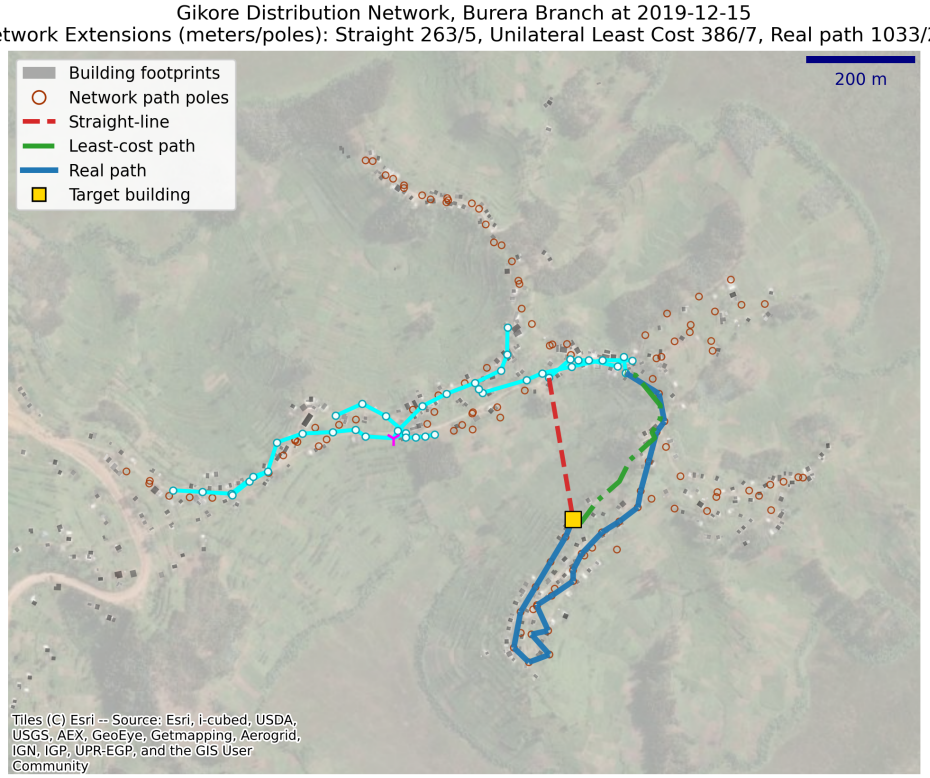
The realized network path $\mathcal{P}_{it}^{\text{path}}(\{i\}, \mathcal{E}_t)$ is not known to customers ex ante, before a neighbor connects, nor are they charged for it. The straight-line and unilateral LCP network extension methods are inspired by the geographic instrumental variable methods often seen in the infrastructure and development literature. For example, Faber (2014) uses straight lines and minimum spanning trees to instrument for highway infrastructure in China, while Lipscomb, Mobarak, and Barham (2013) use a least-cost plan connecting hydroelectric dams to instrument for electricity transmission infrastructure in Brazil. However, while the above papers use network extensions to instrument for realized infrastructure *ex post*, I use network extensions to measure *ex ante* connection charges and, later, the externalities from neighbor connection requests.

5.4. Connection Charges

Customers get charged for the material costs in their "minimum technical solution" or unilateral least-cost plan extension $\mathcal{L}_{it}^{\text{lcp}}(\{i\}, \mathcal{E}_t)$ and $\mathcal{P}_{it}^{\text{lcp}}(\{i\}, \mathcal{E}_t)$. If the utility chooses to deviate from the minimum solution to enhance the long-run effects of i's provision, the utility pays the cost of building that additional infrastructure. Customers can assess connection charges either by asking the utility for a quotation or by applying the connection-charge policy themselves.

Applying Rwanda's connection charge policy function to the poles and lines of the unilateral LCP network extension gives the *scheduled unilateral charge* each consumer faces over time, because they represent the minimum technical solution for i's network expansion. Suppose the unilateral least-cost

Figure 6 — Alternative network-extension methods on hilly switchback terrain



Notes. Grid state at 2019/12/15 of the Gikore distribution network in Burera. The magenta shape is the distribution transformer, cyan lines with white circles are energized poles and lines, gray polygons are Google Open Buildings footprints, brown circles are the raw network-path poles in the data before geospatial cleaning, so duplicates are observable. For an adopting customer i, represented as a yellow square, the figure shows three network-extension methods.

- The red dashed straight-line extension draws the shortest line from the existing grid to the new adopter. This network extension would require 263 m of line and 5 poles for the connection.
- The green dash-dotted unilateral least-cost-planning extension incorporates the effects of terrain and obstacles to simulate the least-cost path of connecting i to the grid. This network extension would require 386 m of line and 7 poles.
- The blue solid network path follows the switchback around the hillside. This is the path the utility actually constructed, and it required 1,033 m of line and 23 poles. The utility built this path to extend the network to en-route customers.

plan to connect i at a given moment t is $\mathcal{L}^{\text{lcp}}_{it}(\{i\}, \mathcal{E}_t)$ and $\mathcal{P}^{\text{lcp}}_{it}(\{i\}, \mathcal{E}_t)$. The total material requirements are $P^m_{it}(Q, \mathcal{E}_t) = |\mathcal{P}^m_{it}(Q, \mathcal{E}_t)|$ poles and $\Lambda^m_{it}(Q, \mathcal{E}_t) = \sum_{\ell \in \mathcal{L}^m_{it}(Q, \mathcal{E}_t)} d_\ell$ meters of line. Then Rwanda's connection-charge policy, described in subsection 3.5 and visualized in Figure 4, charges RWF 2,000 per meter of line and RWF 200,000 per pole, so that the *unilateral charge* is

$$c^{\text{lcp}}_{it}(P, \Lambda) = 2\Lambda^{\text{lcp}}_{it} + 200 P^{\text{lcp}}_{it}$$

in RWF 1000.[32] This assumes full take-up of the available credit for the RWF 56,000 meter and service drop.

5.5. Grand-Coalition Cost Shares

The unilateral charge c^{lcp}_{it} is an upper bound for each adopter. But at each point in time, a customer may also adopt as part of a coalition. I use *grand-coalition Shapley cost shares* $\kappa^{\bar{Q}}_{it}$ to control for the maximal benefit of cooperation. Each share measures the cost if every remaining customer in a neighborhood $\bar{Q}_{it}$ were to adopt at the same time and share the charges of its common infrastructure according to the Shapley rule. When applied to cooperative infrastructure-provision games in settings ranging from airports and irrigation systems to generic rooted trees like this one, the Shapley value divides the cost of each component among its users (Littlechild and Owen 1973; Littlechild and Thompson 1977; Márkus, Pintér, and Radványi 2012). Let n_e be the number of customers sharing each network component $e \in \mathcal{B}_t(\bar{Q})$. Then applying the Shapley rule piecewise and Rwanda's scheduled element charges gives

$$\begin{aligned} \kappa^{\bar{Q}}_{it} \equiv \kappa_i(\bar{Q}_{it}; q^R, \mathcal{E}_t) = &\sum_{\ell \in \mathcal{L}_{it}(\bar{Q}_{it}, \mathcal{E}_t)} \frac{2 d_\ell}{|\{j \in \bar{Q}_{it} : \ell \in \mathcal{L}_{jt}(\bar{Q}_{it}, \mathcal{E}_t)\}|} \\ &+ \sum_{p \in \mathcal{P}_{it}(\bar{Q}_{it}, \mathcal{E}_t)} \frac{200}{|\{j \in \bar{Q}_{it} : p \in \mathcal{P}_{jt}(\bar{Q}_{it}, \mathcal{E}_t)\}|}. \end{aligned} \tag{9}$$

I use $\kappa^{\bar{Q}}_{it}$ as a control because I will later construct and instrument a measure of the incentives for joint adoption following the model. Here, I want to control for cooperation by other means, such as collective bargaining with the utility, coordination of rights-of-way, information sharing, and other unobserved means via which customers may cooperate. Conditional on c^{lcp}_{it}, a smaller $\kappa^{\bar{Q}}_{it}$ implies a stronger cooperation incentive. The unilateral charge c^{lcp}_{it} and the grand-coalition share $\kappa^{\bar{Q}}_{it}$ together provide upper and lower bounds, respectively, for connection charges. Together with customer fixed effects that absorb willingness to pay, they provide static controls for the primitive adoption incentives described in section 2 except the externality $c - c'$.

5.6. Social Peer Exposure

Another identification challenge is that social peer network effects may correlate with grid infrastructure network expansions through common geographic primitives. I use a bilateral inverse-distance network

32. As Rwanda uses a distance-based rather than a material-based policy, c^{lcp}_{it} is the appropriate charge. If instead a utility used a material-based rather than a distance-based charge, customers would pay c^{path}_{it}. This charge would not be known before an engineer's survey revealed the path and its cost. The network-path materials provide the engineering supply-side costs for the utility.

interacted with connection status to control for *social peer exposure*, following Blume et al. (2015). I intend for it to control for spatially local social peer effects, including learning, imitation, status, appliance availability, and electricity sharing. Bernard and Torero (2015) provide a framework for the social peer effects of electricity, separating them into *expectation interactions*, *preference interactions*, and *constraint interactions*. Expectation interactions include peer effects that change how customers value electricity, such as social learning. Preference interactions include social status effects such as conspicuous consumption. Constraint interactions include grid externalities, such as cooperation and free-riding that I will explicitly model, and peer effects that change the demand for connections, such as sharing electric appliances like fridges, which increase the value of non-adoption, or external economies of scale in the supply of complementary electric appliances. Except for the strategic incentives I describe in section 2, the remaining electricity peer-effect channels that Bernard and Torero (2015) catalog affect grid electricity adoption mainly via consumption rather than through the act of connection. While some potential channels can affect connection directly, such as information about the request process, quotations, credit, eligibility, or coordinated trips to the utility branch, these channels affect coordination costs ρ in the theory and therefore determine strategic responses.

Let D_{ij} denote the distance in meters between customers i and j. Throughout the empirical sections, the superscript $\circ(-\zeta)$ denotes an entrywise inverse-distance power, with self-links and distances beyond 1,000 meters assigned zero weight:

$$\left[\mathbf{D}^{\circ(-\zeta)}\right]_{ij} = \begin{cases} D_{ij}^{-\zeta}, & i \neq j, \quad D_{ij} \leq 1000, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

This is an entrywise transformation, not a matrix inverse. The same weights apply to connected-customer exposure $\mathbf{D}^{\circ(-\zeta)}\mathbf{y}_t^*$ and contextual peer characteristics $\mathbf{D}^{\circ(-\zeta)}\mathbf{x}_{-i}$, including their time trends.

I follow the functional form of social peer effects in Bernard and Torero (2015), who assume that social peer effects decline with distance and are a function of the connected set of customers y_{jt}^*,

$$\left[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}_t^*\right]_i = \sum_{j \neq i, D_{ij} \leq 1000} D_{ij}^{-\zeta} y_{jt}^*. \tag{11}$$

As they do, I set $\zeta = 1$ in the baseline, but perform robustness checks using alternative parameterizations of decay. I let social peer effects occur across transformers.

5.7. Event-Study Evidence

I present an event study to motivate the remaining strategic measures. I estimate how adoption hazards vary in response to the first neighbor externality that changes the connection charge each customer faces. Specifically, this is the first week in which the grid endogenously expands toward customer i. I omit adoptions that extend the grid at $t = 1$ because they indicate infrastructure supply rather than endogenous

neighbor provision.[33]

$$\tau_i = \min\{t \in \{2, \ldots, 53\} \mid d_i(\mathcal{E}_t) < d_i(\mathcal{E}_{t-1})\}.$$

In response to a neighbor's adoption request at $\tau_i - 1$, whose network extension is realized between periods $\tau_i - 1$ and τ_i, customer i's distance d_i falls at τ_i.

I regress adoption hazards on bins for each time period $T' = \{-12, \ldots, 12\}$ around a neighbor's first externality. I include customer fixed effects, branch-by-week fixed effects, and optionally cost measures, social peer exposures, and additional controls:

$$y_{it} = \sum_{t' \in T'} \beta_{t'} \mathbf{1}\{t - \tau_i = t'\} + \theta_c c_{it}^{\text{lcp}} + \theta_\kappa \left(c_{it}^{\text{lcp}} - \kappa_{it}^{\bar{Q}} \right) + \theta_D [\mathbf{D}^{\circ(-1)} \mathbf{y}_t^*]_i + \mathbf{w}_{it}' \theta_w + \mu_i + \psi_{b(i)t} + \varepsilon_{it}. \quad (12)$$

In Figure 7a, I show four estimates: one with only fixed effects μ_i and $\psi_{b(i)t}$, one adding the static cost control c_{it}^{lcp}, one adding the cooperation surplus $c_{it}^{\text{lcp}} - \kappa_{it}^{\bar{Q}}$ and social peer exposure $[\mathbf{D}^{\circ(-1)} \mathbf{y}_t^*]_i$, and finally, one that matches the full specification of Equation 12. This includes the vector $\mathbf{w}_{it}$, which contains additional controls in the full specification I add to ensure exclusion of my instrumental variables.[34] Because the event is induced by endogenous neighbor adoption, I consider these estimates a descriptive pattern rather than a causal effect.

Under a null hypothesis of no strategic incentives, customers should adopt electricity based only on static differences in their willingness or ability to pay. The event study would show a step function at event time 0, when a lower charge is realized without anticipation of the event or joint adoption. Under this null, controlling for the connection charge should attenuate the step if behavior were unstrategic.

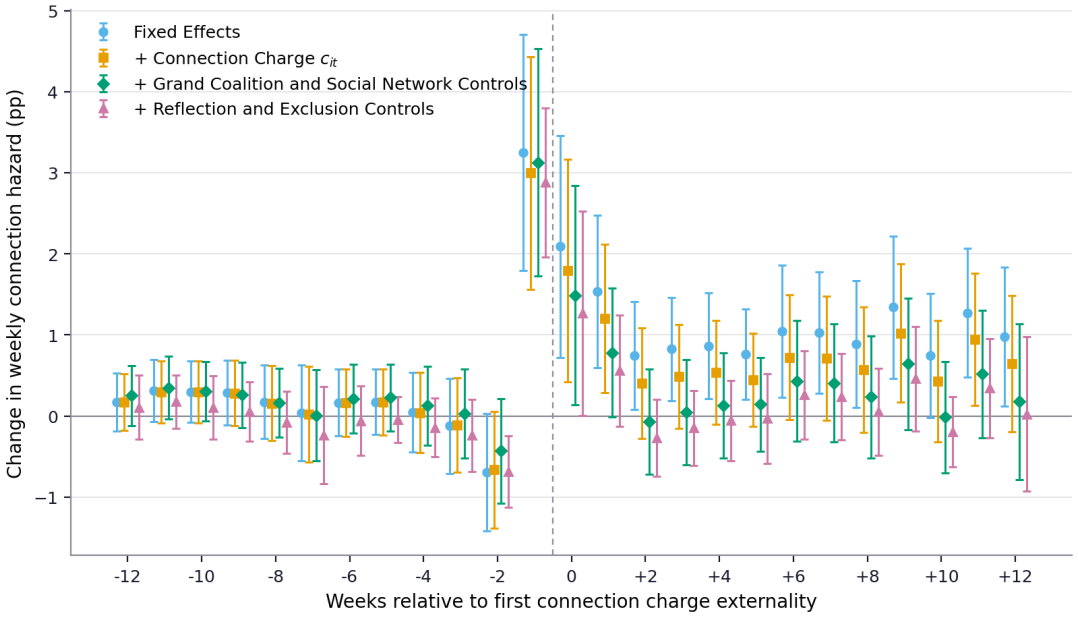
Instead, Figure 7a looks like a wave, a pattern consistent with the theoretical predictions of Proposition 3. Adoption is relatively stable far from the event. It decreases as the event approaches, consistent with the proposition's delay phase. Connection hazards spike as consumers make connection requests that coincide with their neighbor's, consistent with the cooperation the theory predicts. However, the spike could also reflect demand-side economies of scale or supply-side batching, which can mimic joint requests. After the externality occurs, connection hazards settle at a hazard above the pre-event level. Rather than spuriously creating the event-time patterns, adding more controls iteratively tends to attenuate the strategic adoption effects.

Figure 7b does a heterogeneity analysis that descriptively compares events with large and small externalities. It classifies the event occurring at τ_i according to whether the reduction in the connection charge is above or below the median. The same patterns consistent with Proposition 3 occur in both groups, with larger coefficients around larger externalities. These patterns are consistent with potential utility customers adjusting adoption behavior not only in anticipation of an event on the extensive margin but also in response to externality size on the intensive margin.

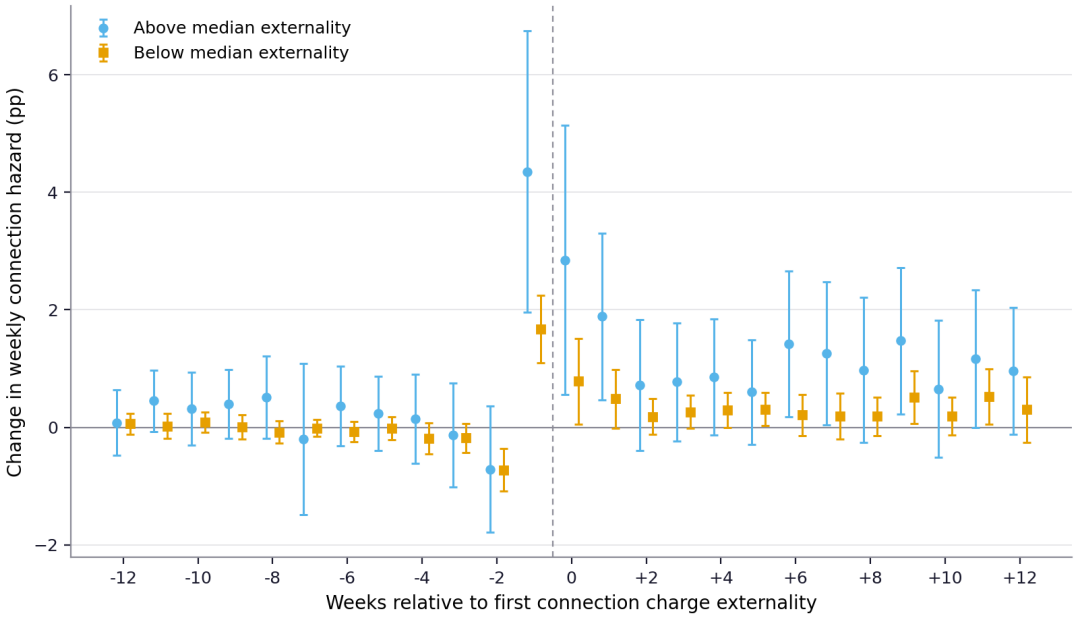
The next section constructs measures that decompose these descriptive patterns before, during, and after neighbor externalities into anticipation, cooperation, and free-riding. Unlike the event study, I aim

33. Transformers are constructed at $t = 0$. Therefore, connections made at $t = 1$ respond to the supply-side initial transformer placement rather than to endogenous externalities in connection charges.

34. Reflection controls include time-varying contextual peer effects and cooperative side-payments, and exclusion controls include exchange-rate exposure and spatial autocorrelation.



(a) First cost-relevant peer externality



(b) Above- and below-median peer externalities

Figure 7 — Connection hazards around cost-relevant peer externalities.

Notes. The event week τ_i is the first week in which a neighbor provides an externality. Standard errors are two-way clustered at the transformer and week level. The reference group is the same set of adopters from periods more than 12 weeks before the event and potential adopters who never receive an externality. Observations more than 12 weeks after the first event are dropped. In panel (a), I show a fixed-effects-only specification and add additional controls iteratively. In the fixed-effects-only specification, joint adoption at event time -1 is about six times the baseline hazard of 0.448%. The second specification adds a connection charge control, the next adds the cooperation surplus and social peer exposure; the final specification adds reflection and exclusion controls discussed in the next section. These include inverse-distance neighbor-characteristic time trends. Panel (b) uses the connection charge and fixed-effects specification, and separates events by whether the externality is above or below the median size. Larger externalities induce larger effects, but these are underestimates. Differences in baseline adoption hazards mask some of the effect, as the baseline hazard is 0.592% for the above-median-externality group and 0.265% for the below-median-externality group. Binwise fixed effects control for baseline adoption hazards.

to use all externalities over the course of the grid rollout, not just the first one. It also describes how I instrument the measures to estimate causal effects.

6 — Strategic Incentives, Identification, and Instruments

The reduced-form section primarily tests empirically whether electricity adoption responds to strategic incentives.

Identifying the strategic incentives described in section 2 is a substantial challenge for four reasons. First, utilities place infrastructure to minimize costs over the whole network, not for any one customer. Infrastructure placement is endogenous to the spatial organization of customer demand. It is also not observed ex ante. This motivates the straight-line and least-cost-planning network extensions that provide boundedly rational measures of how customers anticipate externalities. Second, delayed, joint, and sequential adoption all result from strategic incentives, and these need to be separated. Third, grid externalities may be correlated with social peer effects, which can confound estimation via omitted variable bias. Finally, adoption decisions are made in equilibrium with neighbor actions. Therefore, adoption decisions are simultaneously determined and generate a reflection problem. Combined, the latter three empirical challenges require the empirical strategy to separate seven channels of external network effects from static incentives:

1. endogenous anticipation of future connection charge externalities,
2. endogenous cooperation via contemporaneous connection charge externalities,
3. endogenous free-riding on realized connection charge externalities,
4. endogenous social peer effects,
5. contextual grid peer effects,
6. contextual social peer effects,
7. correlated effects.

Only the first three endogenous peer effects are of theoretical interest in section 2. Disentangling *endogenous, contextual, and correlated* effects constitutes the reflection problem (Manski 1993). Applying Manski's terminology to this context, *endogenous* peer effects occur when adoption behavior depends on neighbors' adoption behavior, *contextual* peer effects occur when behavior depends on neighbors' characteristics, and *correlated* effects occur due to random local shocks. The additional challenge of my setting is that *grid* infrastructure externalities must also be separately identified from *non-grid* social or geographic peer effects, and further, that there are several endogenous responses to strategic incentives I wish to decompose.

In this section, I describe how I construct measures of each of the above *network channels* to identify them separately or control for them. First, I construct an anticipated externality matrix $\mathbf{G}_t(\mathcal{E}_t)$ which encodes users' anticipation of how each of their neighbors would induce infrastructure if they were to adopt. Each cell of the externality matrix simulates a neighbor's path extension from the current grid state

with an LCP or straight-line network extension, and then measures the decrease in distance or connection charge each user receives from it.

Second, I temporally decompose the wave into anticipation, cooperation, and free-riding using Proposition 3 and combine neighbor adoptions with the externality matrix to construct three measures: F_{it}, S_{it}, and L_{it}. In advance of the event at $t' < -1$, reduced adoption associated with a future externality is measured by future leads of the externality F_{it}. In the neighbor's installation week at $t' = -1$, same-week adoption is measured by contemporaneous externalities S_{it}. After the externality is realized, at $t' \geq 0$, sequential adoption is measured by lagged differences in the connection charge L_{it}. Separate demand-side peer-of-peer Z_{it}^F and supply-side shift-share instrumental variables Z_{it}^S instrument F_{it} and S_{it} respectively.

Third, in addition to the externality matrix, as previously described, $\mathbf{D}^{\circ(-\zeta)}$ provides an alternative network structure. Its goal is to control for social peer effects that may be correlated with grid externalities via common geographic primitives. After introducing these variables, I proceed to present the estimating equation, and then state how the construction of each variable maps to specific network channels. I then turn to identification assumptions and explain the construction, relevance, and exclusion of the instruments.

6.1. Externality Matrices

The theory in subsection 2.1 predicts that a positive externality $c_i - c_i'$ upon neighbor adoption would lower i's incentives to adopt relative to a static willingness-to-pay model. With more than two players, an index is needed for each neighbor j's hypothetical network extension. In the present notation, the realized externality that i would receive if j were to adopt can be written $c_i^{\text{lcp}}(\mathcal{E}_t) - c_i^{\text{lcp}}(\mathcal{E}_t \cup \text{path}_{\mathscr{G}}(j))$. As in the strategic game, the positive externality affects i's behavior via the prospect of whether j may adopt in the future via backward induction, rather than only ex post upon its realization.

Some further restrictions on the externality become relevant in an $N = |\mathcal{I}|$-player game. By definition, customers cannot receive externalities from themselves: $i \neq j$. And connection charges are non-negative, so the maximum externality reduces the connection charge to zero. With these restrictions, the *realized externality* if j were to adopt at time t and provide an externality to i is:

$$G_{ijt}^{\text{path,c}}(\mathscr{G}, \mathcal{E}_t) = \begin{cases} \dfrac{c_i^{\text{lcp}}(\mathcal{E}_t) - c_i^{\text{lcp}}(\mathcal{E}_t \cup \text{path}_{\mathscr{G}}(j))}{c_i^{\text{lcp}}(\mathcal{E}_t)}, & c_i^{\text{lcp}}(\mathcal{E}_t) > 0 \text{ and } i \neq j \\ 0, & c_i^{\text{lcp}}(\mathcal{E}_t) = 0 \text{ or } i = j. \end{cases} \tag{13}$$

Equation 13 uses relative shares of i's current connection charge that would be cross-subsidized if j adopted unilaterally. Using shares is the simplest way to ensure that strategic incentives do not also measure static ones. It allows me to control for the static cost without collinearity, so that variables based on Equation 13 can be pure measures of strategic effects.

I stack the elements $G_{ijt}^{\text{path,c}}(\mathscr{G}, \mathcal{E}_t)$ across each column inducer j and each row receiver i within each distribution transformer in a square network-path *realized externality matrix* $\mathbf{G}_t^{\text{path,c}}$. Each row of the realized externality matrix summarizes the inward externalities that customer i would receive from each of its neighbors, and each column describes the outward externalities that adopter j would confer on each of its neighbors through unilateral adoption. The externality matrix is inspired by Elliott and Golub (2019)

and can be viewed as an applied version of their adjacency matrix for networked public goods.

The realized externality matrix can be sparse. It contains zeros for four reasons. First, the diagonal of $\mathbf{G}_t^{\text{path,c}}$ is zero because a customer cannot confer an externality on itself. Second, the rows and columns corresponding to customers in the connected set Y_t^* are zero because connected customers can no longer receive or confer connection-charge externalities through subsequent connections. Third, the rows are zero for unconnected customers under a service drop already facing the minimum connection charge, as their adoption would induce no public infrastructure. Finally, the externality matrix is block-hierarchical, inheriting this property from the adjacency matrix of a tree graph. Customers on different branches of the tree that do not share any unconstructed infrastructure, such as customers on opposite sides of the transformer, never provide realized externalities to each other.

The realized externality matrix inherits the properties of the externality itself; it is non-negative, local, directed, dynamic, and policy-dependent. It is also asymmetric because externalities are expressed as shares of recipients' current charges. A more downstream customer with a higher $c_i^{\text{lcp}}(\mathcal{E}_t)$ receives a smaller relative externality from an upstream neighbor than the upstream customer would receive from the downstream neighbor because the shared infrastructure makes the numerator approximately equal, but the downstream customer's denominator is larger. Some theoretical results of Elliott and Golub (2019) apply to this setting, but not all, because here the externality matrix is neither differentiable nor necessarily irreducible.[35]

An *anticipated externality matrix* uses straight-line and unilateral least-cost-planning extension rules instead of the realized network path. These omit potential endogeneity in the neighbor's extension path $\text{path}_{\mathscr{G}}(j)$ by simulating unilateral network extensions, and are defined in cost or distance measures as:

$$G_{ijt}^{\text{lcp,c}}(\mathscr{G},\mathcal{E}_t)=\begin{cases}\dfrac{c_i^{\text{lcp}}(\mathcal{E}_t)-c_i^{\text{lcp}}\left(\mathcal{E}_t\cup\mathcal{P}_{jt}^{\text{lcp}}(\{j\},\mathcal{E}_t)\right)}{c_i^{\text{lcp}}(\mathcal{E}_t)}, & c_i^{\text{lcp}}(\mathcal{E}_t)>0 \text{ and } i\neq j\\ 0, & c_i^{\text{lcp}}(\mathcal{E}_t)=0 \text{ or } i=j.\end{cases} \tag{14}$$

$$G_{ijt}^{\text{lcp,d}}(\mathscr{G},\mathcal{E}_t)=\begin{cases}\dfrac{d_i(\mathcal{E}_t)-d_i\left(\mathcal{E}_t\cup\mathcal{P}_{jt}^{\text{lcp}}(\{j\},\mathcal{E}_t)\right)}{d_i(\mathcal{E}_t)}, & c_i^{\text{lcp}}(\mathcal{E}_t)>0 \text{ and } i\neq j\\ 0, & c_i^{\text{lcp}}(\mathcal{E}_t)=0 \text{ or } i=j.\end{cases} \tag{15}$$

$$G_{ijt}^{\text{straight,c}}(\mathscr{G},\mathcal{E}_t)=\begin{cases}\dfrac{c_i^{\text{lcp}}(\mathcal{E}_t)-c_i^{\text{lcp}}\left(\mathcal{E}_t\cup\mathcal{L}_{jt}^{\text{straight}}(\{j\},\mathcal{E}_t)\right)}{c_i^{\text{lcp}}(\mathcal{E}_t)}, & c_i^{\text{lcp}}(\mathcal{E}_t)>0 \text{ and } i\neq j\\ 0, & c_i^{\text{lcp}}(\mathcal{E}_t)=0 \text{ or } i=j.\end{cases} \tag{16}$$

$$G_{ijt}^{\text{straight,d}}(\mathscr{G},\mathcal{E}_t)=\begin{cases}\dfrac{d_i(\mathcal{E}_t)-d_i\left(\mathcal{E}_t\cup\mathcal{L}_{jt}^{\text{straight}}(\{j\},\mathcal{E}_t)\right)}{d_i(\mathcal{E}_t)}, & c_i^{\text{lcp}}(\mathcal{E}_t)>0 \text{ and } i\neq j\\ 0, & c_i^{\text{lcp}}(\mathcal{E}_t)=0 \text{ or } i=j.\end{cases} \tag{17}$$

The first uses the unilateral LCP network extension to measure what poles would be constructed if j were to adopt, and the corresponding relative cost change on i as a result of their extension. The second does the

35. Indeed, it is neighborhood-wise reducible.

same but proxies the impact on i's connection charge with the reduction in distance. The remainder are analogous connection charge and distance measures, but instead of using the LCP network extension, they base the externality on the geometric union of the poles in the grid state and the straight line from the closest pole in that graph to j. Similarly to $\mathbf{G}_t^{\text{path,c}}$, these measures are stacked across i and j into matrices $\mathbf{G}_t^{\text{lcp,c}}$, $\mathbf{G}_t^{\text{lcp,d}}$, $\mathbf{G}_t^{\text{straight,c}}$, and $\mathbf{G}_t^{\text{straight,d}}$.

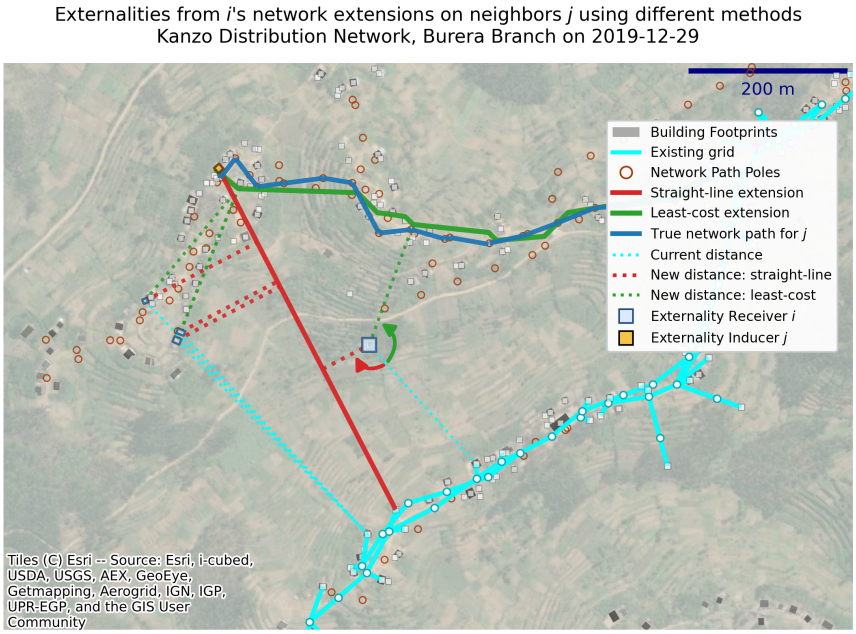
The main specifications of the paper use the anticipated externality matrix $\mathbf{G}_t^{\text{straight,d}}$ to reflect what customers know as they make decisions. While the utility's least-cost plan $\mathscr{G}$ minimizes total infrastructure costs and therefore incorporates not only i's demand but every other potential demand in the area, customers do not observe planning documents. To correctly project neighbor externalities, they must understand the policy, anticipate who will adopt in the village and when they do so, estimate how the utility optimizes infrastructure over space to project the geographic route the utility would extend the grid in response to each neighbor's request, calculate how each extension would affect their distance to the grid and subsequent connection charge, and best respond to neighbors' mixed strategies. This is a demanding computational requirement. Network expansions are not observed at state $\mathcal{E}_t$ and are only anticipated.

Further, for potential utility customers, anticipating the geographic realization of network expansions is difficult because of asymmetric information. Each customer only adopts electricity once and observes limited utility network extensions in their neighborhood. Understanding how the utility extends the network requires potential customers also to understand to what extent the utility constructs infrastructure to incorporate unrealized demand, what information the utility has on potential customers, and what weights they place on their anticipated customers as they build out the grid, and how fluctuations in multilateral funding, supply of materials, and exchange rates may affect distribution network construction. I consider this too challenging. Instead, I consider a boundedly rational strategy in this local interaction game, following a tradition emanating from Simon (1955) and Sadler (2014). I use anticipated externality matrices $\mathbf{G}_t^{\text{straight}}$ and $\mathbf{G}_t^{\text{lcp}}$ as a boundedly rational heuristic for the realized externality matrix $\mathbf{G}_t^{\text{path}}$. The heuristic assumes that unilateral network extensions encode a reasonable expectation that if j adopts, the utility builds the minimum technical solution to reach j, which I measure using the LCP network extension or a straight line. It assumes that i does not count on the utility to necessarily extend the network towards them. Instead, i considers how much closer they would get to the grid if the utility built a network extension directly to j and considers how it would affect their connection charge. I compare these strategies visually in Figure 8.

In the primary specification of the paper, I use the straight-line-distance-based externality matrix $\mathbf{G}_t^{\text{straight,d}}$ to construct F_{it} and S_{it}, use the network-path externality matrix $\mathbf{G}_t^{\text{path,c}}$ to construct O_{it}, and use $\mathbf{G}_t^{\text{lcp,d}}$ and $\mathbf{G}_t^{\text{lcp,c}}$ in robustness tests to allow for richer expectations over network expansion. I opt to use distance rather than the connection charge partially for the same boundedly rational reason, but also because a charge-based externality matrix is sensitive to measurement error in the geospatial network data and simulated network extensions, as there are non-linearities in the connection charge. That said, the straight-line network extension is more naive than the unilateral LCP network extension, as it also does not incorporate the effects of roads or terrain.

The network path externality is in general slightly larger than the straight-line externality. The mean ratio of G_{ijt}^{path} to its corresponding $G_{ijt}^{\text{straight}}$ is 1.16, while the median ratio is 1.02. I deflate estimates using

Figure 8 — Externality measurement with network path, unilateral straight-line, and unilateral least-cost plan network extensions



Notes. This picture shows how adopting an externality inducer j, pictured by the yellow diamond, would extend the grid. At this state, the existing grid is colored in neon. The induced network extensions in red and green represent the boundedly rational, hypothetical, and unilateral straight-line and least-cost network extensions j would induce. Eventually, if j connects, the utility uses the blue path. The picture focuses on four customers in blue squares who receive externalities. While their initial straight-line distances from the grid are in dotted neon, their new distances are in dotted red or green. The differences in length from the neon to the red or green dotted lines are $G_{ijt}^{\text{straight,d}}(\mathscr{G}, \mathcal{E}_t)$ and $G_{ijt}^{\text{lcp,d}}(\mathscr{G}, \mathcal{E}_t)$ respectively. While the three receivers on the left have similar externalities under both methods, the central receiver has different externalities, as indicated by the red and green arrows.

$G_{ijt}^{\text{straight}}$ by 1.16, to ensure that they match the size of the real externality, while also ensuring that the estimates reflect ex ante anticipation and contemporaneous exposure to strategic incentives.

6.2. Measuring Anticipation, Cooperation, and Free-Riding

From the externality matrices, I derive three adoption incentives for the main reduced-form specification from their rows, and another measure from their columns.

$$F_{it}(\beta, H) \equiv \max_j \max_{1 \le h \le H} G_{ijt}^{\text{straight,d}} \beta^h y_{j,t+h}, \tag{18}$$

$$S_{it} \equiv \max_j G_{ijt}^{\text{straight,d}} y_{jt}, \tag{19}$$

$$L_{it}(H) \equiv \frac{c_i^{\text{lcp}}(\mathcal{E}_{t-H}) - c_i^{\text{lcp}}(\mathcal{E}_t)}{c_i^{\text{lcp}}(\mathcal{E}_{t-H})}, \tag{20}$$

$$O_{it} \equiv \sum_j \left[G_{jit}^{\text{path,c}} \, c_j^{\text{lcp}}(\mathcal{E}_t) \right] = \sum_{j \neq i} \left[c_j^{\text{lcp}}(\mathcal{E}_t) - c_j^{\text{lcp}}(\mathcal{E}_t \cup \mathcal{B}_{\mathscr{G}}(it)) \right]. \tag{21}$$

Current inward externalities S_{it} and *future inward externalities* $F_{it}(\beta, H)$, which I refer to collectively as *in-externalities*, are the primary variables of interest in the reduced form. They are each motivated by the wave in Proposition 3 and the event study in Figure 7a. The in-externalities measure the externality that i receives from neighbors j in the current period y_{jt} or in the next H periods $y_{j,t+h}$. They interact neighbor adoption with the resulting externality that each neighbor adoption would induce, and take the best neighbor overall.

Future in-externalities F_{it} measure the strategic *anticipation* response to future neighbor externalities, corresponding to the anticipable externalities $y_{j,t+h}$ from j that occur in period $t+h$. They correspond to the reaction to the upcoming event in periods $t' \in [-H, -2]$ in the event study, and to the delay and freeze phases of Proposition 3. A negative response to this would be consistent with the strategic delay of electricity adoption. The externality $G_{ijt}^{\text{straight,d}} y_{j,t+h}$ is discounted weekly by β^h and truncated at horizon H to measure the present value benefit from waiting for a neighbor to connect in the next H periods. Parameters β and H encode how much customers discount future adoptions and how far in the future they can anticipate neighbor adoption. A discount factor $\beta < 1$ would mean that i discounts externalities further in the future, while a finite horizon H would encode limited foresight. In my baseline specifications, I assume $\beta = 1$ and that $H = 8$. The former imposes no weekly discounting to benchmark anticipation and weights all H future weeks equally. The latter appears to be non-binding as the event study in Figure 7a shows no effect more than eight weeks ahead of the externality.[36] A larger H has an empirical cost because it censors observations in the last weeks of the panel, as adoptions past the last adoption in the data are not observed.

There is one form of reflection that F_{it} does not control for. When i adopts, their extension may induce j to adopt at $t+h$, so realized future exposure can contain feedback from current own adoption inducing others to adopt. This creates attenuation bias against the theoretical prediction that anticipated

36. Also note that myopic response to neighbor play is rational when discounting is strong relative to the arrival of revision opportunities (Blume 1993).

neighbor externalities reduce adoption hazards because F_{it} would be artificially large and would increase contemporaneous adoption rather than decrease it.

Current in-externalities S_{it} are an analogous measure but only include externalities in the joint adoption phase of Proposition 3 or during the spike in adoption at $t' = -1$ in Figure 7a. They measure the *joint-adoption cooperative* response to strategic incentives in neighbor externalities. They test the reduced-form demand-side coordination economies of scale and infrastructure cost-sharing incentives that enter ρ in the theory, aggregating coordination costs and benefits.

Both future and current in-externalities are constructed using a *maximum* over neighbors $j \neq i$ because, at a fixed state, neighbors provide *mutually exclusive* externalities to potential adopter i.[37] For example, consider a potential adopter in between a Y-shaped fork along a road. If neighbors on both branches of the Y extend the grid to the top of both branches, the realized connection charge for those in the middle depends on the minimum distance to the grid, and only on whichever side's infrastructure is realized to be closer to them. Therefore, if multiple connecting neighbors create alternative routes to the grid, i's remaining distance and charge are determined by the ultimate minimum connection charge to the nearer extension and the maximum cumulative externality.[38] If multiple inducers j and k adopt sequentially and both provide externalities to i, the aggregate effect is also as the closer extension at time t, induced by the more downstream of j and k.

As both measures depend on contemporaneous and future adoption, both F_{it} and S_{it} are endogenous. I instrument them using beginning-of-period-t information to only use the adoptions anticipable at that time. The former uses demand-side censored peer-of-peer effects to avoid the reflection problem, while the latter uses a supply-side shift-share instrument to estimate joint adoption. I return to the instrumental variables after presenting the estimating equation and controls.

The *past externality* $L_{it}(H) = \frac{c_i^{\text{lcp}}(\mathcal{E}_{t-H}) - c_i^{\text{lcp}}(\mathcal{E}_t)}{c_i^{\text{lcp}}(\mathcal{E}_{t-H})}$ measures the total externalities that were realized in the last H weeks. It measures the strategic *free-riding* response to a neighbor externality in the war of attrition described in Proposition 3, and the post-event decline to a new equilibrium in Figure 7a. After the connection charge c_{it}^{lcp} controls for static incentives to adopt, L_{it} measures who has delayed their adoption in order to free-ride. I construct it analogously to $G_{ijt}^{\text{path,c}}$ but using realized rather than prospective adoption, since the past is observable to prospective adopters at the beginning of week t. Under a null hypothesis of no strategic behavior, unlike the other measures, it would still produce a positive coefficient because it would capture selection effects (i.e., adopters in the new baseline war in the proof of Proposition 3); it is therefore neither identified nor a primary test of strategic incentives. I currently use it only to decompose how often a customer would free-ride or cooperate conditional on a neighbor's adoption in a secondary dyadic regression.

The theory encodes cooperation as joint adoption, but joint adoption is not the only way potential grid adopters may cooperate. In particular, cooperation in the act of connecting to electricity can occur through unobserved side payments such as informal loans, contributions of money, materials, or labor, and reciprocal transfers, which would internalize the positive externalities provided by inducers. I control

37. The condition $j \neq i$ is not explicit here because $G_{iit} = 0$ for all externality matrices.

38. As multiple neighbors y_{jt} would increase F_{it} by the most downstream adopter with largest G_{ijt}, this is reflected in F_{it}.

for potential side payments with the *outward externality* or out-externality O_{it}, which measures the total benefits that customer i's hypothetical adoption would confer on its neighbors. I construct it by interacting the ith column of the realized externality matrix with neighbors' connection charges and summing, because side payments should be related to the monetary size of the externality that i confers on others, operating via sums instead of maxima because side payments can be received from multiple downstream neighbors. Any inferential conclusions stemming from O_{it} are indirect because it does not directly measure side payments, and the measure does not depend on endogenous adoption behavior and is predetermined at the beginning of week t, so it will not need to be instrumented.

Together, S_{it}, O_{it}, and $\kappa_{it}^{\bar{Q}}$ decompose three ways that customers may cooperate with neighbors to adopt electricity: joint adoption, transfers and side payments, and aggregate gains from grand-coalition adoption.[39]

6.3. Estimating Equation

I estimate the adoption hazard on the set of remaining unconnected customers. Each potential adopter contributes one observation per week until it connects, then exits the panel. Customers who adopt in the first week of a distribution network contribute only one observation and are implicitly dropped in estimation because they have singleton fixed effects. After the sample selection filters in Table B1, the panel consists of customer-week observations with complete data on the instruments and controls.

The main specification is the linear probability model,

$$\begin{aligned} y_{it} =& \theta_c c_{it}^{\text{lcp}} + \theta_\kappa (c_{it}^{\text{lcp}} - \kappa_{it}^{\bar{Q}}) + \theta_F F_{it}(\beta, H) + \theta_O O_{it} + \theta_S S_{it} + \theta_{Dy}[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}_{-it}^*]_i \\ &+ \theta_{Dx}[t\mathbf{D}^{\circ(-\zeta)}\mathbf{x}_{-i}]_i + \theta_{Gx}[\mathbf{G}_t\mathbf{x}_{-i}]_i + \theta_{\text{FX}}\text{FX}_{it} \\ &+ \mu_i + \psi_{b(i)t} + \varepsilon_{it}. \end{aligned} \tag{22}$$

The first line includes the aforementioned variables. The second line includes controls for instrumental variables, and time-trend contextual effects. The final line adds customer and branch-by-week fixed effects and the error term: own characteristics such as willingness to pay, static characteristics x_i, and static peer effects of all kinds are controlled for by customer fixed effects μ_i. Branch-by-week fixed effects $\psi_{b(i)t}$ control for correlated shocks, where $b(i)$ denotes the utility branch containing customer i. Standard errors are two-way clustered by transformer and by week, because customers on the same distribution network share transformer-specific variation in the externality matrix.

Each of the terms in the first and second lines targets how specific grid and non-grid endogenous and contextual peer effects affect adoption incentives via network channels:

- Static, non-strategic incentives are measured by the connection charge c_{it}^{lcp}.
- Static individual non-pecuniary adoption costs and benefits are measured by customer fixed effects μ_i.
- Endogenous anticipation is measured by future inward externalities $F_{it}(\beta, H)$.

39. I struggle to think of other cooperation channels that would not be captured by these measures, but I do not necessarily think that this is exhaustive.

- Endogenous cooperation is measured by current inward externalities S_{it}, outward externalities O_{it}, and potential gains from a grand-coalition wave of adopters $c_{it}^{\text{lcp}} - \kappa_{it}^{\bar{Q}}$.
- Endogenous free-riding is optionally measured by past externalities in L_{it}.
- Endogenous social peer effects are controlled for by inverse-distance-weighted exposure to connected customers $[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}_{-it}^*]_i = \sum_{j \neq i, D_{ij} \leq 1000} D_{ij}^{-\zeta} y_{jt}^*$.
- Contextual grid peer effects are controlled for by externality-weighted peer characteristics $[\mathbf{G}_t \mathbf{x}_{-i}]_i = \sum_j G_{ijt} x_{jt}$. They are assumed not to vary with time except via grid expansion, and support the peer-of-peer instrument's exclusion restriction.
- Contextual social peer effects that are static are controlled for by customer fixed effects μ_i,[40] while time-varying contextual peer effects are controlled for by $[t\mathbf{D}^{\circ(-\zeta)}\mathbf{x}_{-i}]_i = \sum_{j \neq i, D_{ij} \leq 1000} t D_{ij}^{-\zeta} \mathbf{x}_j$.
- Static correlated effects are captured by customer fixed effects μ_i.
- Correlated shocks are partially absorbed by branch-by-week fixed effects $\psi_{b(i)t}$.
- Own procurement exposure FX_{it} supports the shift-share instrument's exclusion restriction.

The primary hypothesis test of strategic incentives is $F_{it}(\beta, H)$. An estimated $\theta_F < 0$ would indicate that anticipated endogenous positive externalities from neighbors decrease connection charges and delay current adoption. Contemporaneous inward externalities S_{it} combine cooperation and within-week free-riding incentives, so the sign of θ_S is theoretically ambiguous, but on net I hypothesize $\theta_S > 0$, noting that it may be imprecise because of multiple equilibria.

Cooperation variables O_{it} and $c_{it}^{\text{lcp}} - \kappa_{it}^{\bar{Q}}$ only indirectly observe realized transfers and the surplus from grand-coalition formation. Nevertheless, if neighbors internalize outward externalities via side payments or cooperate in other ways aside from joint adoption, then $\theta_O > 0$ and $\theta_\kappa > 0$ should be expected. Downward-sloping demand implies $\theta_c < 0$, and the remaining measures are pure controls.

The most involved control is for endogenous social peer effects. Using $[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}_{-it}^*]_i$ as a control is based on a two-network identification logic from Blume et al. (2015). They show that when endogenous and contextual social peer effects are transmitted via different networks, including both sources of network variation can help identify those specific network channels.[41] In my setting, I want to control for endogenous non-grid social and geographic peer effects with $[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}_{-it}^*]_i$, and contextual social and geographic peer effects with $[t\mathbf{D}^{\circ(-\zeta)}\mathbf{x}_{-i}]_i = \sum_{j \neq i, D_{ij} \leq 1000} t D_{ij}^{-\zeta} \mathbf{x}_j$. The inverse distance matrix proxies for social and geographic peer networks, specifically in how they would correlate with the utility's infrastructure network through common geographic primitives. As geography is a common primitive of social network formation and grid infrastructure, controlling for the endogenous geographic component of social networks is necessary to prevent omitted variable bias.

Where social peer effects exist that affect joint cooperation costs or benefits, such as demand-side economies of scale, they are measured in ρ, which directly affects cooperation costs and benefits, so they

40. Contextual social peer effects such as $[\mathbf{D}^{\circ(-\zeta)}\mathbf{x}_{-i}]_i = \sum_{j \neq i, D_{ij} \leq 1000} D_{ij}^{-\zeta} \mathbf{x}_j$ are static.

41. See Theorem 7 of Blume et al. (2015).

do not pose a threat to my identification strategy and should be captured by S_{it}. Creating a hypothesis for the sign of θ_{Dy} is outside the motivation of this paper because its job is to control for the expectation, preference, and non-grid constraint peer effects described in subsection 5.6. Nevertheless, most of the peer-effect mechanisms described by Bernard and Torero (2015) imply that electricity consumption should increase the demand for electricity connections, so one should expect $\theta_{Dy} > 0$ as a gut check.

The remaining controls in Equation 22 support instrumental variable exclusion restrictions. Externality-weighted peer characteristics $[\mathbf{G}_t\mathbf{x}_{-i}]_i$ control for grid contextual peer effects that operate through the infrastructure network or are correlated with it (Bramoullé, Djebbari, and Fortin 2009). Own procurement exposure FX_{it} absorbs the direct effect of an exchange-rate-based shifter on customer i's connection cost. I describe the logic and construction of these controls after explaining their corresponding instruments below.

Sans forthcoming instrumental variables and their respective controls, the primary identification assumption for Equation 22 is sequential exogeneity (Kuersteiner and Prucha 2020).[42] Sequential exogeneity allows for non-independence across the cross-section of unconnected adopters within the panel, so that adopter behavior can be jointly determined and affected by geography, past adoption, and anticipated neighborhood demand. It requires that the grid state and the set of already-connected neighbors at the beginning of week t are uncorrelated with new idiosyncratic shocks to customer i's adoption decision during period t, after conditioning on controls and remaining in the panel. The timing of the model fits sequential exogeneity because the grid state and the set of unconnected customers are predetermined at the beginning of each period. Neither current nor future adoption retroactively changes the beginning-of-week connection charge c_{it}^{lcp}, grand-coalition Shapley charge $\kappa_{it}^{\bar{Q}}$, outward externality O_{it}, inverse-distance exposure to connected peers $[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}^*_{-it}]_i$, externality matrix $\mathbf{G}_t$, or neighbor characteristics $\mathbf{x}_j$ and their time trends $t\mathbf{x}_j$. Nor will adoption affect any of the instruments Z_{it} and Z_{it}^S.

The current and future inward externality measures, S_{it} and $F_{it}(\beta, H)$, however, remain sequentially endogenous, because they depend on neighbors' equilibrium adoption decisions. Recall that joint adoption happens contemporaneously in S_{it}, while anticipation of future neighbor externalities happens in $F_{it}(\beta, H)$. These are mutually exclusive because future inward externalities $F_{it}(\beta, H) = \max_j \max_{1\leq h\leq H} \beta^h G_{ijt} y_{j,t+h}$ are constructed to omit contemporaneous connection measured in $S_{it} = \max_j G_{ijt} y_{jt}$. As both measures depend on the externality matrix G_{ijt}, they pose an additional identification challenge because they are correlated.

Common primitives in the externality matrix $\mathbf{G}_t$ mean that identification of either θ_F or θ_S requires a joint instrumental variable strategy. If I solely instrumented F_{it}, the remaining uninstrumented variation would be correlated with S_{it}, creating omitted variable bias in the estimate of θ_S. If I solely instrumented S_{it}, the opposite would also occur. Both must be jointly instrumented with distinct sources of variation to avoid spilling over variation from one term to the other, and further, both instruments must be sequentially exogenous. I instrument them jointly with a peer-of-peer instrument Z_{it}^F and a shift-share instrument Z_{it}^S. The peer-of-peer instrument primarily predicts anticipated future exposure through demand-side variation. Meanwhile, the shift-share provides supply-side variation in contemporaneous exposure. The overall joint

42. Future versions of the paper may add interactive fixed effects, as in Kuersteiner and Prucha (2020), to control for correlated shocks.

IV is just identified, with one instrument per endogenous variable: Z_{it}^F for F_{it} and Z_{it}^S for S_{it}.

The IV reduced form and first-stage regressions are:

$$\begin{aligned} q_{it} = {} & \pi_F^q Z_{it} + \pi_S^q Z_{it}^S + \gamma_c^q c_{it}^{\text{lcp}} + \gamma_\kappa^q (c_{it}^{\text{lcp}} - \kappa_{it}^{\bar{Q}}) + \gamma_O^q O_{it} \\ & + \gamma_{\text{D}y}^q [\mathbf{D}^{\circ(-\zeta)} \mathbf{y}_{-it}^*]_i + \gamma_{\text{D}x}^q [t\mathbf{D}^{\circ(-\zeta)} \mathbf{x}_{-i}]_i + \gamma_x^q [\mathbf{G}_t \mathbf{x}_{-i}]_i + \gamma_{\text{FX}}^q \text{FX}_{it} + \mu_i^q + \psi_{b(i)t}^q + u_{it}^q \end{aligned} \tag{23}$$

Each regression includes one of the three endogenous variables $q_{it} \in \{y_{it},\ F_{it}(\beta, H),\ S_{it}\}$, controls, fixed effects, and indirect measures from Equation 22.

6.4. Instrumenting Anticipated Externalities

Testing whether predictable future neighbor adoption changes own adoption is a fundamental challenge because of reflection. I interact the externality matrix with itself to create a second-order externality instrument. This method adapts the social-network peer-of-peer approach developed by Bramoullé, Djebbari, and Fortin (2009) to externalities that operate via maxima. Although I apply this method to externalities, I retain the *peer-of-peer instrument* label for familiarity instead of calling it an unwieldy and unintuitive "externality-of-externality instrument." Their approach instruments a linear-in-means peer effect $[\mathbf{G}_t \mathbf{y}_t]_i = \sum_j G_{ijt} y_{jt}$ with the summed second-order peer contextual effect, or "peer-of-peer characteristics" $[\mathbf{G}_t^2 \mathbf{x}_t]_i = \sum_j G_{ijt} \sum_k G_{jkt} x_{kt}$. Throughout, I denote the focal customer i, the peer customer j, and the peer-of-peer customer k, noting that $\{i, j, k\} \subset \mathcal{I}$ without loss of generality. Empirically, x_k is the area of k's remotely sensed building footprint, which is an imperfect proxy for willingness to pay.

I make two modifications to the peer-of-peer instrument of Bramoullé, Djebbari, and Fortin (2009). The first modification is to use maxima as in F_{it} and S_{it}, instead of linear-in-means. The max analog of their peer-of-peer instrument is $Z^F = \max_j G_{ijt} \sum_k G_{jkt} x_{kt}$. Here, the outer externality is a maximum, matching the $\max_j$ term in F_{it} over neighbor externalities j to adopter i. Meanwhile, the inner term is a sum, because it uses contextual peer effects rather than endogenous peer effects. I have no reason to expect contextual effects occur via maxima.[43] A second modification strengthens the exclusion restriction of the standard peer-of-peer instrument. In standard peer-of-peer applications, Bramoullé, Djebbari, and Fortin (2009) argue that an identifying assumption that requires linear independence of I, G, and G^2 is sufficient for identification. Informally, however, an intuitive application of the exclusion restriction would require that peer-of-peer k's characteristics affect focal customer i's behavior only through peer j.

I show that one can censor the instrument by imposing this intuition on G^2. When k's adoption would provide a grid externality to i, this is recorded in the matrix cell $G_{ikt} \neq 0$. Censoring the instrument with $\mathbf{1}\{G_{ikt} = 0\} \circ \mathbf{G}^2$ removes peer-of-peers k who have a direct network linkage to i, removing customers who would directly provide an externality to i.[44] Together with the max analog, the *exclusion-censored*

43. Recall the customer in the middle of a fork, where the grid is endogenously expanded on either side in subsection 6.1. If the grid were not expanded, it would be contextual because then arguably either downstream fork neighbor could provide a contextual peer effect.

44. It does not fully impose the exclusion restriction because non-grid channels may still allow k to affect i. I include non-grid endogenous and contextual peer-effect controls for these channels.

peer-of-peer instrument for F_{it} I use is

$$Z_{it} = \max_j G_{ijt} \sum_{k \neq i,\, G_{ikt}=0} \frac{G_{jkt}}{\deg^i(j)} x_k, \qquad \deg^i(j) \equiv \sum_{k \neq i,\, G_{ikt}=0} \mathbf{1}\{G_{jkt} > 0\}, \tag{24}$$

where the exclusion-censored degree $\deg^i(j)$ counts the peer-of-peers who satisfy the censoring condition. The censored instrument limits its variation to peer-of-peers whose adoption would not provide a grid externality to i. The inner sum $\sum_{k \neq i,\, G_{ikt}=0} \frac{G_{jkt}}{\deg^i(j)} x_k$ takes the weighted average of the characteristics x_k of peer-of-peers who are not known to violate the exclusion restriction ex ante. When $\deg^i(j) = 0$, no peer-of-peers of j satisfy the censoring condition and so the instrument is set to zero. Censoring prevents variation from peer-of-peers who violate the exclusion restriction from entering the instrument. It does not impose an exclusion restriction, as k may affect i in other ways, but it prevents the instrument from including k's variation when I know that k directly affects i.

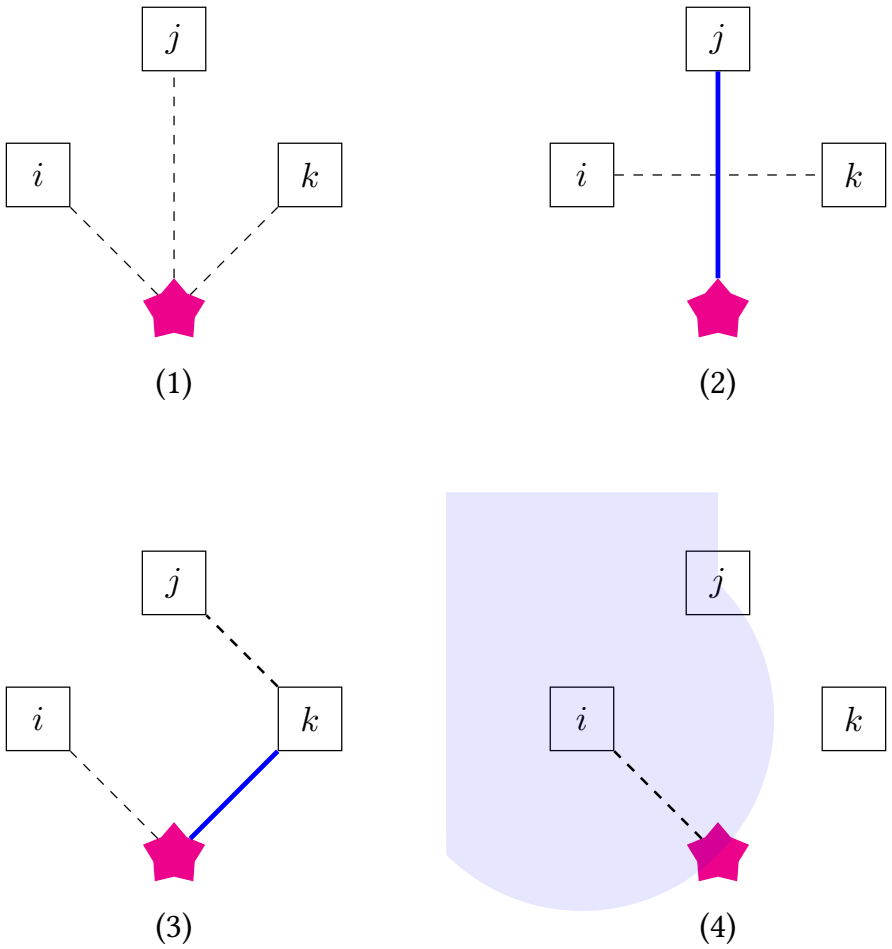
Figure 9 illustrates the instrumental variable logic and the censoring. In panel 1, the magenta transformer is placed, and customers i, j, and k estimate their connection charges based on their straight-line distances to the existing grid indicated by the dashed lines. In panel 2, j connects. This changes the minimum distance for i to connect to the grid, so j's adoption is relevant to i's externality. In panel 3, suppose that k connects first instead. This changes the minimum distance for j to connect to the grid but not for i, which reflects $G_{jk} > 0$. Peer-of-peer k is relevant to j's adoption decision, but as they do not reduce i's distance, $G_{ik} = 0$, so they do not violate the exclusion restriction through G. With relevance established and the direct grid-charge channel shut down, x_k can instrument for the effect of j's endogenous adoption y_j on i, as long as k does not affect i's adoption via other uncontrolled non-grid channels. Panel 4 shows a shaded region that represents building locations that are en route to or downstream of i's hypothetical network extension. Customers in these regions all violate the exclusion restriction, as their adoption would decrease i's connection charge. These customers fail $G_{ik} = 0$, and are censored in the inner sum of Z_{it}.

The exclusion-censored peer-of-peer instrument is relevant to F_{it} because peer-of-peer characteristics x_k affect the endogenous peer behavior y_j, which in turn affects i's decisions. For example, k may have a high willingness to pay, making j more likely to free-ride. So i may then adopt sooner because they cannot free-ride on j, because the best response for i would be to increase adoption if i's and j's adoptions are strategic substitutes, or decrease adoption if they are strategic complements. The instrument is externality-weighted on both legs because j may be more or less likely to free-ride on k based on the size of the potential externality G_{jk}, with the same logic for i free-riding on j.

The exclusion restriction requires x_k to affect i's adoption only through j. The instrument Z_{it} censors k when their adoption would affect i's charge path, but k's characteristics may still affect i without grid extensions, so censoring does not impose an exclusion restriction by itself. This censoring strategy is limited to cases in which $G_{ik} = 0$ is known. The network must be fully observed, which may be easier in geographic network settings than in social network settings, where data are often censored (Griffith 2022).

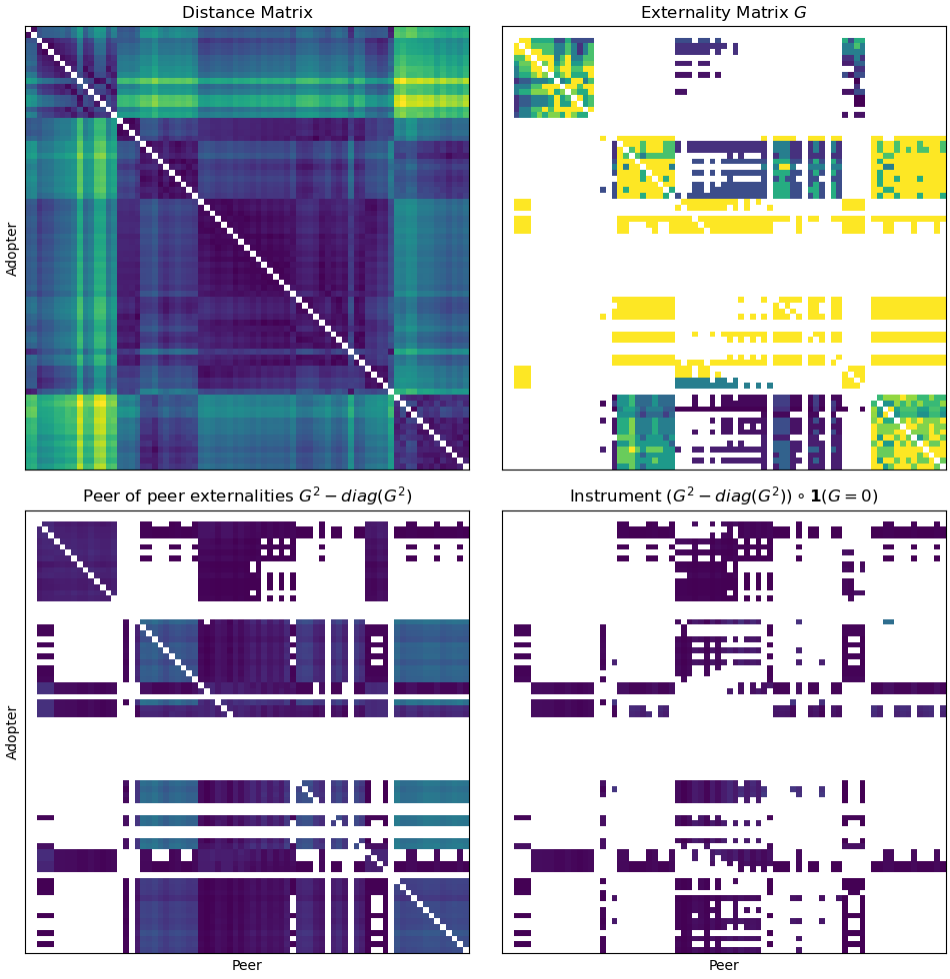
I use two measures to control for remaining grid and non-grid channels through which k may affect i. Following standard practice, *grid contextual peer effects* are controlled for with $[\mathbf{G}_t \mathbf{x}_{-i}]_i$ (Bramoullé, Djebbari, and Fortin 2009). In this infrastructure network setting, $[\mathbf{G}_t \mathbf{x}_{-i}]_i$ controls for spatial autocorrelation in the

Figure 9 — Exclusion-censored peer-of-peer instrumental variable logic



Notes. This figure illustrates the logic of the peer-of-peer instrument. Panel 1: Initial distances to the existing grid. Panel 2: Endogenous adoption of j that provides an externality to i by decreasing their distance to the grid. Panel 3: Relevance and direct exclusion of k. Relevance via providing j with an externality. Neighbor k does not affect i's initial distance, so they do not provide them with an externality. Panel 4: Censored locations of neighbors who would violate the exclusion restriction. See the text for details.

Figure 10 — Censoring the externality matrix



Notes. Top-left: Distance matrix between customer i and customer j, a static and symmetric matrix. Top-right: Externality matrix between customer i and customer j, representing the change in distance to the grid for customer i if customer j were to adopt grid electricity. It is asymmetric when i and j do not use the same infrastructure. Bottom-left: Square of the externality matrix. It measures second-order peer-of-peer exposure. Removing the diagonal removes reflected variation. Bottom-right: Instrument after censoring cells with a grid path from customer k to customer i, $1\{G_{ik} = 0\}$. All matrices use the same ordering of i and j in both rows and columns. These customers are sorted in the network-path-based optimal leaf order.

utility's remaining least-cost plan. If a component of k's characteristics, such as building area, is correlated with those of i's immediate neighbors, then the instrument Z_{it} could partly capture first-order neighborhood composition rather than only the effect of k on j's adoption. If buildings were systematically larger in one direction, spatial autocorrelation would make the instrument correlated with errors. The measure $[\mathbf{G}_t\mathbf{x}_{-i}]_i$ also captures non-grid contextual peer effects that happen to be correlated with infrastructure network externalities. I also use $[t\mathbf{D}^{\circ(-\zeta)}\mathbf{x}_{-i}]_i$, as described above, which also includes x_k, controlling for *non-grid contextual peer effects* that may not operate via the grid.

I visualize the distance matrix $\mathbf{D}$; the externality matrix $\mathbf{G}$; an uncensored peer-of-peer externality matrix $\mathbf{G}^2 - \text{diag}(\mathbf{G}^2)$, which removes own effects by setting $G^2_{ii} = 0$; and an exclusion-censored peer-of-peer externality matrix $\left(\mathbf{G}^2 - \text{diag}(\mathbf{G}^2)\right) \circ \mathbf{1}\{\mathbf{G} = 0\}$ with cells $\sum_j G_{ij}G_{jk}\,1\{G_{ik} = 0\}$ in Figure 10. The censored peer-of-peer externality matrix is considerably sparser where $\mathbf{G}$ indicates a violation of the exclusion restriction, showing how limiting this variation trades potential bias in the instrument from exclusion-restriction-violating peers for variance.

6.5. Instrumenting Contemporaneous Externalities

While future inward externalities are instrumented with demand-side peer-of-peer characteristics, current inward externalities S_{it} are instrumented with a supply-side procurement-based shift-share, so they use separate identifying variation. I map supply-side variation to S_{it} and demand-side variation to F_{it} because of an informal argument. Regardless, I will include both instruments in both first stages and will show empirical relevance via first-stage conditional F-statistics (Sanderson and Windmeijer 2016).

Current inward externalities S_{it} measure the net effect of two strategic responses generated by incentives created by an adopting coalition $\mathcal{Q}_t$. First, customer i may join the coalition, making it $\mathcal{Q}_t \cup \{i\}$. The incentive to do so is the difference between i's static connection cost less the cost share of the realized coalition, $c^{\text{lcp}}_{it} - \kappa^{\mathcal{Q}_t\cup\{i\}}$, and i's bargaining costs with that group of neighbors, $\rho^{\mathcal{Q}_t\cup\{i\}}$. Second, i may also free-ride on the coalition $\mathcal{Q}_t$'s network extension. The size of this incentive depends on β and how much infrastructure i shares with the group. Both incentives depend on the supply of shared infrastructure, so supply-side variation shifts the realizations of network extensions S_{it} via pass-through costs, timing, or utility choices. By contrast, future inward externalities F_{it} primarily depend on which neighbors i can anticipate will adopt in future weeks, and the externalities they would give to i via their spatial configuration relative to i and the existing grid. Therefore, demand-side variation in the peer-of-peer instrument is more relevant to anticipating extensions in F_{it}, and supply-side variation is more relevant to S_{it}.

Specifically, the supply-side shift-share instrument is

$$Z^S_{it} = \underbrace{\xi_{b(i),t-78}}_{\text{Shift}} \underbrace{\max_j \left\{ \frac{M^{\text{straight}}_{ij0}}{d^{\text{straight}}_{i0} + d^{\text{straight}}_{j0}} \mathcal{P}^{\text{path}}_{j0} \right\}}_{\text{Shares}}. \tag{25}$$

I justify the shifts and the shares in turn.

The shifts are a mouthful: lagged foreign exchange rates weighted by within-utility-branch LV pole

supplier composition. Specifically, let the shifts for branch b be a supplier-weighted procurement-cost shock

$$\xi_{bt} \equiv \sum_{m \in \mathcal{M}} w_{bm,0} \left(\frac{e_{mt}}{e_{m0}} - 1 \right) \tag{26}$$

where $w_{bm,0}$ is branch b's pre-2019 pole-manufacturer composition share $\sum_m w_{bm,0} = 1$ over pole-manufacturing countries $m \in \mathcal{M}$, and e_{mt} is the Rwandan/foreign exchange rate for m's currency in period t. The normalization $\frac{e_{mt}}{e_{m0}} - 1$ indexes exchange rates to the July 2019 reference month within the October 2018–November 2021 study period.

Branch-weighted exchange rates are used because the Rwandan utility imports a substantial share of the poles that it uses, so these are relevant to infrastructure provision. Using country-of-manufacture metadata on the utility's LV poles, I find that 30% are manufactured in Rwanda, with the rest unknown or imported. The imported poles primarily come from Uganda, South Africa, and Tunisia, with a small share from developed countries as well as India and Egypt. Supplier composition varies substantially across REG branches. For example, Gatsibo uses approximately 85% poles manufactured in South Africa, and Gicumbi uses approximately 63% poles manufactured in Uganda. Branch-specific supplier-weighted shifts create variation across branches to the extent that they have specific pole supply arrangements. There are many distribution networks within each branch b, so the shift is uniform not only across all players within each network g, but also across all distribution networks within the same branch b. Foreign exchange rates are assumed to be on an 18-month (78-week) lag. I chose this lag by comparing the pole's year of manufacture to its estimated installation date. The lag between manufacture and installation has a median of 1.4 years and a mean of two years in the data. As procurement sits between manufacture and installation, I select an 18-month lag in exchange rates to measure exposure to the shift.

Shares scale the proportion of infrastructure that i and j would be expected to share by the number of poles $\mathcal{P}_{j0}^{\text{path}}$ on customer j's planned network path when the transformer is placed at $t = 0$. Via Proposition 1, the more they share, the larger the strategic complements. In $\left\{ \frac{M_{ij0}^{\text{straight}}}{d_{i0}^{\text{straight}} + d_{j0}^{\text{straight}}} \mathcal{P}_{j0}^{\text{path}} \right\}$, the numerator $M_{ij0}^{\text{straight}} = \frac{1}{2} \left(G_{ij0}^{\text{straight,d}} \, d_{i0}^{\text{straight}} + G_{ji0}^{\text{straight,d}} \, d_{j0}^{\text{straight}} \right)$ measures the anticipated shared infrastructure savings of i and j, a boundedly rational measure of the shared infrastructure savings M_{ij} in subsection 2.1. Each term $G_{ij0}^{\text{straight,d}} \, d_{i0}^{\text{straight}}$ multiplies the externality share by i's initial distance to recover the reduction in meters of i's straight-line distance to the existing grid if j were to adopt at time $t = 0$. Then it takes the bilateral average and divides it by the sum of their initial distances $d_{i0}^{\text{straight}} + d_{j0}^{\text{straight}}$, to give the ex ante network expansion weighted average of how much shared infrastructure would be expected to lie on i and j's individual paths. These shares are bilateral-pair specific and relevant to how likely each customer is to cooperate with each neighbor j via their shared infrastructure prior to any network expansion either performed by the utility or induced by anyone in the game.

The shift-share instrument is relevant because lagged procurement shocks affect utility network expansions and procurement. I control for own exposure to the same exchange-rate shift in $\text{FX}_{it} = \mathcal{P}_{i0}^{\text{path}} \xi_{b(i),t-78}$, which follows the same formula for own exposure to poles.[45] After controlling for FX_{it}

45. I also test $\text{FX}_{it} = \mathcal{P}_{it}^{\text{path}} \xi_{b(i),t-78}$, which does not change any results.

and including branch-by-week fixed effects, the identifying excluded variation in Z_{it}^S comes from the interaction of bilateral spatial proximity with infrastructure procurement costs. Any branch-level rationing or changes in charges or input prices would not be i-specific and are controlled for in the fixed effects, and any individual infrastructure exposure is controlled for in FX_{it}.

6.6. Hypotheses

I make the following hypotheses following the previous discussion and Proposition 3.

Hypothesis 1 (Anticipation of neighbor adoption)**.** Anticipated inward externalities from future neighbor adoption delay current adoption, $\theta_F < 0$.

Hypothesis 2 (Joint cooperative adoption)**.** Current inward externalities increase contemporaneous adoption, $\theta_S > 0$.

I test both hypotheses against the null of no strategic response, $\theta_F = 0$ and $\theta_S = 0$, with alternative hypotheses $\theta_F < 0$ and $\theta_S > 0$. Signed estimates that reject the null would indicate consistency with the strategic mechanism. Analysis of the selection effects predicted by Proposition 2 and heterogeneity across primitives of Proposition 1 that determine strategic regimes will be descriptive and informal.

7 — Evidence for Anticipation, Cooperation, and Free-Riding

This section presents reduced-form evidence on anticipation, cooperation, and free-riding, which constitute the strategic responses of adopters to neighbor externalities. I first present summary statistics of all the measures described in the last three sections. Next, I present first-stage and reduced-form estimates for the joint instrumental variables in Equation 23. I then evaluate the main hypotheses of the paper and present Ordinary Least Squares (OLS) and Two-Stage Least Squares (2SLS) estimates of Equation 22. Then I consider verification of the mechanism and robustness to parametric choices, including network-extension rule variants, different specifications of the social peer effect control, and a placebo for supply-side economies of scale. Next, I assess selection and price discrimination by relating realized connection charges to wealth and willingness to pay. Finally, I perform a heterogeneity analysis across ex ante measures of the primitives underlying Proposition 1 that, according to theory, determine the degree of strategic substitutes or complements, to assess the efficacy of the mechanism's price discrimination.

7.1. Summary Statistics

Table 3 provides summary statistics for the panel. The first two columns show the initial summary statistics of the full set of buildings upon transformer placement, reflecting the cross-section of customers at the outset of the game. The middle columns show summary statistics for the overall estimation panel of 1,245,910 customer-week observations. The right set of columns shows summary statistics for customers at the beginning of the week in which they connect, reflecting buildings that select into adopting electricity.

Close to one-third of the observed eventual adopters connect in the first week of their distribution network. They are dropped due to singleton fixed effects. On average across the panel, the remaining

Table 3 — Summary statistics for the reduced-form estimation panel

	Cross-section at $t = 0$		All customer weeks Full panel		Adopters just before adoption $t \mid y_{it} = 1$		
Variable (Units)	Mean	SD	Mean	SD	Mean	SD	Median
Grid state and connection incentives							
First-week share of eventual adopters (pp)	32.5	46.8	–	–	–	–	–
Weekly connection hazard $\bar{y}_{it}$ (pp)	–	–	0.38	6.14	100	0.000	100
Ex post path distance to grid d_{it}^{path} (m)	464	286	333	297	294	355	148
Ex post path poles to grid p_{it}^{path}	10.2	6.20	7.08	6.50	6.07	7.51	3.00
Ex ante LCP planned line to grid d_{it}^{lcp} (m)	447	322	1,530	1,437	1,572	1,513	1,018
Ex ante LCP planned poles to grid p_{it}^{lcp}	10.3	7.08	24.8	20.4	24.2	21.0	18.0
Ex ante straight distance to grid d_{it}^{straight} (m)	351	214	233	220	196	240	77.5
Ex ante straight poles to grid p_{it}^{straight}	5.81	4.25	3.65	4.19	3.02	4.56	0.000
Supply-side connection cost (10^3 RWF)	8,694	5,395	6,042	5,636	1,373	2,674	529
Unilateral connection charge c_{it}^{lcp} (10^3 RWF)	1,461	1,581	879	1,369	1,037	2,104	79.2
Grand coalition surplus $c_{it}^{\text{lcp}} - \kappa_{it}^{R}$ (10^3 RWF)	1,342	1,510	780	1,303	830	1,965	21.0
Outward externality O_{it} (10^5 RWF)	3,298	4,114	1,901	2,756	1,096	2,299	203
Strategic variables							
Future inward externality $F_{it}^{d,\text{straight}}(1,8)$ (%)	28.3	38.8	12.5	28.9	21.8	35.8	0.000
Future inward externality $F_{it}^{c,\text{straight}}(1,8)$ (%)	31.2	42.6	13.9	32.4	23.4	39.7	0.000
Future inward externality $F_{it}^{d,\text{lcp}}(1,8)$ (%)	26.4	36.3	13.4	28.2	20.4	33.1	0.000
Future inward externality $F_{it}^{c,\text{lcp}}(1,8)$ (%)	31.2	42.4	15.4	33.9	22.6	39.1	0.000
Current inward externality $S_{it}^{d,\text{straight}}$ (%)	45.6	38.8	2.62	13.9	57.5	44.4	84.7
Current inward externality $S_{it}^{c,\text{straight}}$ (%)	51.2	43.4	2.92	15.7	53.6	49.0	96.5
Current inward externality $S_{it}^{d,\text{lcp}}$ (%)	43.4	36.8	2.99	14.1	44.9	40.5	46.6
Current inward externality $S_{it}^{c,\text{lcp}}$ (%)	51.9	43.4	3.42	17.0	47.6	47.8	28.9
Lagged externality $L_{it}(8)$ (%)	0.000	0.000	13.8	32.5	33.2	44.1	0.000
Peer-of-peer instrument Z_{it}^{F}	0.10	0.067	0.20	0.30	0.21	0.40	0.080
Supply shift-share instrument Z_{it}^{S}	0.026	0.13	0.015	0.13	0.010	0.18	0.007
Consumer and location characteristics							
Building footprint area (m^2)	45.9	37.8	46.5	38.1	57.1	38.8	51.4
Building height (m)	2.42	1.24	2.38	1.23	2.72	1.24	2.77
Nighttime lights, 2019	0.026	0.16	0.027	0.15	0.039	0.20	0.000
Terrain ruggedness index	3.86	3.99	3.63	3.76	4.57	4.97	3.97
Elevation (m)	1,722	309	1,691	292	1,859	316	1,838
Constructed by 2019	0.91	0.28	0.91	0.28	0.93	0.25	1.00
Peer effect and IV controls							
Endogenous non-grid adoption $D^{\circ(-1)}y_{it}$ (m^{-1})	0.070	0.29	0.18	0.30	0.48	1.50	0.14
Contextual non-grid $D^{\circ(-1)}x_j t$: log building area	0.000	0.000	180	176	79.3	128	16.2
Contextual non-grid $D^{\circ(-1)}x_j t$: log nighttime lights	0.000	0.000	0.82	4.34	1.42	8.50	0.000
Contextual non-grid $D^{\circ(-1)}x_j t$: log terrain ruggedness	0.000	0.000	63.9	61.3	29.5	47.5	6.84
Contextual non-grid $D^{\circ(-1)}x_j t$: log elevation	0.000	0.000	367	360	161	257	33.1
Contextual non-grid $D^{\circ(-1)}x_j t$: log building height	0.000	0.000	57.0	56.7	25.3	41.5	5.30
Contextual non-grid $D^{\circ(-1)}x_j t$: constructed by 2019	0.000	0.000	45.4	44.6	19.8	31.9	3.99
Contextual grid GX: log building area	1.74	0.57	1.51	0.81	1.44	1.01	1.58
Contextual grid GX: log nighttime lights	0.010	0.054	0.009	0.050	0.019	0.098	0.000
Contextual grid GX: log terrain ruggedness	0.69	0.29	0.58	0.37	0.62	0.48	0.62
Contextual grid GX: log elevation	3.57	1.12	3.09	1.63	2.91	2.01	3.22
Contextual grid GX: log building height	0.56	0.20	0.48	0.27	0.46	0.34	0.48
Contextual grid GX: constructed by 2019	0.44	0.14	0.38	0.21	0.36	0.26	0.39
Own exchange rate exposure FX_{it}	0.043	0.23	0.025	0.23	0.021	0.32	0.009
Observations	38,826		1,245,910		6,992		
Customers	38,826		38,826		6,992		
Distribution networks	177		177		177		

Notes. Left two columns refer to initial values upon transformer placement. Middle two columns refer to the full panel. Right three columns report values conditional on adoption. The first-week adopter share uses all observed adopters as its denominator, rather than all potential buildings, and reports the share who connect upon transformer placement ($2{,}273/6{,}992 = 32.5\%$). Conditional on a connection event, the hazard is mechanically 100 percentage points. Straight, LCP, and Path refer to straight-line, least-cost plan, and network-path extensions, respectively. The LCP line and pole rows report physical quantities from the route planner. Distance measures are denoted $d(.)$ and connection charges $c(.)$. Peer effects are abbreviated; contextual non-grid refers to time-varying contextual non-grid peer effects, while contextual grid peer effects are assumed to be static.

customers have an adoption hazard of 0.379 percentage points in a given week, defined as the probability that someone adopts conditional on survival.[46] By definition, adopters have a 100% adoption hazard conditioned on adopting.

In grid state and connection incentives, we see that at the outset of the game, unilateral connection costs and connection charges are extremely high, but these fall through the panel until the connection event. The utility builds more poles than necessary beyond the minimum technical requirement, as it uses 6.07 poles to connect people to the grid when only 4 may have been necessary, spending 529,000 RWF ($\approx$ USD 529) on the median connection and 1,373,000 RWF ($\approx$ USD 1,373) on the mean connection. Just before connection, the median unilateral connection charge observed is 79,200 RWF ($\approx$ USD 79), while the mean is 1,037,000 RWF ($\approx$ USD 1,037). These are unilateral charges and do not reflect cost sharing. The grand coalition surplus is substantial for those who share a lot of infrastructure but small for those who do not: connection-event-time customers have a grand coalition surplus of 21,000 RWF. Outward externalities are large, in the hundreds of thousands of USD. This reflects that providing infrastructure can benefit hundreds of downstream potential adopters.

Under strategic variables and instruments, we see that future inward externalities are large upon transformer placement and decline to an average of 12.5% during the panel. When customers connect, some could have received connection events in the future, but they adopt earlier. Fewer than half of connection events have a neighbor connection that follows, so the median is zero. Current inward externalities are very high at $t = 0$ because, by construction, the first adopters on the network occur at $t = 1$. Their construction halves the connection charge and reduces the distance by 45.6% for everyone who could be connected to the network. In an average week, however, current distances decline by 2.62% and charges by 2.92%. When customers adopt, the 57.5% mean and 84.7% median inward externalities indicate many joint adoptions. Lagged externalities are zero at baseline by definition. During the panel, customers on average received a 13.8% reduction in connection charge over the previous eight weeks. During connection events, this tends to be larger, as expected, but the zero median appears to reflect the empty ballroom predicted by the war of attrition (Bliss and Nalebuff 1984). Peer-of-peer instruments are externality-weighted log building-footprint areas, so they are always positive. The supply-side shift-share instrument is an index of supplier exchange rates by branch, multiplied by poles on the path, and these are positive because Rwanda's currency declines throughout the period.

The consumer and location characteristics show that adopters have larger building footprints and taller buildings, higher pre-electrification nighttime lights, are on more rugged terrain, live at higher elevations, and are more likely to occupy buildings constructed by 2019 than the cross-section. Endogenous non-grid adoption can be positive at $t = 0$ because of customers on other grids. At $t = 0$, time-varying contextual non-grid peer effects are zero by definition. The contextual grid controls decline over time because the externality matrix becomes more sparse as customers adopt.

46. Survival means that they have not yet adopted.

7.2. *Instrumental Variable Validity*

I estimate the reduced form and first stages of the instrumental-variable system following Equation 23 and report the results in Table 4. I present two panels: one without social peer effect controls, following Blume et al. (2015), and one including them, my preferred specification. In the top half of each panel, each endogenous variable, y_{it}, F_{it} and S_{it}, is regressed individually on both instruments Z_{it}^F and Z_{it}^S. The main specification is jointly instrumented and just-identified with two instruments; therefore, I report both conditional and standard two-way cluster-robust Wald F-statistics because there are two endogenous variables (Andrews, Stock, and Sun 2019). In the bottom half of each panel, I regress F_{it} and S_{it} on single instruments, so that each cell is a separate regression of F_{it} and S_{it} on a single instrument with the same controls and fixed effects. These regressions help me to assess how well each instrument matches its intended endogenous variable.

Recall that the censored peer-of-peer building-area instrument Z_{it}^F is meant to target future exposure to externalities, while the shift-share instrument Z_{it}^S is meant to target current exposure to externalities. Table 4 shows that the instruments match their intended purpose: Z_{it}^F instruments F_{it} well and Z_{it}^S instruments S_{it} well. The single-instrument specifications (Panel B) show that the instruments are primarily relevant for their intended endogenous variables, with diagonal (intended) F-statistics of 15.2 and 12.2, and cross-F-statistics (unintended) of 1.4 and 3.6. Without social controls (Panel A), the conditional F-statistic for Z_{it}^S falls to 11.2, while that for Z_{it}^F rises to 19.5. The diagonal pattern of single-instrument F-statistics does not change. The conditional F-statistic for the Z^S shift-share instrument is 15.4, while that of the peer-of-peer instrument is 17.5 when S_{it} and F_{it} are jointly instrumented. While the joint Wald F-statistics of 7.8 and 9.0 for S_{it} and F_{it} appear weak, recall that adding a second instrument with no predictive power approximately halves a singly instrumented F-statistic. Because individual instruments are intended to match individual endogenous regressors, the joint Wald statistic being approximately half the conditional F-statistic is by design.

The joint-IV first stage for the demand-side peer-of-peer instrument Z_{it}^F reports an estimate of -0.190 on F_{it}, 0.019 on S_{it}, and 0.221 on y_{it}. These estimates are stable across panels. The first coefficient means that a one-standard-deviation increase in the instrument reduces anticipated externalities by approximately 15% relative to their mean. This is consistent with the mean instrumented peer j delaying adoption to free-ride on peer k with a larger building area x_k, resulting in a negative coefficient on F_{it}. As a result, i no longer expects j to connect in the near term and increases adoption in response, resulting in the +0.221 reduced-form coefficient. This is a sizeable response, as a standard deviation in Z_{it}^F increases the adoption hazard by 58% of the baseline rate. It has no significant loading on S_{it}, so Z_{it}^F cleanly instruments future neighbor adoption in the delay phase of the wave.

The pole-based shift-share Z_{it}^S reports an estimate of 0.489 on F_{it}, 0.355 on S_{it}, and -0.130 on y_{it}. Only the estimate on S_{it} is statistically significant, indicating that the pole-based shift-share increases current inward externalities by approximately $1.4\times$ their mean. The sign of the reduced form depends on whether the social peer effect is controlled: it is $+0.097$ in Panel A but -0.142 once $[\mathbf{D}^{\circ(-1)}\mathbf{y}^*_{-it}]$ is included (Table D1). For those interpreting exchange rates as a pure supply-side cost shifter, it would appear to have the wrong sign on S_{it}. However, two countervailing forces matter. Recall that Proposition 1 shows

Table 4 — Instrumental-variable reduced form and first stages

	Reduced form	First stage	First stage
Dependent variable	y_{it}	S_{it}	$F_{it}(1,8)$
Units	pp	10% reduction in current distance	
Panel A: No social peer effect, no social contextual trends			
Joint instrumental variable regressions			
Demand-side peer-of-peer Z_{it}^F (1 s.d.)	+0.213 (0.081)	+0.017 (0.017)	-0.211 (0.054)
Supply-side shift-share Z_{it}^S (1 s.d.)	+0.097 (0.111)	+0.315 (0.094)	+0.423 (0.251)
Dependent variable mean	0.379	0.26	1.25
Conditional F-statistic	–	11.2	19.5
Joint Wald F-statistic	–	5.6	9.7
Single instrumental variable regressions			
Standard Z_{it}^S F-statistic	–	11.2	2.7
Standard Z_{it}^F F-statistic	–	1.1	14.8
Panel B: Social peer effect $[\mathbf{D}^{\circ(-1)}\mathbf{y}^*_{-it}]$ *and social contextual trends* $\mathbf{D}^{\circ(-1)}\mathbf{X}t$			
Joint instrumental variable regressions			
Demand-side peer-of-peer Z_{it}^F (1 s.d.)	+0.221 (0.076)	+0.019 (0.017)	-0.190 (0.054)
Supply-side shift-share Z_{it}^S (1 s.d.)	-0.130 (0.139)	+0.355 (0.091)	+0.489 (0.254)
Dependent variable mean	0.379	0.26	1.25
Conditional F-statistic	–	15.4	17.5
Joint Wald F-statistic	–	7.8	9.0
Single instrumental variable regressions			
Standard Z_{it}^S F-statistic	–	15.2	3.6
Standard Z_{it}^F F-statistic	–	1.4	12.2
Observations		1,245,910	
Distribution networks		177	
Potential adopters		38,826	

Notes. Panel A omits the social peer effect $[\mathbf{D}^{\circ(-1)}\mathbf{y}^*_{-it}]$ and the social contextual trends $\mathbf{D}^{\circ(-1)}\mathbf{X}t$; Panel B is the main specification, which includes both. Both panels use the same estimation sample. Within each panel, in the top half, each column is a separate regression following Equation 23. The first column is the reduced form, the second the first stage for S_{it}, and the final column the first stage for $F_{it}(1,8)$. Standard errors are two-way clustered at the transformer and calendar-week level. The reduced-form outcome is measured in percentage points, while the first-stage estimates are scaled to an externality that decreases i's current distance to the grid by 10%. Conditional and joint Wald F-statistics are reported below the means (Sanderson and Windmeijer 2016; Andrews, Stock, and Sun 2019). Under single instrumental variable regressions, each cell reports the F-statistic of the endogenous variable on a single instrument with all controls.

that more shared infrastructure leads to more joint adoption under strategic complements. As S_{it} measures joint cooperative adoption, more expensive connection charges should lead to strategic complements, resulting in the positive sign indicated. An alternative interpretation is that the utility procures poles from foreign suppliers approximately eighteen months in advance using multilateral funding and uses current exchange rates at the time of procurement to pass pole costs on to consumers. Fixing pole costs to procurement-time supply costs means that Rwandan consumers' purchasing power for connection charges counterintuitively increases relative to other goods as the Rwandan franc depreciates smoothly over time. The regression does not distinguish between these possibilities.

7.3. Full Specification Estimates

The main specification in Equation 22 is reported in Table 5. The table adds one element at a time. The first two columns omit S_{it} to show that delay estimates in F_{it} are not spuriously created via correlation with S_{it}; Columns 3–4 add S_{it}; Columns 5–6 add the social peer effect, and Columns 7–8 add the social contextual trends, giving the preferred specification. The F_{it} and S_{it} coefficients and standard errors are divided by 1.16 to convert to realized-externality units. The table also reports coefficients for the static cost control c_{it}, grand cooperation surplus $c_{it} - \kappa_{it}^{\bar{Q}}$, outward externality O_{it}, and endogenous social peer effect control $[\mathbf{D}^{\circ(-1)}\mathbf{y}^{*}_{-it}]_i$.

Signs are consistent across specifications and match the empirical design. The 2SLS estimate of F_{it} is stable as each element is added, from -0.881 to -0.910. I focus on the preferred estimates in Columns 7 and 8. Start with future inward externalities $F_{it}(1, 8)$. The OLS and IV estimates are both negative and significant, as both reject zero and therefore reject the null of Hypothesis 1. The IV magnitude is about $12\times$ the OLS magnitude, which is consistent with substantial simultaneity and reflection. In the OLS, if customers are waiting for each other in the chicken game, then the reflected effect is absorbed by customer fixed effects. The 2SLS removes most of the reflection by design, so it estimates the impact of a neighbor providing an externality without the impact of the customer on the neighbor. One might notionally consider the OLS estimate the marginal equilibrium effect and the 2SLS estimate the initial best response. The 2SLS estimates that a 10% anticipated decline in distance over the next eight weeks reduces the adoption hazard by 0.910 percentage points, $2.4\times$ the mean weekly hazard. The adoption hazard conditional on adoption on the transformer is 3.909, so 0.910/3.909 is 23.3% of the intensive margin for a 10% charge reduction. The estimates indicate that potential grid adopters substantially delay adoption when they can anticipate neighbor adoption that would decrease their connection charge.

Current inward externalities S_{it} create a 0.655-percentage-point increase in the adoption hazard in the OLS and a 0.937-percentage-point increase in the IV. These correspond to $1.7\times$ and $2.5\times$ the baseline hazard, and 16.8% and 24.0% of the conditional active-transformer-week hazard, respectively. The OLS is significant, but the IV is imprecise and does not reject zero. The instrument for S_{it} is a shift-share instrument, where the share is the proportion of infrastructure that customers share with neighbors, and the shift is foreign exchange rates weighted by supplier composition before the study. As S_{it} measures cooperation, it does not remove reflection, as joint adoption requires both parties to participate in the adopting coalition. Notionally, one might consider the difference to represent whether i is marginal to the coalition. The imprecise 2SLS estimates may reflect that the endogenous variable measures cooperation

Table 5 — Main estimates

	F		$F + S$		$+ \mathbf{D}^{\circ(-1)}\mathbf{y}^*$		$+ \mathbf{D}^{\circ(-1)}\mathbf{X}t$	
	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Future externality $F_{it}(1, 8)$	-0.112	-0.881	-0.104	-0.770	-0.069	-0.771	-0.076	-0.910
(10% reduction in distance)	(0.025)	(0.406)	(0.024)	(0.313)	(0.020)	(0.316)	(0.021)	(0.372)
Current externality S_{it}			+0.607	+1.299	+0.653	+0.759	+0.655	+0.937
(10% reduction in distance)			(0.111)	(0.692)	(0.104)	(0.737)	(0.104)	(0.838)
Current unilateral cost c_{it}	-0.720	-0.397	-0.763	-0.529	-0.739	-0.461	-0.718	-0.383
(10^5 RWF $\approx$ 100 USD)	(0.270)	(0.251)	(0.286)	(0.246)	(0.272)	(0.225)	(0.268)	(0.227)
Grand coalition surplus $c_{it} - \kappa_{it}^R$	+0.688	+0.426	+0.728	+0.543	+0.701	+0.478	+0.684	+0.417
(10^5 RWF $\approx$ 100 USD)	(0.276)	(0.234)	(0.293)	(0.241)	(0.278)	(0.217)	(0.274)	(0.214)
Outward externality K_{it}	-0.014	+0.194	-0.089	+0.006	-0.053	+0.112	-0.059	+0.116
(10^5 RWF $\approx$ 100 USD)	(0.072)	(0.125)	(0.079)	(0.138)	(0.073)	(0.126)	(0.069)	(0.137)
Endogenous social peer effects $[\mathbf{D}^{\circ(-1)}\mathbf{y}^*_{-it}]$					+0.946	+0.728	+0.951	+0.716
(weighted peer counts)					(0.079)	(0.140)	(0.078)	(0.146)
Observations	1,245,910							
Potential adopters	38,826							
Distribution networks	177							
Mean weekly hazard (pp)	0.379							
Fixed effects	Customer and branch-week							
Additional controls	Grid Contextual Peer Effects; IV exclusion controls							
Social contextual trends $\mathbf{D}^{\circ(-1)}\mathbf{X}t$	No	No	No	No	No	No	Yes	Yes
Overall R^2	0.185	–	0.203	–	0.218	–	0.219	–
RMSE	5.543	–	5.483	–	5.431	–	5.427	–
F-statistic for F_{it}	–	14.8	–	19.5	–	20.0	–	17.5
F-statistic for S_{it}	–	–	–	11.2	–	14.0	–	15.4

Notes. Each column presents a separate regression. Column pairs add one object or control at a time. Columns 1–2 include F_{it} only, and Columns 3–4 add S_{it}; neither includes social controls. Columns 5–6 add the social peer effect $[\mathbf{D}^{\circ(-1)}\mathbf{y}^*_{-it}]$, and Columns 7–8 add the social contextual trends $\mathbf{D}^{\circ(-1)}\mathbf{X}t$. In Column 2, the 2SLS estimate uses only the single peer-of-peer instrument. Column 8 is the preferred specification in Equation 22, where S_{it} and F_{it} are instrumented with first stages on Z^S and Z^F from Equation 23. The future- and current-externality coefficients report the change in the weekly adoption hazard caused by a future or current externality that decreases the customer's current connection cost by 10% in realized terms. Grid contextual peer effects and social contextual peer effects interact $\mathbf{G}_t^{\text{straight,d}}$ and $\mathbf{D}^{\circ(-\zeta)}t$ with logged variables x_j: building footprint area, 2019 nighttime lights, terrain ruggedness index, elevation, building height; and with an indicator for whether the building was constructed by 2019. The IV exclusion controls include grid contextual peer effects but also own exposure to exchange rate shocks. Standard errors are two-way clustered at the transformer and calendar-week level.

and, therefore, strategic complements. Jumps between multiple equilibria are inherently noisy and hard to predict or instrument. I will return to this variable in the heterogeneity analysis.

Own unilateral-cost coefficients are approximately halved from the OLS to the IV specifications. A RWF 100,000 ($\approx$ USD 100) reduction in unilateral cost increases the weekly hazard by 0.718 percentage points ($1.9\times$ the mean hazard) in the OLS and 0.383 percentage points ($1.0\times$ the mean hazard) in the preferred 2SLS specification. The decline suggests customers are more elastic when joint adoption is allowed and fully accounted for in the 2SLS. The grand-coalition-surplus coefficients have signs opposite to the unilateral-cost coefficients, with similar magnitudes that tend to be smaller in the OLS and slightly larger in the IV. This is consistent with the instrument measuring joint adoption better: as the instrument identifies more of the instantaneous cooperation incentive, the overall cooperation control declines in magnitude. Finally, the outward externality O_{it} measures how much unilateral adoption in a particular state would benefit neighbors. If consumers believed that the positive externalities they induced would be compensated through side payments or internalized via some other measure, I would expect outward externalities to increase adoption. All 2SLS estimates of the outward externality are positive but insignificant. The social-peer-effect control is positive and precisely estimated in Columns 5–8. In the preferred IV column, its coefficient is 0.716 for each unit of weighted peer exposure, consistent with electricity consumption raising adoption via the social network. For scale, it implies one connected peer 100 meters away would raise adoption hazards by 0.007 percentage points.

Overall, I take these results as strong evidence for strategic delay in advance of neighbor adoption in F_{it}, with weaker evidence for cooperation. The effect sizes are large: an anticipated or current 10% reduction in distance can decrease or increase adoption, respectively, by about $2.5\times$ the baseline adoption rate. I conclude from the above that, conditional on the forthcoming robustness and verification checks, strategic incentives strongly affect the *timing* of adoption.

7.4. Ruling Out Alternative Explanations

I now turn to checking robustness to various assumptions in the pipeline and verifying that alternative explanations could not explain the anticipation, cooperation, or free-riding results. First, I use alternative network-extension rules to consider how least-cost pathing that incorporates geography, terrain, and nonlinearities in the connection charge could change the results. Second, I consider alternative specifications of the social network controls by varying how social peer effects decline with distance. Finally, I use a placebo to address the threat that supply-side economies of scale confound estimation.

7.4.1 Alternative Network Expansion Measures

The baseline externality matrix is constructed using the straight-line method for simulating ex ante anticipated network extensions. Figure 6 shows that a straight-line network extension can differ substantially from the actual network extension, and Figure 8 shows that the resulting externalities can also differ in magnitude. Because the preferred straight line does not take into account geography, terrain, or the endogenous placement of other customers, it does not necessarily simulate the realized network extensions particularly well. In this subsection, I assess robustness to their construction by varying how I construct

the endogenous variables F_{it} and S_{it}. Let $\chi \in \{d(.), c(.)\}$ denote distance or cost and $\nu \in \{\text{straight}, \text{lcp}\}$ denote the straight-line or LCP network extension. Then define analogous endogenous variables to the baseline F_{it} and S_{it} constructions, which use $d(.), \text{straight}$, as:

$$F_{it}^{\chi,\nu}(\beta, H) \equiv \max_j \max_{1 \leq h \leq H} G_{ijt}^{\chi,\nu} \beta^h y_{j,t+h}, \qquad S_{it}^{\chi,\nu} \equiv \max_j G_{ijt}^{\chi,\nu} y_{jt}. \tag{27}$$

I replace F_{it} and S_{it} in the preferred specification with these network extension variants. As above, I convert the coefficients to realized-externality units using the ratio of the externality on the realized path to the straight-line and LCP externalities. This is 1.16 for the straight-line extension and 0.81 for the least-cost-path extension. The charge-based columns are not converted and are in anticipated units. I report these results in Table 6.

Table 6 replicates the preferred specification in the first column. In the second, fourth, and fifth columns, I use $(\chi, \nu) \in \{(c(.), \text{straight}), (d(.), \text{lcp}), (c(.), \text{lcp})\}$ respectively. In the third column, I add a second censor to the baseline peer-of-peer instrument. In addition to censoring the peer effects of customers who have a direct linkage to i, $G_{ik} \neq 0$, in column 3, I censor peers k of peers j that are within the service drop, a 37-meter radius of the focal adopter i. This limits the instrument's variation to peers of peers k who are located both outside the grid network and the immediate neighborhood of either i or j:

$$Z_{it} = \max_j G_{ij} \sum_{k: G_{ik}=0 \cap d_{ik} \geq 37} \frac{G_{jk}}{\sum_k \mathbf{1}(G_{ik} = 0 \cap d_{ik} \geq 37)} X_k. \tag{28}$$

Proximity censoring addresses the concern that k could affect i not only through the grid network but also more generally in space, without this effect already being controlled for in contextual or endogenous social peer effects $[\mathbf{D}^{\circ(-\zeta)}\mathbf{x}_{-i}t]_i$ and $[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}^*_{-it}]_i$. Specifically, there may be a scenario in which the externality matrix captures the minimum network path overlap rather than geographic proximity: a peer-of-peer k might satisfy $G_{ik} = 0$ even when they are immediate neighbors because their closest points to the grid are different. It simply ensures that k is not spatially close to i, even if their adoption would not affect i's distance to the grid.

The results of Table 6 show that the results are robust to the construction of the endogenous variable and the instrument. If anything, effect sizes seem to increase when adding proximity censoring or using the least-cost-planning route, but decrease when the externality is measured in connection charges.

7.4.2 Distinguishing Infrastructure from Social Peer Effects

The empirical design relies on controlling for endogenous and contextual social peer effects that may be correlated with infrastructure provision externalities. The main specification uses the weight $\zeta = 1$ and the functional form $[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}^*_t]_i = \sum_{j \neq i, D_{ij} \leq 1000} D_{ij}^{-\zeta} y^*_{jt}$ from Bernard and Torero (2015) to control for social and geographic peer effects. I test robustness to this parameterization by changing the denominator's functional form over a grid of $\zeta \in [0.5, 2]$ in increments of 0.1. Lower values of ζ imply a flatter geographic decay and higher weight on distant peers. In comparison, higher values imply a steeper decay in social peer effects and more weight on proximal neighbors. I show a $2 \times 3 \times 15$ grid of estimates for future, current, and social peer effects. The top row varies ζ in both $[\mathbf{D}^{\circ(-\zeta)}\mathbf{x}_{-i}t]_i$ and $[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}^*_{-it}]_i$. The bottom row

Table 6 — Robustness to network extension method

Network extension method	Straight-line			Least-cost planning route	
Charge measure	Distance 2SLS	Cost 2SLS	Distance 2SLS	Distance 2SLS	Cost 2SLS
	(1)	(2)	(3)	(4)	(5)
Future externality $F_{it}(1,8)$	-0.910	-0.807	-1.144	-1.553	-1.019
(10% reduction in charge measure)	(0.372)	(0.322)	(0.499)	(0.678)	(0.458)
Current externality S_{it}	+0.937	+0.928	+1.256	+1.978	+1.338
(10% reduction in charge measure)	(0.838)	(0.835)	(1.070)	(1.402)	(1.037)
Current unilateral cost c_{it}	-0.383	-0.406	-0.305	-0.362	-0.305
(10^5 RWF $\approx$ 100 USD)	(0.227)	(0.221)	(0.256)	(0.217)	(0.239)
Grand coalition surplus $c_{it} - \kappa_{it}^R$	+0.417	+0.426	+0.355	+0.411	+0.330
(10^5 RWF $\approx$ 100 USD)	(0.214)	(0.213)	(0.233)	(0.198)	(0.222)
Outward externality K_{it}	+0.116	+0.107	+0.139	+0.120	+0.176
(10^5 RWF $\approx$ 100 USD)	(0.137)	(0.120)	(0.167)	(0.165)	(0.173)
Endogenous social peer effects $[\mathbf{D}^{\circ(-1)}\mathbf{y}^*_{-it}]$	+0.716	+0.797	+0.682	+0.812	+0.823
(weighted peer counts)	(0.146)	(0.141)	(0.179)	(0.136)	(0.144)
Conditional F for Z_{it}^F	17.5	22.0	9.7	9.6	9.8
Conditional F for Z_{it}^S	15.4	15.6	15.2	13.2	14.3
Proximity Censoring in Z_{it}^F	No	No	Yes	No	No
Direct Linkage Censoring in Z_{it}^F			Yes		
Observations			1,245,910		
Potential adopters			38,826		
Distribution networks			177		
Mean weekly hazard (pp)			0.379		
Fixed effects			Customer and branch-week		
Additional controls			Grid and Social Contextual Peer Effects; IV exclusion controls		

Notes. Columns differ by network extension method and whether externalities use connection charges or distance. Instruments Z_{it}^F and Z_{it}^S are held fixed. The F and S coefficients correspond to a 10-percent reduction in exposure; cost, surplus, and outward-externality coefficients correspond to 10^5 RWF; the social-peer row is in weighted peer counts. Standard errors in parentheses are clustered by transformer and week.

removes $[\mathbf{D}^{\circ(-\zeta)}\mathbf{x}_{-i}t]_i$ as a control and similarly varies ζ in $[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}^*_{-it}]_i$. This shifts F_{it} and S_{it} towards zero, and $[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}^*_{-it}]_i$ away from zero. The columns report the estimates for F_{it}, S_{it}, and $[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}^*_{-it}]_i$ respectively with 95% confidence bands.

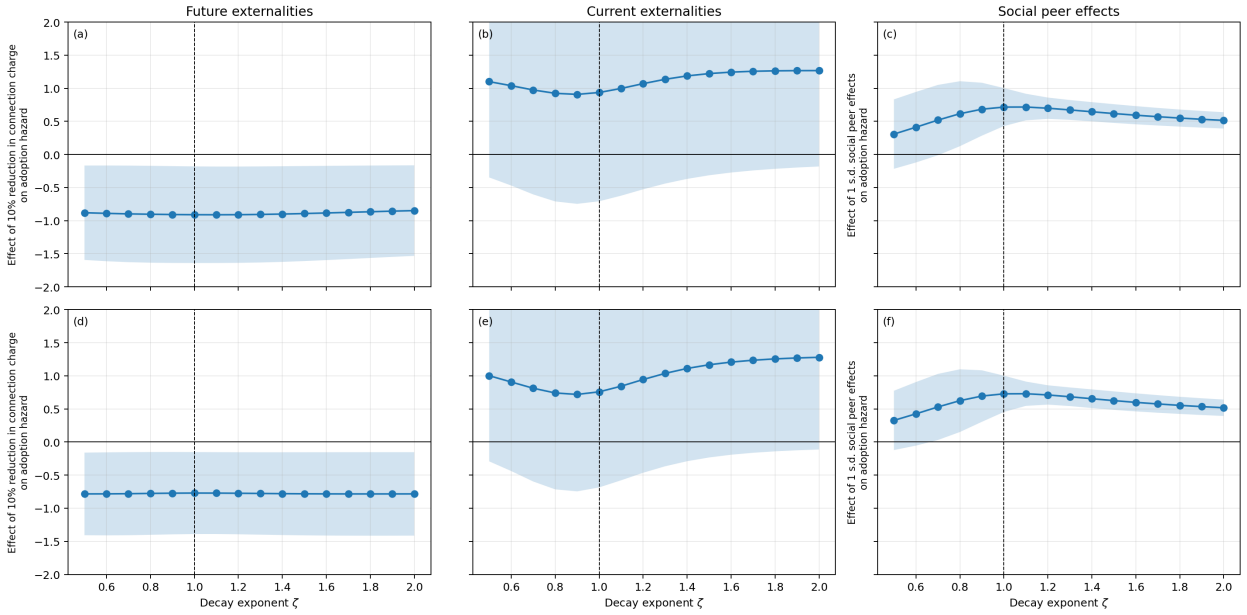


Figure 11 — Sensitivity of estimates to assumed decay in social peer effects ζ

Notes. Panels (a)–(c) use a common ζ for $[\mathbf{D}^{\circ(-\zeta)}\mathbf{y}^*_{-it}]_i$ and $[\mathbf{D}^{\circ(-\zeta)}\mathbf{x}_{-i}t]_i$; panels (d)–(f) omit the contextual peer controls $[\mathbf{D}^{\circ(-\zeta)}\mathbf{x}_{-i}t]_i$. All other controls and variables follow Table 6.

The anticipation estimate changes little across the distance-decay grid. The current-exposure estimate is minimized near $\zeta = 1$, while the estimated social-peer effect peaks near that value. This indicates not only that social peer effects do not drive the results, but also that the main specification attributes the least amount of explanatory power to the externality mechanism I hypothesize. Contextual peer trends $[\mathbf{D}^{\circ(-\zeta)}\mathbf{x}_{-i}t]_i$ barely affect results across the top and bottom specifications.

7.4.3 Placebo: Grid-Irrelevant Economies of Scale

A substantial threat to identification is supply-side economies of scale in utility operations. Customers may adopt simultaneously or sequentially for reasons other than the strategic incentives in the model. For example, let Assumption 3 be incorrect, so that the utility does not expand networks on its reported timeline. Instead, suppose that it batches local connection requests until they reach a threshold and constructs networks only when they reach a certain scale. This would appear as joint adoption, or even as sequential adoption across consecutive weeks if the construction is large. However, it would reflect utility practices rather than my proposed mechanism. Therefore, I use a placebo to test whether the estimates reflect connection-charge externalities rather than utility-side economies of scale.

Define a *grid-relevant peer* as one who could provide an externality to the consumer at any point during the electrification rollout. Placebo *grid-irrelevant* peers are their complement: they could never provide an externality but sit on the same transformer. Using the externality matrix, this defines the set of

viable placebo peers for i to be $\mathcal{P}_i = \{j \neq i : g(j) = g(i),\ G_{ijt} = 0\ \forall t\}$. As the mechanism I propose in section 2 operates through infrastructure provision $G_{ijt} > 0$, placebo peers should not affect i through the externalities I propose. They may still affect i through utility economies of scale because they remain on the same network, are typically within one mile of each other, and their adoption would bring utility supply teams to the local village. If utility distribution, transportation, and logistics create economies of scale that drive my previous results, then the placebo should have the same sign and attenuate estimates in a horse race.

I construct the placebo on the intensive margin by defining

$$F_{it}^{\text{Placebo}} = \max_{j \in \mathcal{P}_i} \max_{1 \leq h \leq 8} y_{j,t+h} \min\left\{\frac{37}{d_{ij}}, 1\right\}, \qquad S_{it}^{\text{Placebo}} = \max_{j \in \mathcal{P}_i} y_{jt} \min\left\{\frac{37}{d_{ij}}, 1\right\}$$

and then use an extensive margin indicator to define, for example, $\mathbf{1}[F_{it}^{\text{Placebo}} > 0] = \mathbf{1}\left\{\exists j \in \mathcal{P}_i : 1 \leq t_j - t \leq 8\right\}$, so that the variation in F_{it} and F_{it}^{Placebo} can be easily compared, and similarly for S_{it} and S_{it}^{Placebo}. I then replace the baseline F_{it} and S_{it} with the extensive-margin indicator or rescale them to standard deviations, and enter them into a horse race with the analogous placebo measure. The results of the horse race are in Table 7.

Table 7 — Placebo grid-irrelevant neighbor network extensions

	Extensive margin: $\mathbf{1}[F_{it} > 0]$		Intensive margin: F_{it}	
	Baseline IV	IV + placebo	Baseline IV	IV + placebo
	(1)	(2)	(3)	(4)
Placebo future F_{it}^{Placebo}		+1.76 (0.96)		+0.413 (0.259)
Grid future F_{it}	-9.04 (3.73)	-9.85 (4.87)	-2.694 (1.102)	-2.495 (1.117)
Placebo current S_{it}^{Placebo}		+0.369 (0.270)		+0.396 (0.232)
Grid current S_{it}	+1.631 (1.321)	+1.725 (1.575)	+1.447 (1.295)	+1.739 (1.435)
Observations		1,245,910		
Distribution networks		177		
Conditional first-stage $F(F)$	17.15	9.80	17.54	12.80
Conditional first-stage $F(S)$	14.19	11.24	15.38	13.31

Notes. The first and third columns replace the endogenous variables in the preferred 2SLS specification with extensive-margin and standard-deviation-scaled versions of those measures. Columns 2 and 4 add analogous placebo measures to that baseline.

Table 7 shows that in anticipation of a neighbor externality F_{it}, the placebo has the opposite sign and is insignificant. That is, having the grid extended on the other side of the network increases adoption rather than decreasing it. This occurs on both the extensive and intensive margins. For contemporaneous neighbor externalities S_{it}, the placebo matches the sign and is insignificant, but it increases the size of the S_{it} estimate. It indicates that demand-side cooperation may be strengthened when utility-side economies of scale are controlled for. Together, these indicate that utility-side economies of scale do not drive the results.

7.5. Selection and Price Discrimination

Lemma 1 and Proposition 2 predict that high-connection-value or high-willingness-to-pay consumers are more likely to connect earlier, pay higher connection charges, and provide externalities to their neighbors. I assess the empirical magnitude of these predictions by estimating the connection charges of realized adopters as a function of wealth and customer characteristics. These forthcoming regressions are cross-sectional and descriptive, and they reflect game outcomes of the mechanism rather than causal estimates of strategic incentives.

To ensure these cross-sectional results do not arise from spurious bilateral matching between remotely sensed buildings and electricity meters, I also run the same analysis on the World Bank Multi-Tier Framework Survey. This survey focused on energy use in Rwanda and is publicly available (World Bank 2018a, 2022). I use theory-relevant values from the separate rounds conducted in 2016 and 2022 and regress connection charges on them.

The cross-section is limited to customers who adopt an infrastructure connection. In the REG data, I regress the realized connection charge, the log cross-subsidy, adoption week, and adoption rank on log building area, the most predictive customer covariate I can match to meters. In the World Bank data, I regress paid connection charges on dwelling area, dwelling value, electricity consumption, a principal component analysis (PCA) of electric appliance ownership, and Ubudehe category, a social classifier that the Rwandan government assigns to households for social programs.[47] Dwelling area and value were not available in the 2022 dataset. I treat building area and value as a pure income effect, while electric appliances, consumption, and expenditure reflect willingness to pay for electricity specifically, since connections are used to consume and provide electricity services.

I estimate Poisson pseudo-maximum-likelihood regressions of the form

$$\mathbb{E}\left[c_i^R \mid A_i, c_{i0}^{\text{path}}, x_i, g(i)\right] = \exp\left(\iota_{g(i)} + \beta \ln A_i + \gamma \ln\left(c_{i0}^{\text{path}}\right) + \eta x_i\right),$$

and matched OLS regressions of the form

$$\ln\left(c_i^R\right) = \iota_{g(i)} + \beta \ln A_i + \gamma \ln\left(c_{i0}^{\text{path}}\right) + \eta x_i + \varepsilon_i.$$

These specifications allow me to estimate the elasticity of the realized connection charge with respect to building area and other variables determining willingness to pay. The variables are $\iota_{g(i)}$, which denotes transformer fixed effects; c_{i0}, the initial connection charge upon transformer placement; and c_i^R, the realized connection charge. To put the connection charges I estimate on the same footing as the World Bank connection charges, for which there are few zeros in the connection fee, I add the meter connection fee to my public-good connection charges, without assuming, as before, that there is 100% take-up of the credit available for the private component of the connection fee. I use transformer or cell (adm4) fixed effects, and include two additional controls from the utility data: whether the building was constructed by mid-2019, and the initial connection charge upon transformer placement. The former excludes newly constructed

47. It categorizes households into poorest, poor, self-reliant, and affluent categories, with the bottom tier allowing access to certain social programs. These are ordinal numbers 1–4.

buildings from entering the price discrimination metric, and the latter conditions the realization of price discrimination on the initial state of the game.

First, note that under a flat fee, any regression coefficient of the connection charge on an independent variable after conditioning on adoption would mechanically be zero. The results indicate price discrimination in nearly all OLS specifications and most PPML specifications. Panel A shows that building area itself, a weak proxy for wealth, is correlated with the realized charge and cross-subsidy. The larger the building area, the higher the connection charge and the fewer cross-subsidies customers receive from neighbors. Within the sample, this suggests that the connection charge has an elasticity of 0.04 with respect to building area. At the same time, the proposed mechanisms lead customers with 1% larger building area to receive 0.06% less in subsidies. Panel B shows that connection charges increase with dwelling value, an indicator of wealth, with an elasticity of 0.05–0.06. It also indicates that households in a higher Ubudehe category (higher social class) paid larger connection charges in 2022. However, neither Ubudehe category nor dwelling area is correlated with connection charges in the 2016 survey. Across measures of willingness to pay, the elasticities of paid connection charges range from 0.09 to 0.24. Connection charges are most elastic with respect to electricity consumption in 2022; the elasticity is 0.17 for electricity expenditure and 0.10 for the PCA of electrically powered appliances.

Because a flat fee would produce no variation in connection charges by construction, I interpret these regressions as evidence that the proposed mechanism results in modest price discrimination. The results indicate that adopters with a high connection value or willingness to pay, or who are wealthier, adopt earlier, pay higher connection charges, and receive fewer cross-subsidies from their neighbors' provision of shared infrastructure.

7.6. Decomposition of Free-Riding and Cooperation

To close out the reduced-form part of the paper, I finally turn to a heterogeneity analysis to decompose free-riding and cooperation. Proposition 1 predicts that the larger the strategic complements, the more customers should join the same adopting coalition. The stronger the strategic substitutes, the more free-riding should result. When applied to the near-mechanical baseline of Rwanda's connection charge policies, one should imagine two close neighbors toward the leaves of the distribution tree network as having strategic complements, while those sharing a trunk but on different branches have strategic substitutes.

I compare joint adoption $J_{ij} = \mathbf{1}(y_{it} = y_{jt})$, which is defined as i connecting in the same week as j, with sequential adoption, in which i connects between one and eight weeks after j, $L_{ijt} > 0$ or $L_{jit} > 0$, using a logit to estimate the probability that customers jointly connect in a dyadic regression. Each pair is one observation. I take all dyadic pairs that fall into one of the two categories and define the binary variable $\Pr(J_{ij}) = J_{ij}/(J_{ij} + L^8_{ijt} + L^8_{jit})$, which is the probability that customers jointly adopt given that they either jointly or sequentially adopt. This conditions the sample to 56,665 dyadic pairs who adopt within eight weeks of each other. I divide the result of Proposition 1 by c_i, and estimate this probability on bins of the proportion of overlapping infrastructure $M_{ij}/c_i = G_{ij}$, which is the share of i's standalone connection charge that can be saved by sharing infrastructure with j, other terms in the regression, and controls. Specifically, the baseline dyadic regression is

Table 8 — Connection charges, adoption order, and consumer wealth

Dependent variable	Independent variable	Dataset	PPML	OLS	N_{PPML}	N_{OLS}
Panel A: Realized adoption						
Connection charge	Building area	REG	0.037	0.041	5,813	5,813
			(0.026)	(0.020)		
Gross cross-subsidy	Building area	REG	-0.057	-0.058	5,688	5,688
			(0.021)	(0.021)		
Adoption week	Building area	REG	-0.565	-0.532	5,813	5,813
			(0.267)	(0.254)		
Adoption rank	Building area	REG	-0.019	-0.019	5,813	5,813
			(0.006)	(0.006)		
Panel B: WB MTF survey						
Connection charge	Dwelling area	MTF 2016	0.002	0.012	950	950
			(0.034)	(0.028)		
Connection charge	Dwelling value	MTF 2016	0.063	0.047	932	932
			(0.027)	(0.023)		
Connection charge	Electric appliance PCA	MTF 2016	0.108	0.092	982	982
			(0.036)	(0.029)		
Connection charge	Electric appliance PCA	MTF 2022	0.109	0.110	1,480	1,480
			(0.026)	(0.030)		
Connection charge	Electricity consumption	MTF 2016	0.107	0.102	933	933
			(0.040)	(0.034)		
Connection charge	Electricity consumption	MTF 2022	0.170	0.235	1,449	1,449
			(0.053)	(0.047)		
Connection charge	Electricity expenditure	MTF 2022	0.125	0.171	1,449	1,449
			(0.038)	(0.034)		
Connection charge	Ubudehe category (1–4)	MTF 2016	0.080	0.029	898	898
			(0.052)	(0.040)		
Connection charge	Ubudehe category (1–4)	MTF 2022	0.115	0.101	1,471	1,471
			(0.052)	(0.046)		

Notes. Panel A uses the cross-section of realized adoption. Table 3 provides descriptive statistics for these. These use transformer fixed effects and clustered standard errors, with controls for baseline building existence and initial connection charge. Panel B uses the World Bank Multi-Tier Framework Survey 2016 and 2022 datasets (World Bank 2018a, 2022). Fixed effects are at the cell or survey group level, with corresponding clustered standard errors. The PCA is standardized to z-scores; building areas, value, electricity consumption, and spending are logged. PPML and OLS regressions are both reported and rescaled where necessary to report elasticities, except for Ubudehe and the PCA. Standard errors are clustered by transformer in the top panel and by cell (Admin 4-level regions) in the bottom panel. Estimates reflect elasticities conditional on controls, except for adoption week and rank which are level outcomes. The sample is restricted to realized electricity adopters. In the World Bank 2016 data, connection charges are in panel-column C7, area is B13, value is B9, monthly kWh is C22, and Ubudehe is B29. In the World Bank 2022 survey, connection charges are in panel-column C007, monthly electricity spending is C025, and Ubudehe is B13. PCAs are constructed from lighting, media, heating and cooling, computer, and TV-related appliances.

$$\log\left(\frac{\Pr(J_{ijt})}{1-\Pr(J_{ij})}\right) = \beta_0 + \sum_{b=2}^{6}\beta_b \mathbf{1}\left\{\frac{M_{ij}}{c_i} \in B_b\right\} + \rho\frac{c_j}{c_i} + \gamma f(\frac{1}{c_i}) + \theta_\kappa\left(c_{it}^{\text{lcp}} - \kappa_{it}^{\bar{Q}}\right) + \theta_O O_{it} + \theta_{Dy}[\mathbf{D}^{\circ(-1)}\mathbf{y}_{-it}^*]_i + \theta_{\text{FX}}\text{FX}_{it} + \boldsymbol{\theta}'_{Gx}[\mathbf{G}_t^{\text{lcp}}\mathbf{x}_{-i}]_i + \boldsymbol{\theta}'_{Dx}[\mathbf{D}^{\circ(-1)}\mathbf{x}_{-i}t]_i. \tag{29}$$

I exclude fixed effects to avoid incidental parameters. I bin the shared infrastructure savings as $B \in \{[0, .125), [.125, .25), [.25, .50], (.50, .75], (.75, .875], (.875, 1]\}$, where the middle bins are larger to reflect sample sizes.

I report four specifications in two pairs. The first pair uses connection charges $c(.)$ while the second pair uses straight-line distances $d(.)$. Then I consider linear and spline forms for $f(\frac{1}{c_i})$. In a linear specification, $f(\frac{1}{c_i}) = \frac{1000}{c_i}$, while in a spline, I fit a cubic spline to $\frac{1}{c_i}$ for more flexibility. For each interval, I report fitted logit probabilities over characteristics so that $\widehat{\Pr}_b = \frac{1}{N}\sum_{ijt}\Lambda\left(\widehat{\beta}_0 + \widehat{\beta}_b + \widehat{\mathbf{X}}'_{ijt}\widehat{\boldsymbol{\delta}}\right)$ for each bin b, holding the distribution of controls across bins constant.

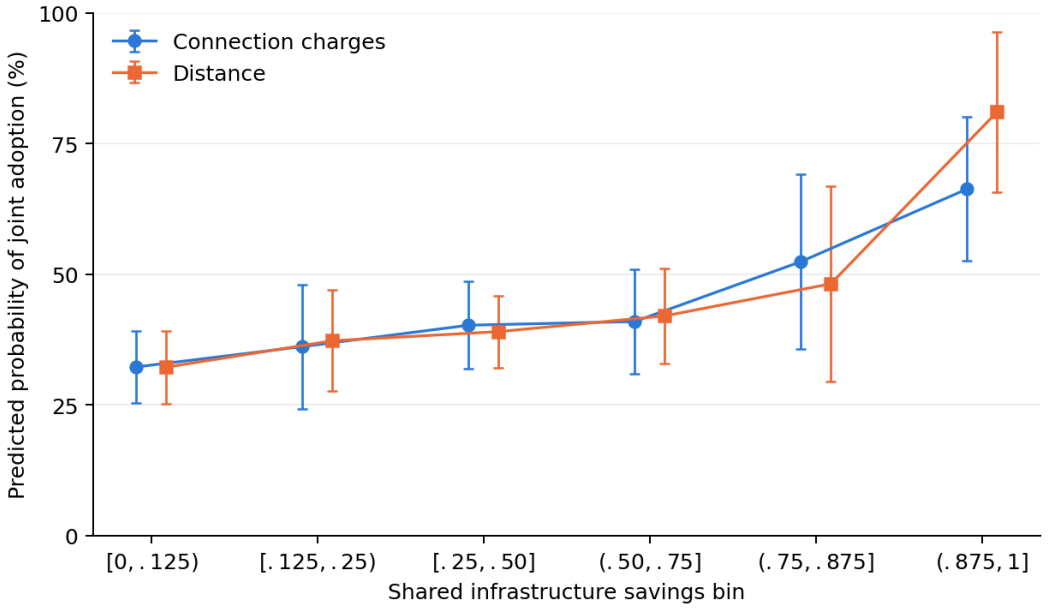
Table 9 — Predicted probability of joint adoption

	$[0, .125)$	$[.125, .25)$	$[.25, .50]$	$(.50, .75]$	$(.75, .875]$	$(.875, 1]$
Cost measure						
Linear	32.2	36.1	40.2	40.9	52.4	66.3
	(3.5)	(6.1)	(4.3)	(5.1)	(8.6)	(7.0)
Splined	32.2	37.4	40.4	42.0	54.6	64.6
	(3.5)	(6.5)	(4.6)	(4.7)	(8.3)	(6.1)
Bilateral Pairs	43,009	2,303	3,106	3,123	1,583	3,541
Distance measure						
Linear	32.2	37.3	39.0	42.0	48.2	81.0
	(3.6)	(4.9)	(3.5)	(4.6)	(9.6)	(7.8)
Splined	32.1	38.3	39.8	42.6	48.0	78.2
	(3.6)	(4.9)	(3.5)	(4.8)	(9.3)	(8.4)
Bilateral Pairs	39,993	3,721	4,895	3,501	2,267	1,903

Notes. Each bin cell reports the predicted probability for dyadic pairs of customers who jointly adopt rather than sequentially do so $\Pr(J \mid J + L^8 = 1)$. Each cell reports predicted probabilities after averaging the logit probability over the entire sample (removing selection effects). I use two-way clustered standard errors by transformer and week and use the delta method to report standard errors in percentage points to match the logit probability.

Table 9 and Figure 12 show that joint adoption becomes much more common as customers share more infrastructure. The figure reports the linear estimates of the table. In the linear specification, the predicted share of active dyadic responses that are joint rises from 32.2% in the lowest shared-infrastructure bin to 66.3% in the highest bin using connection charges, and from 32.2% to 81.0% using distance. The spline estimates are similar, rising from 32.2% to 64.6% for connection charges and from 32.1% to 78.2% for distance. The gradient is relatively gradual across the middle bins and becomes especially steep as customers share

Figure 12 — Predicted probability of joint adoption by shared-infrastructure bin



Notes. Predicted probabilities from the linear specification of Table 9. Delta-method standard errors are reported as 95% confidence intervals in whiskers. The raw SEs were two-way clustered by transformer and week. Cubic-spline estimates and per-bin pair counts are reported in Table 9.

most of their infrastructure. Because these probabilities condition on either joint or sequential adoption, the pattern shows that greater infrastructure sharing shifts strategic responses from free-riding toward joint adoption, consistent with Proposition 1.

The overall reduced-form evidence shows that distance-based charges create local strategic responses consistent with the theory. Potential adopters delay when they anticipate a neighbor-induced reduction in their future charge. Conditional on a neighbor's intended expansion, they either join and cooperate to adopt jointly or free-ride on that neighbor sequentially. Adopters with higher willingness to pay and greater wealth adopt first, provisioning infrastructure that correlates with realized connection charges and willingness to pay. Strategic complements and substitutes result in heterogeneity in the degree of cooperation or free-riding.

8 — Discussion and Conclusion

This paper shows that marginal-cost pricing of infrastructure connections creates a dynamic externality. Whoever connects first pays a high price to provision the infrastructure they need, but their neighbors can then connect to that same infrastructure at a low price. The externality is double-edged. On one hand, the positive externality delays adoption and is an efficiency cost. On the other hand, high-value customers value it more, so they select into provisioning infrastructure that benefits their low-value neighbors, an equity benefit.

Evidence from 177 electricity distribution networks in Rwanda supports the proposed mechanisms. A 10% anticipated reduction in straight-line distance to the grid over the next eight weeks lowers the weekly

adoption hazard by 0.910 percentage points, an effect equal to 2.4 times the unconditional mean weekly hazard. A contemporaneous externality of the same size raises adoption by 0.655 percentage points in OLS; the IV estimate is similar but imprecise. A placebo test for supply-side economies of scale produced no comparable response, so the mechanism appears to be strategic behavior. The probability of joint rather than sequential adoption rises from 32% to as much as 81% as more infrastructure is shared and strategic complements increase in strength. Consistent with selection, customers with larger buildings connect earlier and pay more: relative to a mechanical zero under a flat fee, the elasticity of realized connection charges with respect to building area is 0.04 in the utility data. Similar elasticities with respect to willingness-to-pay proxies range from 0.10 to 0.24 in the World Bank surveys.

The results show a substantive policy trade-off. The selection effects are relatively small, while the delay is large but partly recouped after a neighbor provides infrastructure. Price discrimination may allow the utility to raise revenue from connections, and it may use that revenue to lower minimum connection charges. Equity effects are enhanced by proximal heterogeneity in willingness to pay, which allows high-value neighbors to cross-subsidize lower-value neighbors. Which policy is optimal depends on the government's weights on universal access, utility cost recovery, spatial efficiency, and the incidence of expansion costs relative to the size of strategic delay.

In my review of other countries' connection pricing policies, I found that some clearly use policies that address or mute some of the paper's strategic incentives. For example, Zambia's policy seems to indicate an understanding of strategic incentives, as it conditions connection charges on population density, using density as an ex ante proxy for how public or private an extension is likely to be (Zambia Electricity Regulation Board 2023). Ghana and Côte d'Ivoire use Coasian pseudo-ownership policies to internalize the externality: Consumers who pay for an extension upfront are later reimbursed when other customers connect to the same infrastructure.[48] These policies are consistent with efforts to address the strategic incentives described in this paper.

The paper highlights a broader tension between planning-based and market-based approaches to pricing infrastructure connections. A planning-based approach uses increasingly rich data to predict demand (Lee et al. 2021), asserting that better data will help target infrastructure more effectively toward customers who use it productively, or increase the utility's revenue. A market-based approach instead asks high-value customers to reveal their connection values and provision the grid to cross-subsidize their lower-value neighbors by exposing customers to declining prices over time and seeing when they choose to take up electricity. It uses prices to get customers to reveal preferences instead. As Mission 300 deploys $35 billion toward connecting 300 million people by 2030, connection policy will determine how effective financing will be by changing who pays, how long others wait, and where distribution networks expand in the last mile.

48. Ghana calls this a capital contribution policy, while Côte d'Ivoire calls it "droit de suite" (Compagnie Ivoirienne d'Électricité 2026; Public Utilities Regulatory Commission Ghana 2024).

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A — Rwanda's Connection Costs and Charges

I report sources and figures for each of the costs used in the paper in Table A1 and then discuss them.

Table A1 — Documentary evidence on connection costs, charges, and subsidies

Object	Value	Basis	Source	Location
Connection charges				
Standard connection fee	RWF 56,000	posted	Ministry of Infrastructure (MININFRA) Rwanda (2015)	Annex T.24
Per-pole charge	US$200	discussion, bunching	Blimpo and Cosgrove-Davies (2019) and World Bank (2018a)	microdata
Installed cost per km				
LV (ABC 3×35 / 3×70)	US$15,000 / 17,931	planning catalogue	Ministry of Infrastructure (MININFRA) Rwanda (2019)	T.13–15
LV rehabilitation	∼US$50,000	project tables	Rwanda Energy Group (2024a)	T.54, p. 88
LV in mixed extensions	US$17–21k	project tables	Rwanda Energy Group (2024a)	T.52–53
Cost per connection				
EARP actual	US$411	completion	World Bank (2018b)	p. 56, p. 21
ESSP actual	∼US$533	completion	World Bank (2023)	p. 43
EAQIP appraisal	US$603–758	planned	World Bank (2020)	pp. 41, 35
FY2023/24 implied	∼US$857	derived	Rwanda Energy Group (2024b)	outputs
Implied subsidy				
Lower bound subsidy	∼60–66% of total	derived	—	this appendix
Customer share of total	∼1/10	stated	World Bank (2015)	pp. 22, 9

Note. RWF 1,000 ≈ US$1 over the sample period. Per-km-to-per-pole conversions use 20.4 poles per km (median observed span 49 m over 239,829 real LV spans, consistent with the codified 44–55 m standard, Rwanda Energy Group 2020, Table 18). "Installed" costs include materials, labor, transport, design, margins.

A.1. Connection Fee of RWF 56,000

The online schedule charges a flat RWF 56,000 connection fee, and it is documented in the connection policy (Rwanda Energy Group 2017). This number is also reported in the EARP Connection Policy schedule reproduced in the 2015 sector strategy (Ministry of Infrastructure (MININFRA) Rwanda 2015, Annex Table 24).

A.2. Pole and Line Connection Charges

The paper assumes customers are charged US$2 per meter of line beyond a 37 m service drop and US$200 per pole. The US$200-per-pole figure was shared with me by Rwanda's Ministry of Infrastructure in 2022. However, I did not find this exact number in utility or Ministry of Infrastructure documents. I nevertheless find evidence broadly consistent with this number in the following sources:

- Figure 3.5b in Blimpo and Cosgrove-Davies (2019), which indicates US$189 for poles and wiring in Rwanda alongside other distance-based connection charges; and
- a World Bank survey in which consumers reported the price they paid to connect to the grid, indicating bunching at US$50 intervals as well as shared connections (World Bank 2018a, 2022).

In the World Bank Multi-Tier Framework survey, household-reported connection payments bunch at US$100, US$150, and US$200, round splits of a US$200-per-pole charge among coalition members (World Bank 2018a, 2022).

A.3. Pole and Line Supply Costs and Subsidies

The total *installed* cost of LV extensions is US$15,000–17,931 per km of LV aerial bundled cable in Rwanda's National Electrification Plan's network catalogue (Ministry of Infrastructure (MININFRA) Rwanda 2019, Tables 13–15). The same document shows that the supply cost of an MV/LV transformer ranges from US$3,600 (15 kVA) to US$28,455 (400 kVA). More recent records from the Electricity Distribution Master Plan indicate a total cost of US$17,000–21,000 per km of LV (Rwanda Energy Group 2024a, Tables 52–54). A World Bank document on the Electricity Sector Strengthening Project reports 1,295.4 km of LV and 849.8 km of MV built, at a blended cost of roughly US$20,750 per km (World Bank 2023, results framework, p. 43). At the observed 44-meter average and 55-meter reticulation standard, these crude estimates correspond to raw supply costs of US$660–1,155 per pole-span (Rwanda Energy Group 2020, Table 18).

Conservatively assuming the lower end of material costs, a US$200-per-pole and US$2-per-meter connection charge is approximately a 66% subsidy or one-third pass-through of material, labor, service, and transport costs. This is primarily financed by multilateral development banks and other donors (World Bank 2023, 2020). This rate is conservative and perhaps reflective of the time period of the study; earlier World Bank records indicated that customers contributed only about one-tenth of the total connection cost (World Bank 2015). I assume a two-thirds subsidy rate (equivalently, a one-third pass-through of supply-side costs) throughout the study. I document additional sources behind the cost figures in section 3. Within the sample period, RWF 1,000 $\approx$ US$1.

A.4. Average Costs per Connection

Per-connection totals are the best-documented metric because they appear in World Bank reports. The EARP completion report puts the realized average cost per connection at US$411 against an appraisal baseline of US$1,000 (World Bank 2018b; World Bank Independent Evaluation Group 2019, p. 56, p. 21). A more recent project appraised the average supply cost of a connection at US$603–758 per connection (World Bank 2020, pp. 41, 35). The Electricity Sector Strengthening Project realized roughly US$533 per connection (World Bank 2023), and its appraisal stated that the customer's US$100 contribution covered about one-tenth of the total (World Bank 2015, pp. 22, 9): it appears that different policies were used prior to 2017 that may not have been distance-based. Utility financial reports show that in fiscal year 2023/24, supply costs are about US$857 per connection when solving for them from the reported US$50,000 per km cost, 2,357.7 km of LV, and 137,502 connections added in that year (Rwanda Energy Group 2024b).

B — Data Appendix: Sample Selection

The limiting factor for sample selection is that connection dates were available only in the smaller of the two consumer files when I accessed the data in March 2022: 34,125 records carry a connection date, or 34,087 unique metered customers after deduplication—only 4.96% of the 686,891. The game is unobservable for customers without connection dates because it is dynamic. I report how many consumers and networks I exclude at each step in Table B1.

The top half of the table reports excluded data based on raw geospatial data quality. In the first row, I remove duplicate customer records (174 records, 38 of them dated) across the two datasets, leaving 686,891

Table B1 — Data pipeline and sample selection

Step	Networks dropped	Buildings	Metered customers	Dated customers	Networks remaining
Raw meter records			*687,065*	*34,125*	*7,079*
1. Duplicate records			174	38	7,079
2. Failed network assembly	508		60,690	2,539	6,571
3. Transformer filtering and unmatched meters	4,137		289,321	0	2,434
Reconstructed distribution networks		*951,047*	*336,880*	*31,548*	*2,434*
1. Kigali	359	208,818	84,153	1,768	2,075
2. Date coverage $< 80\%$	1,629	652,346	239,584	16,928	446
3. Outside height-raster footprint	8	396	362	358	438
4. Unclear institutional details	14	4,486	1,489	1,409	424
5. Building beyond 1,200 m	18	7,192	1,327	1,259	406
6. Fewer than three consumers	120	17,482	163	163	286
7. Rollout window < 1 week	109	18,040	2,697	2,671	177
Selected sample		*42,287*	*7,105*	*6,992*	*177*

Notes. Italicized rows report totals at that stage of the pipeline, and numbered rows report counts dropped at that step.

unique metered customers. In the second row, I drop the 508 network components that fail assembly: components without any customer data or without any pole or line data.[49] Although the LCP algorithm could construct networks for customer-only clusters, I keep these out of my sample to minimize the LCP algorithm's impact on the reduced-form results. This drops 60,690 customers, 2,539 of them dated. This row also includes networks with multiple transformers that I could not separate. In the third row, I deduplicate transformers via spatial clustering, filter out step-up and transmission network transformers, remove unmatched meters, and drop distribution networks that have no customers with installation date data. Together, these filters drop 4,137 of the 6,571 network components and 289,321 metered customers.[50]

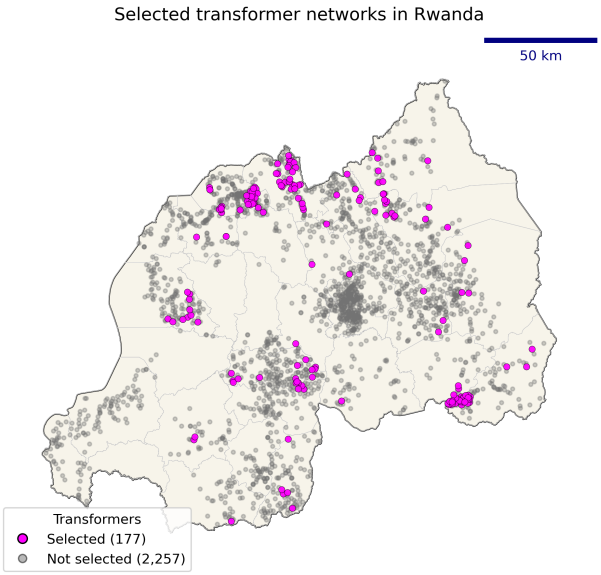
At this point, there are 336,880 observed customer meters, 31,548 of which are dated, and 2,434 transformer networks. I join these meters to Google Open Buildings footprints, and under Assumption 1, there are 951,047 buildings in the vicinity of the reconstructed distribution networks. First, I drop Kigali to focus on rural electrification. Second, I drop distribution networks with less than 80% date coverage among metered customers. This is a critical step: without it, I would not know who induced which pieces of infrastructure or who provided or received externalities. This drops 1,629 distribution networks.

Third, I drop eight networks outside the footprint of Google's 2.5D building-height raster. Fourth, I drop fourteen transformer networks with unclear institutional details: eleven with undated public-service customers, one refugee camp, and two resettlement zones. As previously discussed, the EARP fully subsidizes health, education, and local administration public services, and electrifies everyone en route

49. The transformer dataset is not distribution-specific; it includes transmission network transformers.

50. This number may seem large, but it includes transformer upgrades where additional transformers are added to reinforce the existing network.

Figure B1 — Sample distribution networks relative to all LV transformers



(World Bank Independent Evaluation Group 2019). Because customers on these networks do not pay the full connection charge, their connection decisions may not reveal their willingness to pay the charges implied by distance-based policies. Further, if these connections induced poles earlier than observed, this would substantially inflate the systematic measurement bias discussed at the end of subsection 4.4. The same applies to refugee camps and resettlement zones, which may reflect government policy rather than decentralized consumer demand. Fifth, I drop eighteen distribution networks whose farthest assigned building is more than 1,200 meters from the transformer. These reflect imperfect geospatial network cleaning. Sixth, I drop 120 networks with fewer than three consumers, for which the peer instrument cannot be constructed. Finally, I drop 109 transformer networks with rollout windows shorter than one week. With a weekly time period, these are degenerate in panel regressions, so I include them here for clarity. These sample selection choices leave 177 distribution networks in the sample out of the 2,075 non-Kigali reconstructed networks (2,434 including Kigali). The networks sit in 18 utility branches out of 33 total.

Figure B1 shows the geographic distribution of the selected networks. Table B2 provides sample statistics comparing the sample against non-Kigali reconstructed (but undated) networks, which are the networks remaining after the first row of the second panel of Table B1.

Table B2 — Sample statistics for the selected sample versus all non-Kigali reconstructed networks

Measure	Selected sample	Non-Kigali reconstructed
Totals		
Distribution networks	177	2,075
Branches represented	18	30
Buildings	42,287	742,229
Metered customers	7,105	252,727
Dated customers	6,992	29,780
Mean date coverage	0.98	0.30
Per network		
Metered customers (mean)	40.1	121.8
Metered customers (median)	27	78
Buildings (mean)	238.9	357.7
Building connect rate	0.17	0.34
LV line km (mean)	5.5	9.3
Poles (mean)	119.6	203.8
Synthetic-pole share	0.60	0.70
Max building distance, m (mean)	781	1,046
Distance to Kigali, km (mean)	43.8	45.0
Building covariates (mean)		
Building area (m^2)	46.6	61.3
Building height (m)	2.44	2.88
Nightlights (mean)	0.039	0.648
Terrain ruggedness index	3.99	3.18
New construction share	0.160	0.127
Customer types (share of labeled)		
Labeled customers	6,891	249,336
Residential	0.977	0.888
Commercial	0.020	0.092
Public service	0.002	0.018
Industrial	0.0004	0.001
Apartment	0.000	0.0005

C — Theoretical Proofs

C.1. Derivation of the Shapley Cost Share

Specifically, the Shapley cost shares average each customer's marginal contribution across possible coalitions. Given surplus functions,

$$\begin{aligned} w(\emptyset) &= 0, \\ w(\{i\}) &= V_i - c_i, \\ w(\{j\}) &= V_j - c_j, \\ w(\{i,j\}) &= V_i + V_j - (1+\rho)C, \end{aligned}$$

Shapley payoffs to customer i are

$$\begin{aligned} \phi_i &= \frac{1}{2} w(\{i\}) + \frac{1}{2}\left[w(\{i,j\}) - w(\{j\})\right] \\ &= \frac{1}{2}(V_i - c_i) + \frac{1}{2}\left[(V_i + V_j - (1+\rho)C) - (V_j - c_j)\right] \\ &= V_i - \frac{1}{2}\left[(1+\rho)C + c_i - c_j\right], \end{aligned}$$

and substituting into the cooperation payoff gives Shapley cost shares κ_i.

C.2. Proof of Lemma 1

The strategy profile *(Wait$_i$, Adopt$_j$)* is a pure-strategy Nash equilibrium if and only if

$$V_j - c_j \geq 0 \qquad \text{and} \qquad \beta(V_i - c_i') \geq V_i - \kappa_i.$$

Customer j is willing to pay the unilateral connection charge, and customer i prefers connecting second to adopting jointly.

As i's net surplus is assumed to be higher $V_i - c_i \geq V_j - c_j$, both customers are willing to pay the unilateral connection charge $V_i - c_i \geq V_j - c_j \geq 0$. As the Shapley cost shares from Equation 2 will imply that $\kappa_i - \kappa_j = c_i - c_j$, $(V_i - \kappa_i) - (V_j - \kappa_j) = (V_i - c_i) - (V_j - c_j)$, so the j's free-riding condition implies

$$\beta(V_j - c_j') - (V_j - \kappa_j) \geq \beta(V_i - c_i') - (V_i - \kappa_i) \qquad \geq 0 \tag{C.1}$$

$$\beta(V_j - c_j') \geq V_j - \kappa_j. \tag{C.2}$$

So j prefers to connect second when i adopts. As i is willing to adopt unilaterally and j prefers to wait, *(Adopt$_i$, Wait$_j$)* is also a pure-strategy Nash equilibrium. Therefore, if i has a greater net unilateral surplus $V_i - c_i \geq V_j - c_j$, the parameter region supporting *(Wait$_i$, Adopt$_j$)* is contained in the region supporting *(Adopt$_i$, Wait$_j$)*, so *(Wait$_i$, Adopt$_j$)* will never be a unique Nash equilibrium. □

C.3. *Proof of Lemma 2*

For any $\sigma_j \in [0,1]$, the net gain from adoption is strictly increasing in the customer's connection value $\frac{\partial \Delta_i}{\partial V_i} = \sigma_j(1-\beta) + (1-\sigma_j)\big[1-\beta W'(V_i)\big] > 0$. This implies a cutoff rule where the customer is indifferent between adopting and not adopting when $\Delta_i(V^*(\sigma_j);\sigma_j) = 0$. The neighbor's adoption probability σ_j determines this cutoff. The cutoff need not lie in $[\underline{V}, \overline{V}]$, so there are corner cases in which every type adopts or every type waits. The net gain $\Delta_i(V_i;\sigma_j)$ is continuously differentiable in both parameters and strictly increasing in V_i, so the Implicit Function Theorem gives Equation 8. As just indicated, the denominator is positive. Under strategic substitutes, the numerator is negative $\frac{\partial \Delta_i}{\partial \sigma_j} = \frac{\partial^2 \pi_i}{\partial \sigma_i \partial \sigma_j} < 0$, so the overall term is positive and the cutoff increases with σ_j. Under strategic complements the cutoff decreases with σ_j. □

C.4. *Proof of Proposition 2*

I first establish existence. Without loss of generality, suppose that neighbor j follows a cutoff rule with threshold v, adopting when $V_j \geq v$. Because V_j is an independent draw from the common distribution F, customer i assigns probability $\sigma_j = \Pr[s_j(V_j) = \text{Adopt}] = 1 - F(v)$ to the event that j adopts. Let $\widetilde{V}^*(\sigma_j) \in [\underline{V}, \overline{V}]$ denote customer i's best-response cutoff from Lemma 2 as a function of the neighbor's mixed strategy. The best-response cutoff as a function of the neighbor's cutoff is therefore $\mathrm{BR}(v) = \widetilde{V}^*(1-F(v))$. Because F and $\widetilde{V}^*(\sigma_j)$ are continuous, BR is a continuous map from $[\underline{V}, \overline{V}]$ into itself. Let $g(v) = \mathrm{BR}(v) - v$. Since $\mathrm{BR}(v) \in [\underline{V}, \overline{V}]$, $g(\underline{V}) \geq 0$ and $g(\overline{V}) \leq 0$. The intermediate value theorem therefore gives some $V^* \in [\underline{V}, \overline{V}]$ such that $g(V^*) = 0$, and $V^* = \mathrm{BR}(V^*)$. As the cutoff rule with threshold V^* is a best response to itself, it is a symmetric cutoff equilibrium.

I now show uniqueness under strategic substitutes. By Lemma 2, strategic substitutes imply that $\widetilde{V}^*(\sigma_j)$ is weakly increasing in σ_j. Because $1 - F(v)$ is weakly decreasing in v, $\mathrm{BR}(v) = \widetilde{V}^*(1-F(v))$ is weakly decreasing in v. Hence, for any $v_2 > v_1$, $g(v_2) - g(v_1) = \mathrm{BR}(v_2) - \mathrm{BR}(v_1) - (v_2 - v_1) < 0$. Therefore, g is strictly decreasing and can have at most one zero. As existence shows that it has at least one zero, there is exactly one fixed point and so the symmetric cutoff equilibrium is unique under strategic substitutes. Under strategic complements, BR need not be decreasing, and so there may be multiple fixed points and equilibria.

Finally, consider behavior in any symmetric cutoff equilibrium with threshold V^*. By Lemma 2, a player ℓ adopts if $V_\ell > V^*$, waits if $V_\ell < V^*$, and is indifferent if $V_\ell = V^*$. Consequently, whenever two players' values lie on opposite sides of the cutoff, only the higher-value player adopts first. Further, if a type adopts, then every higher type also adopts, so the adoption decision is weakly increasing in connection value. High-connection-value players therefore select into earlier adoption at the first-mover charge, while low-connection-value players select into waiting for the second-mover charge. □

C.5. *Proof of Proposition 3*

Holding $W(V)$ fixed, a waiting customer can choose between continuing the baseline war of attrition or holding out until T^k. The marginal payoff from increasing their own adoption probability becomes

$$\Delta_i^{T^k}(V;\sigma_j) = \sigma_j\big[(V-\kappa) - \beta(V-c')\big] + (1-\sigma_j)\left[(V-c) - \max\left\{\beta W(V),\, \beta^{T^k-t}(V-c^k)\right\}\right]. \quad \text{(C.3)}$$

The unilateral net benefit of adoption (the second term) is the minimum of $V - c - \beta W(V)$ and $V - c - \beta^{T^k - t}(V - c^k)$. Both terms are strictly increasing in V because $\beta W'(V) < 1$ and $\beta^{T^k - t} < 1$. The minimum is continuous and strictly increasing, including at the point where the maximizing continuation payoff switches branches, but it will not necessarily be differentiable there. Therefore, $\Delta_i^{T^k}$ is continuous and strictly increasing in V, and the best response in the foreseeable-event equilibrium remains a cutoff rule, as in Lemma 2.

1. *Baseline adoption unaffected.* When the baseline continuation value is weakly preferable to waiting for the foreseeable event at the baseline cutoff, $\beta W(V^*) \geq \beta^{T^k - t}(V^* - c^k)$, the foreseeable option value does not change the payoff. Therefore, the net gains across the wars are the same, $\Delta_i^{T^k}(V^*; \sigma^*) = \Delta_i(V^*; \sigma^*) = 0$, where the last equality comes from indifference, so the baseline symmetric cutoff equilibrium V^* is unaffected. Uniqueness from Proposition 2 then implies $V^*_{t,T^k} = V^*$.

2. *Delay into the externality.* The baseline cutoff strategy is no longer optimal when the foreseeable network extension raises the payoff from waiting, so that $\Delta_i^{T^k}(V^*; \sigma^*) < 0$. As $\Delta_i^{T^k}$ is strictly increasing in V, the cutoff best response to the baseline equilibrium strategy is above V^*. Let $r \in \{0, T^k\}$ correspond to the baseline and foreseeable-event wars, respectively, so that $\mathrm{BR}_0(v)$ and $\mathrm{BR}_{T^k}(v)$ denote the cutoff best responses to a neighbor cutoff, and $g_r(v) = \mathrm{BR}_r(v) - v$. As in Proposition 2, strategic substitutes imply that each g_r is strictly decreasing and has a unique zero, so each best-response function has a unique fixed point. The higher payoff from waiting implies a higher cutoff strategy $\mathrm{BR}_{T^k}(v) \geq \mathrm{BR}_0(v)$, so the fixed point of the foreseeable-event war is farther to the right and implies a higher cutoff, $g_{T^k}(V^*) \geq g_0(V^*) = 0 \implies V^*_{t,T^k} \geq V^*$. As F is increasing, foreseeable-event adoption probabilities are lower than baseline, so $\sigma^*_{t,T^k} = 1 - F(V^*_{t,T^k}) \leq 1 - F(V^*) = \sigma^*$.

3. *Frozen adoption hazards (optional phase).* If the other customer waits, holding out for customer k weakly dominates unilateral adoption because $\beta^{T^k - t}(V - c^k) \geq V - c$ in the maximum of Equation C.3. If the other customer adopts, $\beta(V - c') > V - \kappa$ makes free-riding preferable to joint adoption. As waiting dominates adoption, both players wait and current adoption is zero.

4. *Joint adoption with k at time T^k.* At the event, customer i can either join k's connection request and receive the contemporaneous payoff $V_i - \kappa^k$, or wait one more period and receive the discounted post-event payoff $\beta(V_i - c^k)$. The benefit of joining is $(V_i - \kappa^k) - \beta(V_i - c^k) = (1 - \beta)V_i - \kappa^k + \beta c^k$, which is strictly increasing in V_i. Therefore, customer i joins k if and only if

$$V_i \geq V^{\text{joint}}_{T^k} \equiv \frac{\kappa^k - \beta c^k}{1 - \beta}.$$

   Only customers who have not yet adopted at T^k can join. Let $V^{\text{surv}}_{T^k} \equiv \min_{s < T^k} V^*_{s,T^k}$ denote the lowest cutoff faced before the event. A customer reaches T^k if and only if $V_i < V^{\text{surv}}_{T^k}$. The time T^k joint-adoption probability among the remaining customers is consequently $\big(F(V^{\text{surv}}_{T^k}) - F(V^{\text{joint}}_{T^k})\big)/F(V^{\text{surv}}_{T^k})$ when $V^{\text{joint}}_{T^k} < V^{\text{surv}}_{T^k}$, and zero otherwise.

5. *Realization of free-riding after time T^k.* Consider a customer who reached time T^k because they waited

for k's foreseeable extension, which provides externality $c - c^k \geq 0$, but who did not jointly adopt with k. This implies $V_i - \kappa^k < \beta(V_i - c^k)$, or equivalently $V_i < V_{T^k}^{\text{joint}}$. Their option value of waiting for the event was realized. They now observe the post-event charge c^k and the continuation value $W^k(V_i)$ of the war that resumes after the event, and adopt immediately if $V_i - c^k \geq \beta W^k(V_i)$. Customers who delayed adoption to receive neighbor externalities adopt sequentially, immediately after the event.

6. *A higher baseline after the event.* If neither i nor j joined k's coalition or free-rode, they still observe a lower connection charge c^k. Holding $W(V)$, c', and κ fixed, replace the unilateral charge c in Equation 7 by c^k and denote the resulting marginal payoff from own adoption by $\Delta_i^{T^k+1}$. Then for every V and σ_j,

$$\Delta_i^{T^k+1}(V;\sigma_j) - \Delta_i(V;\sigma_j) = (1-\sigma_j)(c - c^k) \geq 0,$$

so the post-event cutoff best response is weakly below the baseline best response. As above, let BR_{T^k+1} be the post-event best response and let $g_{T^k+1}(v) = \text{BR}_{T^k+1}(v) - v$. Under strategic substitutes, g_{T^k+1} is strictly decreasing and has a unique zero. At the baseline cutoff V^*, $g_{T^k+1}(V^*)$ is nonpositive:

$$g_{T^k+1}(V^*) = \text{BR}_{T^k+1}(V^*) - V^* \leq \text{BR}_0(V^*) - V^* = 0,$$

so $V^{*,k} \leq V^*$. As F is increasing, baseline adoption hazards are higher after the event: $\sigma^{*,k} = 1 - F(V^{*,k}) \geq 1 - F(V^*) = \sigma^*$. In the resumed war, the remaining customers adopt exactly when $V_i \geq V^{*,k}$. Therefore, the sequential adopters of the previous phase are those with $V^{*,k} \leq V_i < V_{T^k}^{\text{joint}}$.

□

D — Reduced-Form

Table 4 reports the reduced form and first stages of Equation 23 with and without the social peer effect $[\mathbf{D}^{\circ(-1)}\mathbf{y}^*_{-it}]$ and social contextual trends $\mathbf{D}^{\circ(-1)}\mathbf{X}t$. Table D1 reports an intermediate specification that controls for the endogenous social peer effect but not contextual trends.

Table D1 — Instrumental-variable reduced form and first stages: social peer effect without social contextual trends

	Reduced form	First stage	First stage
Dependent variable	y_{it}	S_{it}	$F_{it}(1,8)$
Units	pp	10% reduction in current distance	
Joint instrumental variable regressions			
Demand-side peer-of-peer Z_{it}^{F} (1 s.d.)	+0.200 (0.074)	+0.019 (0.017)	-0.207 (0.053)
Supply-side shift-share Z_{it}^{S} (1 s.d.)	-0.142 (0.125)	+0.347 (0.095)	+0.500 (0.261)
Dependent variable mean	0.379	0.26	1.25
Conditional F-statistic	–	14.0	20.0
Joint Wald F-statistic	–	6.9	9.9
Single instrumental variable regressions			
Standard Z_{it}^{S} F-statistic	–	13.4	3.5
Standard Z_{it}^{F} F-statistic	–	1.4	14.8
Observations		1,245,910	
Distribution networks		177	
Potential adopters		38,826	

Notes. Each column is a separate regression following Equation 23, estimated on the same sample as Table 4. Controls include the social peer effect $[\mathbf{D}^{\circ(-1)}\mathbf{y}^{*}_{-it}]$ but not the social contextual trends $\mathbf{D}^{\circ(-1)}\mathbf{X}t$. The first column is the reduced form, the second the first stage for S_{it}, and the final column the first stage for $F_{it}(1,8)$. Standard errors are two-way clustered at the transformer and calendar-week level. The reduced-form outcome is measured in percentage points, while the first-stage estimates are scaled to an externality that decreases i's current distance to the grid by 10%. Conditional and joint Wald F-statistics are reported below the means (Sanderson and Windmeijer 2016; Andrews, Stock, and Sun 2019). Under single instrumental variable regressions, each cell reports the F-statistic of the endogenous variable on a single instrument with all controls.